Group Theory

Useful Facts

- Let G be a finite group, and suppose G acts on itself by conjugation. Then, the stabilizer of $x \in G$ is denoted $C_G(x)$, and by the orbit-stabilizer theorem, it follows that the number of *elements* of the conjugacy class of x, $K_G(x)$, is given by $[G : C_G(x)]$. The *class equation* says we may find

  $$|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(x_i)],$$

  where $\{x_1, \dots, x_r\}$ is a family of representatives of conjugacy classes.
- If G acts on its *subgroups* by conjugation, then the *normalizer* of $H \leq G$, denoted $N_G(H)$, is the stabilizer of H under conjugation. Similarly, number of subgroups conjugate to $H \leq G$ is given by $[G : N_G(H)]$.
- Normalizers and centralizers commute with conjugates. That is, if $x = gyg^{-1}$, then $C_G(x) = gC_G(y)g^{-1}$, and if $H = gKg^{-1}$, then $N_G(H) = gN_G(K)g^{-1}$.
- If $H \trianglelefteq G$ is a normal subgroup, then G acts on H by conjugation. The corresponding permutation $\rho\colon G \to \operatorname{Aut}(H)$ has kernel $C_G(H)$, where

  $$C_G(H) = \left\{ g \in G : ghg^{-1} = h \text{ for all } h \in H \right\}.$$

- A subgroup $H \leq G$ is called *characteristic* if $\sigma(H) = H$ for all automorphisms $\sigma \in \operatorname{Aut}(G)$. If $K \leq H \trianglelefteq G$ is a chain of subgroups with K characteristic in H, while H is normal in G, then $K \trianglelefteq G$ is normal.

  The subgroups $[G, G]$ and $Z(G)$ are characteristic in G.
- Every p-group has nontrivial center.
- If G acts on a set X, and H is the stabilizer subgroup of some $x \in X$, then if $\rho\colon G \to \operatorname{Sym}(X)$ is the permutation representation for this action, then $\ker(\rho) \leq H$.
- Cauchy's Theorem: if p is a prime that divides the order of G, then there exists an element of G with order p.
- Sylow Theorems: suppose we may write $|G| = p^k m$ with $\gcd(p, m) = 1$. The following hold.
  (i) **Existence:** There exists a p-Sylow subgroup, i.e. a subgroup of order p^k.
  (ii) **Conjugacy:** The p-Sylow subgroups of G are all conjugate to each other.
  (iii) **Number:** If n_p denotes the number of p-Sylow subgroups, then $n_p = [G : N_G(P)]$ for some p-Sylow subgroup P of G, $n_p \equiv 1 \bmod p$, and $n_p | m$.
    - Existence is best used for problems related to finding p-Sylow subgroups of $\mathrm{GL}_n\left(\mathbb{F}_p\right)$.
    - One can use the number of the p-Sylow subgroups to determine if a group is simple or not. If this is not deterministic, then the next is to use the fact that conjugation yields a faithful action $G \to \operatorname{Sym}(X)$, where X is the set of p-Sylow subgroups. If the order of G does not divide $n_p!$ for any possible n_p, it follows that $|X| = n_p = 1$.
    - When counting elements, use the fact that if $|G| = pm$ with $\gcd(p, m) = 1$, then all the p-Sylow subgroups intersect at the identity.
- Every conjugacy class in the symmetric group S_n is given by the cycle type. Moreover, given an element $\sigma \in A_n$, the conjugacy classes $K_{A_n}(\sigma) = K_{S_n}(\sigma)$ if and only if there is an odd permutation in the centralizer $C_{S_n}(\sigma)$. If there is no odd permutation in $C_{S_n}(\sigma)$, then the conjugacy class $K_{S_n}(\sigma)$ splits into two separate conjugacy classes over A_n.

- Burnside's Lemma: If a finite group G acts on a finite set X, then the number of orbits $|X/G|$ is given by
$$|X/G| = \frac{1}{|G|}\sum_{g\in G}|X^g|,$$
where $X^g = \{x \in X : g\cdot x = x\}$ is the set of fixed points of $g \in G$.
- An automorphism $\varphi \in \operatorname{Aut}(G)$ is called *inner* if there exists $g \in G$ such that $\varphi(h) = ghg^{-1}$ for all $h \in G$. The set of inner automorphisms is denoted $\operatorname{Inn}(G)$. It can be shown that $\operatorname{Inn}(G)$ is isomorphic to $G/Z(G)$.

  In particular, showing that $|\operatorname{Aut}(G)| > |G/Z(G)|$ is sufficient to show that there exists an outer automorphism of G.

Group Actions: Double Cosets

Consider a group G with subgroups H and K. Then, the group $H \times K$ acts on G via $(h,k)\cdot x = hxk^{-1}$, where the inverse emerges from the fact that
$$\begin{aligned}(h_1,k_1)(h_2,k_2)\cdot x &= h_1h_2xk_2^{-1}k_1^{-1}\\ &= (h_1h_2,k_1k_2)\cdot x.\end{aligned}$$
The orbit of $x \in G$ under this action is denoted HxK.

- The double coset HxK is a union of right cosets of H and a union of left cosets of K, written as
$$\begin{aligned}HxK &= \bigcup_{k\in K} Hxk\\ &= \bigcup_{h\in H} hxK.\end{aligned}$$
- Double cosets are either disjoint or coincide; this follows from the definition of the orbit. Similarly, the group G is a disjoint union of the double cosets.
- The stabilizer of the right coset Hxk is the set of all $\ell \in K$ such that $(Hxk)\ell = Hxk$, or the set of all $\ell \in K$ such that $H\left(xk\ell(xk)^{-1}\right) = H$. By the definition of the right coset, this gives $(xk)\ell(xk)^{-1} \in H$, or $\ell \in (xk)^{-1}H(xk)$.
- By the orbit–stabilizer theorem, the number of right H-cosets in HxK is then given by $\left[K : K\cap x^{-1}Hx\right]$. Similarly, the number of left K-cosets in HxK is given by $\left[H : H\cap x^{-1}Kx\right]$. In particular, since these right H-cosets and/or left K-cosets are disjoint, we have
$$\begin{aligned}|HxK| &= |H|\left[H : H\cap x^{-1}Kx\right]\\ &= |K|\left[K : K\cap x^{-1}Hx\right].\end{aligned}$$

Semidirect Products

Suppose N and H are subgroups of a group G, where N is normal and H is such that $N\cap H = \{e\}$. We will obtain a description of the structure of G in terms of the structure of N and H.

A short exact sequence of groups is a sequence of groups and homomorphisms
$$1 \to N \xrightarrow{\varphi} G \xrightarrow{\psi} H \to 1$$
where ψ is surjective and φ identifies N with $\ker(\psi)$. In other words, the sequence is exact if N is normal in G and ψ induces an isomorphism $G/N \cong H$.

Definition: Let N and H be groups. A group G is an *extension* of H by N if there is an exact sequence of groups

$$1 \to N \to G \to H \to 1.$$

The exact sequence is said to split if H can be identified with a subgroup of G, so that $N \cap H = \{e\}$.

If N is normal, then observe that any subgroup H of G acts on N by conjugation; furthermore, conjugation determines a homomorphism

$$\iota\colon H \to \operatorname{Aut}(N)$$

given by $h \mapsto \iota_h$, where $\iota_h(n) = hnh^{-1}$.

Now, we consider a certain tautology. If $G = NH$ with $N \cap H = \{e\}$ and N normal, then observe that

$$n_1 h_1 n_2 h_2 = \left(n_1\left(h_1 n_2 h_1^{-1}\right)\right)(h_1 h_2).$$

In particular, if we know the conjugation action of H on N, then we can recover the operation in G from this information and the operations in N and H.

This allows us to construct semidirect products by considering any two arbitrary groups N and H with an arbitrary group homomorphism

$$\begin{aligned} \theta\colon H &\to \operatorname{Aut}(N)\\ h &\mapsto \theta_h. \end{aligned}$$

Then, we may define an operation on $N \times H$ by taking

$$(n_1, h_1)\cdot(n_2, h_2) = \left(n_1 \theta_{h_1}(n_2), h_1 h_2\right).$$

Observe that in the special case where $N, H \leq G$ with $G = NH$, N normal, and $N \cap H = \{e\}$, the operation θ can be given by the map ι. The group $(N \times H, \cdot)$ is denoted $N \rtimes_\theta H$, and is called the semidirect product of N and H with respect to θ. Observe that the ordinary direct product is a semidirect product where θ is the trivial map.

We will discuss some examples.

(i) If G has a normal subgroup K of order n and a subgroup H of order m, with $HK = G$, $H \cap K = \{e\}$, and $\gcd(m, |\operatorname{Aut}(K)|) = 1$, then $G \cong H \times K$.

(ii) Suppose H is an abelian group of arbitrary order, and let $K = \mathbb{Z}/2\mathbb{Z}$. Define $\varphi\colon K \to \operatorname{Aut}(H)$ by inversion. Then, $G = H \rtimes K$ contains the subgroup H of index 2, with $xhx^{-1} = h^{-1}$ for all $h \in H$ and $e \neq x \in K$.

When $H \cong \mathbb{Z}/n\mathbb{Z}$, this yields the dihedral group D_n of order $2n$, while if $H \cong \mathbb{Z}$, we have the infinite dihedral group D_∞.

(iii) If H is a group, and $K = \operatorname{Aut}(H)$, then $\varphi\colon K \to \operatorname{Aut}(H)$ given by the identity yields the semidirect product $H \rtimes \operatorname{Aut}(H)$, denoted the *holomorph* of H.

We now discuss a special case of a semidirect product, known as a *wreath product*.

Definition: Let D and Q be groups. Let $Q \curvearrowright \Omega$ act on a finite set. Write $K = D^\Omega = \prod_{\omega\in\Omega} D$. Then, the *wreath product* of D by Q, written $D \wr Q$, is given by $Q \ltimes_\varphi K$, where

$$\varphi(q)\cdot(d_\omega)_\omega = \left(d_{q\cdot\omega}\right)_\omega.$$

If $D \curvearrowright \Lambda$ is another action of a finite set, then we may naturally take an action $(D \wr Q) \curvearrowright (\Lambda \times \Omega)$ by taking, for each $d \in D$ and $\omega \in \Omega$, a permutation d_ω^* by taking

$$d_\omega^*(\lambda, \omega') = \begin{cases} (d\lambda, \omega') & \text{if } \omega' = \omega,\\ (\lambda, \omega') & \text{if } \omega' \neq \omega. \end{cases}$$

We have $d_\omega^* e_\omega^* = (de)_\omega^*$ for any $d, e \in D$. Defining

$$D_\omega^* = \{d_\omega^* : d \in D\},$$

we obtain that D_ω^* is a subgroup of $\mathrm{Sym}(\Lambda \times \Omega)$ and for each ω, the map $d \mapsto d_\omega^*$ defines an isomorphism $D \cong D_\omega^*$.

For each $q \in Q$, we may define a permutation q^* of $\Lambda \times \Omega$ by taking

$$q^*(\lambda, \omega') = (\lambda, q\omega'),$$

and take $Q^* = \{q^* : q \in Q\}$. Similarly, we have that the map $q \mapsto q^*$ defines an isomorphism $Q \to Q^*$.

Solvable Groups

Definition: A series of subgroups G_i of a group G is a decreasing sequence of subgroups

$$G = G_0 \triangleright G_1 \triangleright G_2 \supsetneq \cdots,$$

where the length of the series is the number of strict inclusions.

A normal series is a series of subgroups where $G_{i+1} \trianglelefteq G_i$ for each i. The number $\ell(G)$ is the length of a maximal normal series.

A *composition series* is a normal series

$$G = G_0 \triangleright G_1 \triangleright \cdots \triangleright G_n = \{e\}$$

where the successive quotients G_i/G_{i+1} are simple.

Theorem (Jordan–Hölder Theorem): Let G be a group, and suppose there are two composition series

$$\begin{gathered} G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\} \\ G = G_0' \supsetneq G_1' \supsetneq G_2' \supsetneq \cdots \supsetneq G_m = \{e\}. \end{gathered}$$

Then, $m = n$, and the quotient groups $H_i = G_i/G_{i+1}$ and $H_i' = G_i'/G_{i+1}'$ agree up to isomorphism and permutation.

Proof. Considering the following composition series

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\}$$

we induct on n. If $n = 0$, then G is trivial and there is nothing to prove.

Suppose $n > 0$, and consider another composition series

$$G = G_0' \supsetneq G_1' \supsetneq G_2' \supsetneq \cdots \supsetneq G_m' = \{e\}.$$

If $G_1 = G_1'$, then the result follows from the induction hypothesis since G_1 has a composition series of length $n-1$.

Therefore, we may assume that $G_1 \neq G_1'$. We observe that $G_1 G_1' = G$, since $G_1 G_1'$ is normal in G and $G_1 \supsetneq G_1 G_1'$, but there are no proper normal subgroups between G_1 and G since G/G_1 is simple.

Consider the group $K = G_1 \cap G_1'$. The distinct subgroups $G_i \cap K$ determine a composition series

$$K \supsetneq K_1 \supsetneq K_2 \supsetneq \cdots \supsetneq K_r = \{e\}.$$

By the Second Isomorphism Theorem, we have

$$\frac{G_1}{K} = \frac{G_1}{G_1 \cap G_1'}$$

$$\begin{aligned}&\cong \frac{G_1G_1'}{G_1'}\\&= \frac{G}{G_1'}\end{aligned}$$

and $G_1'/K \cong G/G_1$. These quotients are simple, so we obtain two new composition series

$$\begin{gathered}G \supsetneq G_1 \supsetneq K \supsetneq K_1 \supsetneq \cdots \supsetneq \{e\}\\ G \supsetneq G_1' \supsetneq K \supsetneq K_1 \supsetneq \cdots \supsetneq \{e\}.\end{gathered}$$

These composition series only differ at the first step. The two series have the same length and the same quotients. Next, we claim that the first of these series has the same length and quotients of the original series. Since

$$G_1 \supsetneq K \supsetneq K_1 \supsetneq K_2 \supsetneq \cdots \supsetneq K_r = \{e\}$$

is a composition series for G_1, the induction hypothesis gives that it must have the same length and quotients as the composition series

$$G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\}.$$

Therefore, applying the induction hypothesis, it follows that the series

$$G_1' \supsetneq K \supsetneq K_1 \supsetneq K_2 \supsetneq \cdots \supsetneq K_r = \{e\}$$

has the same length and quotients as

$$G_1' \supsetneq G_2' \supsetneq \cdots \supsetneq G_m' = \{e\}.$$

□

Proposition: Let G be a group, and let N be a normal subgroup of G. Then, G has a composition series if and only if N and G/N have composition series. Furthermore,

$$\ell(G) = \ell(N) + \ell(G/N),$$

and the composition factors of G consist of the composition factors of N and of G/N.

Proof. If G/N has a composition series, then the subgroups appearing in it correspond to subgroups of G containing N with isomorphic quotients. Therefore, if both G/N and N have composition series, concatenating produces a composition series for G.

For the converse, suppose G has a composition series

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\},$$

and that N is a normal subgroup of G. Taking intersections with N gives a sequence of subgroups

$$N = G \cap N \supseteq G_1 \cap N \supsetneq \cdots \supsetneq \{e\} \cap N = \{e\}$$

such that $G_{i+1} \cap N$ is normal in $G_i \cap N$ for all i. We claim that this is a composition series upon deleting repetitions. For this, we will establish that the quotients $(G_i \cap N)/(G_{i+1} \cap N)$ are either trivial or isomorphic to G_i/G_{i+1}. Consider the homomorphism

$$G_i \cap N \hookrightarrow G_i \to \frac{G_i}{G_{i+1}}.$$

The kernel is $G_{i+1} \cap N$, so by the first isomorphism theorem, we have an injective homomorphism

$$\frac{G_i \cap N}{G_{i+1} \cap N} \hookrightarrow \frac{G_i}{G_{i+1}},$$

where we identify the image with a subgroup of G_i/G_{i+1}. This subgroup is normal since N is normal in G, and since G_i/G_{i+1} is simple, we get our desired result.

As for G/N, we have the composition series

$$\frac{G}{N} \supseteq \frac{G_1N}{N} \supseteq \frac{G_2N}{N} \supseteq \cdots \supseteq \{e_{G/N}\}.$$

such that $(G_{i+1}N)/N$ is normal in $(G_iN)/N$. By the third isomorphism theorem, the quotients

$$\frac{(G_iN)/N}{(G_{i+1}N)/N} \cong \frac{G_iN}{G_{i+1}N}$$

are isomorphic. Considering the homomorphism

$$G_i \hookrightarrow G_iN \to \frac{G_iN}{G_{i+1}N},$$

we have that this homomorphism is surjective, and the subgroup G_{i+1} of the source is sent to the identity of the target. By the first isomorphism theorem, there is a surjective homomorphism

$$\frac{G_i}{G_{i+1}} \to \frac{G_iN}{G_{i+1}N},$$

so that $G_iN/G_{i+1}N$ is either trivial or isomorphic to it as G_i/G_{i+1} is simple. $\square$

Composition series are the first major step in understanding the structure of groups. Next, we consider a special series known as the *derived series*.

Definition: If G is a group, then the commutator subgroup, denoted G' or $[G,G]$, is the subgroup generated by all the commutators $[g,h] = ghg^{-1}h^{-1}$.

The following follows directly from the definitions of the commutator subgroup.

Theorem: If G is a group, and G' is its commutator subgroup, then

(i) G' is normal in G (in fact, G' is *characteristic* in G);

(ii) the quotient G/G' is commutative;

(iii) G' yields the "maximal abelian quotient," in the sense that if $N \trianglelefteq G$ is another normal subgroup with G/N commutative, then $G' \leq G$.

Definition: The *derived series* is a descending sequence of subgroups

$$G \supseteq G' \supseteq G'' \supseteq G''' \supseteq \cdots.$$

Note that if G is commutative, then the sequence automatically hits $\{e\}$, and if G is simple and noncommutative, then $G = G' = G'' = \cdots$.

Definition: A group is called *solvable* if its derived series terminates in the identity.

We call a normal series abelian (resp. cyclic) if all the quotients are abelian (resp. cyclic).

Proposition: For a finite group G, the following are equivalent:

(i) all composition factors of G are cyclic;

(ii) G admits a cyclic series ending in $\{e\}$;

(iii) G admits an abelian series ending in $\{e\}$;

(iv) G is solvable.

Proof. Observe that (i) implies (ii) and (ii) implies (iii) follow immediately from the definition. Furthermore, (iii) implies (i) by refining an abelian series to a composition series into a family of cyclic p-groups. Furthermore, (iv) implies (iii) immediately from the fact that the derived series is an abelian series.

Therefore, we must show that (iii) implies (iv). Consider an abelian series

$$G = G_0 \supsetneq G_1 \supsetneq G_1 \supsetneq \cdots \supsetneq G_n = \{e\}.$$

Then, we claim that $G^{(i)} \subseteq G_i$, where $G^{(i)}$ denotes the ith iterated commutator subgroup.

This follows by induction. Since G/G_1 is commutative, it follows that $G' \subseteq G_1$. Since G_i/G_{i+1} is abelian, it follows that $G_i' \subseteq G_{i+1}$, so that

$$\begin{aligned} G^{(i+1)} &= \left(G^{(i)}\right)' \\ &\subseteq G_i' \\ &\subseteq G_{i+1}. \end{aligned}$$

Therefore, we have that $G^{(n)} \subseteq G_n = \{e\}$, so that the derived series terminates at $\{e\}$. □

We will now discuss an application of solvability to understanding the structure of certain subgroups of the symmetric group. This has applications to Galois theory.

Proposition: If $G \leq S_p$ is a transitive solvable subgroup of the symmetric group S_p (where p is prime), then G is contained in the normalizer of some p-Sylow subgroup. The normalizer of any p-Sylow subgroup is a Frobenius group of order $p(p-1)$.

Proof. We start by showing that the normalizer of any p-Sylow subgroup of S_p is a Frobenius group of order $p(p-1)$. For this, observe that the p-Sylow subgroups of S_p are precisely the p-cycles. Given a p-Sylow subgroup $\langle\sigma\rangle$, we start by determining the number of conjugate subgroups of $\langle\sigma\rangle$. Counting, we have $(p-1)!$ elements of S_p in the conjugacy class of σ (i.e., the number of p-cycles), and there are $(p-1)$ elements of the form σ^k that remain in the group $\langle\sigma\rangle$. That is, there are $(p-2)!$ conjugate subgroups of $\langle\sigma\rangle$, so the normalizer of $\langle\sigma\rangle$ is a group of order $p(p-1)$.

Since the automorphisms of the form $\tau\sigma\tau^{-1} = \sigma^k$ for $1 \leq k \leq (p-1)$ are contained in $N_{S_p}(\langle\sigma\rangle)$, it follows that if we let $\mathbb{Z}/p\mathbb{Z} \rtimes \operatorname{Aut}(\mathbb{Z}/p\mathbb{Z})$ denote this isomorphic subgroup, then $N_{S_p}(\langle\sigma\rangle) \geq \mathbb{Z}/p\mathbb{Z} \rtimes \operatorname{Aut}(\mathbb{Z}/p\mathbb{Z})$, whence $N_{S_p}(\langle\sigma\rangle) = \mathbb{Z}/p\mathbb{Z} \rtimes \operatorname{Aut}(\mathbb{Z}/p\mathbb{Z})$, or that $N_{S_p}(\langle\sigma\rangle)$ is a Frobenius group of order $p(p-1)$.

If G is a solvable subgroup, then there is a minimal normal subgroup N of G such that $[N,N] = \{e\}$ (i.e., N is abelian).

We now look at a fundamental group theory fact. Let $N \trianglelefteq G$ be any normal subgroup of a finite group acting on a finite set. Consider an orbit $N \cdot \alpha$. If $g \in G$ is any element, then since N is normal, we have

$$\begin{aligned} g \cdot (N \cdot \alpha) &= (gN) \cdot \alpha \\ &= (Ng) \cdot \alpha \\ &= N \cdot (g \cdot \alpha). \end{aligned}$$

In particular, this means that G acts on the set of N-orbits. Returning to our discussion, observe that if G acts transitively, then N partitions $\{1, \dots, p\}$ into orbits of size m; if there are t such orbits, then we have $mt = p$, so that $m = 1$ or $m = p$. If it were the case that $m = p$, then N would be transitive. Else, if $m = 1$, then it would follow that every N-orbit is a single point (i.e., N fixes all points); yet, since S_p acts faithfully, this would imply that $N = \{e\}$, which would contradict the fact that N is a minimal (nontrivial) normal subgroup. Thus, $m = p$, meaning that N is transitive.

Observe now that N is an abelian group acting transitively on the set $\{1, \dots, p\}$. Since abelian groups that act transitively act by regular permutations, it follows that $|N| = p$. In particular, this means that, since G normalizes N, we have that $G \leq N_{S_p}(\langle\rho\rangle)$ for some p-cycle ρ in S_p. □

Nilpotent Groups

Definition: Let G be a group. Define the following subgroups inductively:

$$\begin{aligned}
Z_0(G) &= \{e\}\\
Z_1(G) &= Z(G)\\
Z_{i+1}(G)/Z_i(G) &= Z(G/Z_i(G)).
\end{aligned}$$

That is, $Z_{i+1}(G)$ is the preimage of $Z(G/Z_i(G))$ under the canonical projection. The series of subgroups

$$Z_0(G) \leq Z_1(G) \leq Z_2(G) \leq \cdots$$

is called the *upper central series* of G.

A group G is called *nilpotent* if $Z_c(G) = G$ for some integer c. The smallest such c is called the *nilpotence class* of G.

Proposition: Let p be a prime, and let P be a group of order p^a. Then, P is nilpotent with nilpotence class at most $a-1$.

Proof. For each $i > 0$, we have that $P/Z_i(P)$ is a p-group, so we know that if $|P/Z_i(P)| > 1$, then $Z(P/Z_i(P)) \neq 1$ by the properties of p-groups.

If $Z_i(P) \neq P$, then $|Z_{i+1}(P)| \geq p|Z_i(P)|$, meaning that $|Z_{i+1}(P)| \geq p^{i+1}$. In particular, $|Z_a(P)| \geq p^a$, so $P = Z_a(P)$, so P is nilpotent of class at most a.

If P were of nilpotence class a, then this would hold if and only if $|Z_i(P)| = p^i$ for all i. Yet, this would give that $Z_{a-2}(P)$ would have index p^2 in P, so $P/Z_{a-2}(P)$ would be order p^2, hence abelian, meaning that $P/Z_{a-2}(P)$ would equal its center and $Z_{a-1}(P)$ would be equal to p. Therefore, P is nilpotent of class at most $a-1$. □

Proposition: The direct product of a finite number of nilpotent groups is nilpotent.

Proof. Suppose $G = H \times K$ (the rest follows inductively). Furthermore, inductively, we assume that $Z_i(G) = Z_i(H) \times Z_i(K)$, where the base case follows from the definition of the direct product. Let $\pi_H \colon H \to H/Z_i(H)$ be the canonical projection, and let π_K be defined analogously.

Then, we have that $\varphi \colon G \to G/Z_i(G)$ is given by

$$\begin{aligned}
G &= H \times K\\
&\xrightarrow{\pi_H \times \pi_K} H/Z_i(H) \times K/Z_i(K)\\
&\xrightarrow{\psi} \frac{H \times K}{Z_i(H) \times Z_i(K)}\\
&= \frac{H \times K}{Z_i(H \times K)}\\
&= G/Z_i(G).
\end{aligned}$$

For shorthand, write $\pi = \pi_H \times \pi_K$. Then, we have

$$\begin{aligned}
Z_i(G) &= \varphi^{-1}(Z(G/Z_i(G)))\\
&= \pi^{-1} \circ \psi^{-1}(Z(G/Z_i(G)))\\
&= \pi^{-1}(Z(H/Z_i(H) \times K/Z_i(K)))\\
&= \pi^{-1}(Z(H/Z_i(H)) \times Z(K/Z_i(K)))\\
&= \pi_H^{-1}(Z(H/Z_i(H))) \times \pi_K^{-1}(Z(K/Z_i(K)))\\
&= Z_{i+1}(H) \times Z_{i+1}(K).
\end{aligned}$$

Therefore, we have $Z_i(G) = Z_i(H) \times Z_i(K)$ for all i. Thus, G is nilpotent with nilpotence class equal to the larger of the nilpotence class of H and the nilpotence class of K. □

Proposition: If H is a proper subgroup of a nilpotent group G, then H is a proper subgroup of its normalizer $N_G(H)$.

Proof. Let $Z_0(G) = \{e\}$, and let n be the largest index such that $Z_n(G) < H$; such an n exists since H is a proper subgroup and G is nilpotent. Select $a \in Z_{n+1}(G)$ with $a \notin H$. Then, for every $h \in H$, we have $ahZ_n(G) = (aZ_n(G))(hZ_n(G))$, and since $aZ_n(G)$ is in the center of $G/Z_n(G)$ by the definition of $Z_{n+1}(G)$, it follows that $ahZ_n(G) = haZ_n(G)$. In particular, we have $ah = h'ha$ for some $h' \in Z_n(G) < H$, whence $aha^{-1} \in H$ with $a \in N_G(H)$. Since $a \notin H$, it follows that H is a proper subgroup of $N_G(H)$. $\square$

Proposition: A finite group is nilpotent if and only if it is the direct product of its Sylow subgroups.

Proof. If G is a direct product of its p-Sylow subgroups, then G is nilpotent since it is the direct product of a family of nilpotent groups.

Now, suppose G is nilpotent and P is a p-Sylow subgroup. Either, $P = G$, in which case G is a p-group and we are done, or G is a proper subgroup of G. If P is a proper subgroup of G, then P is a proper subgroup of $N_G(P)$ by the previous proposition. Since the normalizer of $N_G(P)$ is itself, it follows that $N_G(P) = G$ since it cannot be a proper subgroup, so P is normal in G, hence it is the unique p-Sylow subgroup of G.

In the general case, let $|G| = p_1^{n_1}\cdots p_k^{n_k}$, where p_i are distinct primes and $n_i > 0$. Let $P_1, \dots, P_k$ be the corresponding proper normal Sylow subgroups corresponding to p_i. Since $|P_i| = p_i^{n_i}$ for each i, it follows that $P_1 P_2 \cdots \widehat{P_i} \cdots P_k$ is a subgroup in wihch every element has order dividing $|G|/p_i^{n_i}$.

Consequently, $P_i \cap \left(P_1 \cdots \widehat{P_i} \cdots P_k\right) = \{e\}$, so inductively, it follows that $P_1 \cdots P_k = P_1 \times \cdots \times P_k$. Therefore, since the order of G equals the order of the direct product, it follows that $G = P_1 \times \cdots \times P_k$. $\square$

We consider another, equivalent characterization of the nilpotent groups.

Definition: Let G be a group. Define the groups G_i inductively by $G_1 = G$ and $G_{i+1} = [G_i, G]$. The series

$$G = G_1 \geq G_2 \geq \cdots$$

defines the *lower central series* of G.

Lemma: Let $K \trianglelefteq G$ and $K \leq H \leq G$. Then, $[H, G] \leq K$ if and only if $H/K \leq Z(G/K)$.

Proof. If $h \in H$ and $g \in G$, then $hKgK = gKhK$ if and only if $[h, g]K = K$ if and only if $[h, g] \in K$. $\square$

In particular, this means that $G_i/G_{i+1} \leq Z(G/G_{i+1})$.

Theorem: Let G be a group. Then, G is nilpotent of nilpotence class c if and only if $G_{c+1} = \{e\}$. Moreover,

$$G_{i+1} \leq Z_{c-i}(G)$$

for all i.

Proof. Suppose $Z_c(G) = G$. We will prove that the inclusion holds by induction on i. If $i = 0$, then $G_1 = Z_c(G) = G$. Inductively, we have

$$\begin{aligned} G_{i+2} &= [G_{i+1}, G] \\ &\leq [Z_{c-i}(G), G] \\ &\leq Z_{c-i-1}(G), \end{aligned}$$

where the last inclusion emerges from the definition of $Z_k(G)$ and the previous lemma. Therefore, we have $G_{c+1} \leq Z_0(G) = \{e\}$.

Now, suppose $G_{c+1} = \{e\}$. We will prove that $G_{c+1-j} \leq Z_j(G)$ by induction on j. If $j = 0$, then $G_{c+1} = Z_0(G) = \{e\}$. By the third isomorphism theorem, we have a surjective homomorphism $G/G_{c+1-j} \to G/Z_j(G)$. Since $\left[G_{c-j}, G\right] = G_{c+1-j}$, we have that $G_{c-j}/G_{c+1-j} \leq Z\left(G/G_{c+1-j}\right)$. Passing to the image of the homomorphism, we have $G_{c-j}Z_j(G)/Z_j(G) \leq Z\left(G/Z_j(G)\right) = Z_{j+1}(G)/Z_j(G)$. Therefore, we have $G_{c-j} \leq G_{c-j}Z_j(G) \leq Z_{j+1}(G)$. $\square$

A Simple Group of Order 168

We will work to understand the existence and uniqueness of a simple group of order $168 = 2^3 \cdot 3 \cdot 7$ by understanding the structure of its Sylow subgroups.

First, observe that $|G|$ does not divide $6!$, so G has no proper subgroup of index less than 7, as otherwise the action of G on the cosets of this subgroup would give an action into some S_n with $n \leq 6$. Since G is simple, this action would necessarily be injective, implying that the order of G would divide the order of this S_n.

Next, since G is simple and the Sylow theorems give that $n_7 = 1$ or $n_7 = 8$, it follows that $n_7 = 8$. Since the number of Sylow subgroups is the index of the normalizer, this gives that the normalizer of a 7-Sylow subgroup is of order 21. Furthermore, no order 2 element normalizes a 7-Sylow subgroup, so G has no elements of order 14.

We will show now that G has no elements of order 21. Suppose x had order 21. Then, there is a 3-Sylow subgroup contained in $\langle x\rangle$, implying that the normalizer of this 3-Sylow subgroup would have order divisible by 7. Since we know that $n_3|8$, it follows that $n_3 = 4$ since $n_3 \equiv 1 \bmod 3$, which would contradict that G has no subgroups of index less than 7. Therefore, $n_3 = 7$ or $n_3 = 28$.

Now, we will show that $n_3 = 28$. Suppose $n_3 = 7$. Let P be a 3-Sylow subgroup, and let T be a 2-Sylow subgroup of $N_G(P)$, where we note that $|N_G(P)| = 24$ (the order of T is 8). Since a 3-Sylow subgroup normalizes some 7-Sylow subgroup R of G, it follows that for every $t \in T$, since $tPt^{-1} = P$, we have P normalizes tRt^{-1}. Since T acts by conjugation on the eight 7-Sylow subgroups of G, and no element of order 2 in G normalizes a 7-Sylow subgroup, we must have that T acts transitively — that is, P normalizes all the 7-Sylow subgroups. Then, P is contained in the intersection of the normalizes of all the 7-Sylow subgroups; yet, since this intersection is a proper normal subgroup of G, it is trivial. Therefore $n_3 = 28$.

Next, we use a lemma.

> **Lemma:** In a finite group G, if n_p is not congruent to 1 modulo p^k, then there are distinct p-Sylow subgroups P and R of G such that $P \cap R$ has index at most p^{k-1} in both P and R.

> *Proof.* Let P act by conjugation on the set of all the p-Sylow subgroups. Suppose p^k divides $[P : P \cap R]$ for all p-Sylow subgroups R not equal to P. Then, each orbit has size divisible by p^k. Since the set of p-Sylow subgroups is equal to the sum of all the lengths of the orbits, then we have $n_p = 1 + \ell p^k$ for some integer ℓ. Thus, there is some R such that $[P : P \cap R] \leq p^{k-1}$. □

Turning our attention to the 2-Sylow subgroups, we observe that $n_2 = 7$ or $n_2 = 21$. In particular, this means that n_2 is not congruent to 1 mod 8, so there are 2-Sylow subgroups T_1 and T_2 with nontrivial intersection such that $U = T_1 \cap T_2$ is of maximum order.

Now, we claim that U is a Klein 4-group with $N_G(U) \cong S_4$. Define $N = N_G(U)$. We have that $|U| = 2$ or $|U| = 4$, and N permutes the non-identity elements of U by conjugation. Observe that a subgroup of order 7 in N would then commute with some element of order 2 in U, implying that an element of order 2 would normalize a 7-Sylow subgroup, which cannot happen. Therefore, the order of N is not divisible by 7.

> **Remark:** In the setting of the lemma, we observe that if $P_0 = P \cap Q$ is an intersection of p-Sylow subgroups, then $P_0 < P$ and $P_0 < R$, so since p-groups are nilpotent, we have $P_0 < N_P(P_0)$ and $P_0 < N_Q(P_0)$. In the special case where $R = P \cap Q$ is maximal, then $R < N_P(R) \leq N_G(R)$ and $R < N_Q(R) \leq N_G(R)$.
>
> Now, consider a p-Sylow subgroup H of $N_G(R)$. Then, there is a p-Sylow subgroup S of G such that $R \leq H \leq S$. Suppose it were the case that there were separate p-Sylow subgroups of $N_G(R)$ H_1 and H_2 that were both contained in S. Then, since $H_i \leq N_G(R)$, it follows that $H_i \leq N_S(R)$. Notice that $N_S(R)$ is a p-group that is a subgroup of $N_G(R)$; by maximality, it follows that $N_S(R) \leq H_i$ as the H_i are p-Sylow subgroups, so that $N_S(R) = H_1 = H_2$, a contradiction. Therefore, if $H_1 \leq S_1$ and $H_2 \leq S_2$ are distinct p-Sylow subgroups of G that contain $N_G(R)$, we have that $R \leq H_1 \cap H_2 \leq S_1 \cap S_2$. Yet, by maximality, it follows that $S_1 \cap S_2 = R$.

> Suppose now toward contradiction that $N_G(R)$ has a unique p-Sylow subgroup H. Notice that $N_P(R) \leq H$ and $N_Q(R) \leq H$, and there is some p-Sylow subgroup $S$of G such that $H \leq S$. This gives that $R < N_P(R) \leq P \cap S$ and $R < N_Q(R) \leq Q \cap S$. If (without loss of generality) $S = P$, then $N_Q(R) \leq S$ and $N_Q(R) \leq Q$, so $N_Q(R) \leq P \cap Q = R$, yet $R < N_P(R)$, a contradiction. If S is not equal to S or P, it follows that P and S are distinct p-Sylow subgroups of G with $N_G(R) \leq P \cap S$, whence $|P \cap S| > |R|$.
>
> In particular, this means that if $P \cap Q$ is maximal, we have that $N_G(P \cap Q)$ has more than one p-Sylow subgroup.

In the case of the group $N = N_G(U)$, we have automatically that N has more than one 2-Sylow subgroup, so $|N| = 2^a \cdot 3$ with $a = 2$ or $a = 3$. Suppose P is a 3-Sylow subgroup of N. Since P is a 3-Sylow subgroup of G, it follows that the group $N_N(P)$ has order 3 or 6 (since the order of $N_G(P)$ is 6). Therefore, by the Sylow theorems, N must have 4 3-Sylow subgroups that are permuted transitively by conjugation. Observe that any group of order 12 must have either a normal 2-Sylow subgroup or a normal 3-Sylow subgroup, but since N has more than one 2-Sylow subgroup, it follows that $|N| = 24$.

Let K be the kernel of N acting by conjugation on its 4 3-Sylow subgroups. Then, K is the intersection of the normalizers of the 3-Sylow subgroups of N. If $K = \{e\}$, then $N \cong S_4$. Else, suppose that $K \neq \{e\}$. Since we have that $K \leq N_N(P)$, it follows that K has order dividing 6, and since P does not normalize any other 3-Sylow subgroup, we cannot have $P \leq K$, so $|K| = 2$. Yet, N/K is then a group of order 12 that has more than one 2-Sylow subgroup and has 4 3-Sylow subgroups. Therefore, $N \cong S_4$. Now, we know that S_4 has a unique nontrivial normal 2-subgroup, which is the Klein 4-group, giving that U is a Klein 4-group.

Since $N \cong S_4$, it follows that N contains a 2-Sylow subgroup of G, and that $N_N(P) \cong S_3$, so that $N_G(P) \cong S_3$ and the 2-Sylow subgroups of G are isomorphic to the dihedral group D_4 of order 8. Furthermore, G has no elements of order 6.

We thus find that any element of order 2 does not commute with an element of order either 3 (else we would have an element of order 6) or 7 (since the order of the normalizers of the 7-Sylow subgroups is 21). If T is a 2-Sylow subgroup, then $T \cong D_4$, so $Z(T) \cong \langle z \rangle$ for some element z of order 2. In particular, this means $T \leq C_G(z)$; since $|C_G(z)|$ has no odd prime factors, it follows that $C_G(z) = T$. Furthermore, any element normalizing T normalizes its center, hence commutes with z, meaning that the 2-Sylow subgroups of G are self-normalizing, so that $n_2 = 21$.

Now, since $|C_G(z)| = 8$, the element z has 21 conjugates. Additionally, G has one conjugacy class of elements of order 4 by the fact that the 2-Sylow subgroups are isomorphic to D_4 (so the elements of order 4 are conjugate to each other in their subgroup), and this conjugacy class has 42 elements. Furthermore, there are 48 elements of order 7 and 56 elements of order 3. Summing these gives all 167 non-identity elements. In particular, this means that all the elements of order 2 are conjugate to each other.

Let T be a 2-Sylow subgroup, and let $U \leq T$ be the aforementioned Klein 4-group in T, and let W be the other Klein 4-group. From the Sylow theorems, since U and W are not conjugate in the normalizer $N_G(T) = T$, it follows that U and W are not conjugate in G.[I] Next, we claim that $N_G(W) \cong S_4$. For this, suppose we have $W = \langle z, w \rangle$ where $Z(T) = \langle z \rangle$. Since w is conjugate in G to z (both have order 2), it follows that $C_G(w) = T_0$ is another 2-Sylow subgroup of G containing W different from T. In particular, this means that $W = T \cap T_0$, so since U is an arbitrary maximal intersection of 2-Sylow subgroups of G, the same argument that gives $N_G(U) \cong S_4$ also gives $N_G(W) \cong S_4$.

All this analysis gives the following.

> **Proposition:** If G is a simple group of order 168, then the following hold:
>
> (i) $n_2 = 21$, $n_3 = 7$, and $n_7 = 8$;
>
> (ii) the 2-Sylow subgroups of G are isomorphic to the dihedral group D_4, while the 3 and 7-Sylow subgroups are cyclic;
>
> (iii) G is isomorphic to a subgroup of A_7, and G has no subgroup of index strictly less than 7;

[I] This follows from from "Frattini's Argument," which says that if H is a normal subgroup of G and P is a p-Sylow subgroup of H, then $N_G(P)H = G$.

(iv) the following are the conjugacy classes of G

- $\{e\}$;
- 2 classes of elements of order 7 with 24 elements per class;
- 1 class of elements of order 3 with 56 elements;
- 1 class of elements of order 2 containing 21 elements;

(v) if T is a 2-Sylow subgroup, and U and W are the Klein 4-subgroups of T, then U and W are not conjugate in G, and have $N_G(U) \cong N_G(W) \cong S_4$;

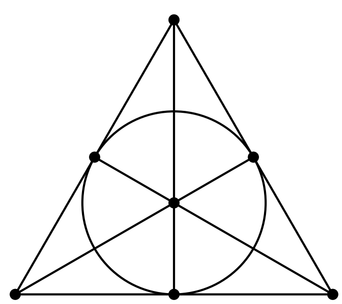
(vi) G has 3 conjugacy classes of maximal subgroups, two of which are isomorphic to S_4 and one of which is isomorphic to the non-abelian group $\mathbb{Z}/7\mathbb{Z} \rtimes \mathbb{Z}/3\mathbb{Z}$ of order 21.

However, none of this shows existence. To find this existence, we will construct a special geometric object that emerges from the action of G on its subgroup structure.

Let $U_1, \dots, U_7$ be the conjugates of U, and let $W_1, \dots, W_7$ be the conjugates of W. We will call the U_i points, and the W_j lines. Next, we will define an incidence relation by saying that U_i is on line W_j if and only if U_i normalizes W_j. Equivalently, U_i is on the line W_j if and only if $U_i W_j \cong D_8$,[II] which occurs if and only if W_j normalizes U_i.

Inside each point (or line) stabilizer (isomorphic to S_4) there is a unique normal 4-subgroup V, and three other non-normal 4-subgroups which we will refer to as B_1, B_2, and B_3. The groups VB_i are the 3 2-Sylow subgroups of S_4. Therefore, we have that each line contains exactly 3 points and every point lies on exactly 3 lines. Since any two non-normal 4-subgroups of S_4 generate S_4 (and determine the other two Klein 4-subgroups of that copy of S_4), it follows that any two points on a line uniquely determine the line. Finally, by counting, every pair of points lies on a unique line, and each pair of lines intersects at a unique point.

This geometry is known as the *projective plane of order* 2, or the *Fano Plane*, denoted $\mathcal{F}$, and is depicted below.



Now, we observe that every $g \in G$ acts by conjugation on the set of points and lines, and preserves the incidence relation, with only the identity element in G that fixes all the points. Therefore, G is isomorphic to a subgroup of $\operatorname{Aut}(\mathcal{F})$. Any automorphism of $\mathcal{F}$ that fixes any two points on a line and fixes a third point not on the line will necessarily fix all points, meaning that any automorphism of $\mathcal{F}$ is determined by its action on any three non-collinear points. There are 168 such triples, so $\mathcal{F}$ has at most 168 automorphisms, so that if the simple group G exists, then it is unique and isomorphic to $\operatorname{Aut}(\mathcal{F})$.

To determine the automorphisms of $\mathcal{F}$, we consider a different construction. Let $V = \mathbb{F}_2^3$ be the 3-dimensional vector space over $\mathbb{F}_2$. Call the 7 1-dimensional subspaces of V the points and the 7 2-dimensional subspaces the lines. Say a point is incident to a line via subspace inclusion. Observe that $\mathrm{GL}(V)$ acts faithfully on the points and lines and preserves incidence, so $\operatorname{Aut}(\mathcal{F})$ has order at least 168, so $\operatorname{Aut}(\mathcal{F}) \cong \mathrm{GL}(V) = \mathrm{GL}_3(\mathbb{F}_2)$, with $\operatorname{Aut}(\mathcal{F})$ having order 168.

[II] There are two Klein 4-groups in the normalizer $N_G(W_j)$, and since U_i is not conjugate to W_j, it follows that U_i normalizes W_j if and only if U_i is the other Klein 4-group in $N_G(W_j)$.

Finally, we will prove that $\mathrm{GL}(V)$ is simple. Suppose H is a proper nontrivial normal subgroup of $\mathrm{GL}(V)$. Suppose Ω is the set of points, and N is the stabilizer in $\mathrm{GL}(V)$ of a point in Ω. The intersection of all the conjugates of N fixes all points, so this intersection is the identity. Therefore, $H \not\leq N$, and we have $\mathrm{GL}(V) = HN$.

By the second isomorphism theorem, $[H : H \cap N] = [HN : N]$, so 7 divides $|H|$. Furthermore, since $\mathrm{GL}(V)$ is isomorphic to a subgroup of S_7 and the normalizers of 7-Sylow subgroups of S_7 have order 42, it follows that $\mathrm{GL}(V)$ does not have a normal 7-Sylow subgroup, so $n_7(\mathrm{GL}(V)) = 8$.

A normal 7-Sylow subgroup of H is characteristic in H, hence normal in $\mathrm{GL}(V)$, meaning that H does not have a unique 7-Sylow subgroup. Since $n_7(H) \equiv 1 \bmod 7$, and $n_7(H) \leq n_7(\mathrm{GL}(V)) = 8$, it follows that $n_7(H) = 8$. Thus, $|H|$ is divisible by 8, so 56 divides H, and since H is proper, $|H| = 56$. By applying a Sylow-style counting argument to H, we see that H has a normal (hence characteristic) 2-Sylow subgroup, which is thus normal in $\mathrm{GL}(V)$. Yet, this would mean that $\mathrm{GL}(V)$ has a unique 2-Sylow subgroup. Note that the upper and lower triangular matrices are subgroups of $\mathrm{GL}_3(\mathbb{F}_2)$ with order 8, so we achieve a contradiction.

Ring Theory

Localization

If R is a (commutative, unital) ring, then we may consider a subset $S \subseteq R$ such that $1 \in S$, $0 \notin S$, and for any $s_1, s_2 \in S$, we have $s_1 s_2 \in S$ (such an S is called a *multiplicative set*). Then, we may construct the *ring of fractions* $S^{-1}R$ as the set of equivalence classes in $R \times S$ with $(r_1, s_1) \sim (r_2, s_2)$ if and only if there exists $t \in S$ such that $t(r_1 s_2 - r_2 s_1) = 0$. We write $[(r,s)] = \frac{r}{s}$, and define addition and multiplication by

$$\frac{r_1}{s_1} + \frac{r_2}{s_2} = \frac{r_1 s_2 + r_2 s_1}{s_1 s_2}$$
$$\frac{r_1}{s_1}\frac{r_2}{s_2} = \frac{r_1 r_2}{s_1 s_2}.$$

There is a natural embedding ι of R into $S^{-1}R$ by taking $r \mapsto \frac{r}{1}$.

Theorem: Let R be a commutative ring with unity, S a multiplicatively closed subset, and $\phi\colon R \to T$ a ring homomorphism that maps S to units in T. Then, there is a unique homomorphism $\varphi\colon S^{-1}R \to T$ such that $\varphi \circ \iota = \phi$.

$$\begin{tikzcd} R \arrow[r, "\iota"] \arrow[dr, "\phi"'] & S^{-1}R \arrow[d, "\varphi"] \\ & T \end{tikzcd}$$

Proof. Define

$$\varphi\left(\frac{r}{s}\right) = \phi(r)\phi(s)^{-1}.$$

□

Proposition: Let R be a commutative ring with identity. Any ideal in $S^{-1}R$ is of the form $S^{-1}J$ for some ideal J of R.

Proof. Consider a proper ideal I of $S^{-1}R$, and let $J = I \cap R$. We claim that $J \cap S = \emptyset$. If not, let $u \in J \cap S$, so that $\frac{u}{1} \in J \cap S$. Since $J \cap S$ is an ideal, it follows that $\frac{u}{1}\frac{1}{u} \in J \cap S$, so $1 \in J \cap S$, so $1 \in J$, so $1 \in I$, meaning I is not proper.

Since $J \cap S = \emptyset$, it follows that $S^{-1}J$ is well-defined. We will start by showing that $I \subseteq S^{-1}J$. Consider an element $\frac{a}{b} \in I$. Then, $\frac{a}{1} = \frac{a}{b}\frac{b}{1}$, so $\frac{a}{b} \in S^{-1}J$. Additionally, if $\frac{a}{b} \in S^{-1}J$, then $\frac{a}{b} = \frac{1}{b}\frac{a}{1}$, so since $\frac{a}{1} \in I$ as $\frac{a}{1} \in I \cap R$, it follows that $\frac{a}{b} \in I$. □

Proposition: There is a one-to-one correspondence between prime ideals of $S^{-1}R$ and prime ideals of R that do not intersect S.

Proof. Let $\iota\colon R \to S^{-1}R$ be the embedding $r \mapsto \frac{r}{1}$. We claim that the maps

$$\begin{aligned} \mathfrak{p} &\mapsto S^{-1}\mathfrak{p} \\ \mathfrak{q} &\mapsto \iota^{-1}(\mathfrak{q}) \end{aligned}$$

are mutually inverse.

First, we observe that for any ideal $I \subseteq S^{-1}R$, then

$$\iota^{-1}\left(S^{-1}I\right) = \{r \in R : sr \in I \text{ for some } s \in S\},$$

known as the *saturation* of the ideal $I \subseteq R$. This follows from the fact that if $sr \in I$, then $\frac{r}{1} = \frac{sr}{s} \in S^{-1}I$, meaning that $r \in \iota^{-1}\left(S^{-1}I\right)$, and if $\frac{r}{1} = \frac{a}{s}$ for some $a \in I$ and $s \in S$, then there is some $t \in S$ such that $t(rs - a) = 0$, so that $(ts)r = ta \in I$. Since $ts \in S$, it follows that r belongs in the right-hand set.

Next, we observe that $S^{-1}I = S^{-1}R$ if and only if $I \cap S \neq \emptyset$. This follows from the fact that if $s \in I \cap S$, then $\frac{s}{s} = \frac{1}{1} \in S^{-1}I$, while if $\frac{1}{1} \in S^{-1}I$, then we may write $\frac{1}{1} = \frac{a}{s}$ for some $a \in I$ and $s \in S$, with $t(s-a) = 0$, so $ts = ta \in I$, whence $ts \in I \cap S$.

First, we show that the map $\mathfrak{q} \mapsto \iota^{-1}(\mathfrak{q})$ sends primes in $S^{-1}R$ to primes in R. Set $\mathfrak{p} = \iota^{-1}(\mathfrak{q})$. This follows from the fact that the primage of a prime under a ring homomorphism is still prime.

Next, we show that if $\mathfrak{p}$ is prime with $\mathfrak{p} \cap S = \emptyset$, then $S^{-1}\mathfrak{p}$ is prime. First, observe that

$$\iota^{-1}\left(S^{-1}\mathfrak{p}\right) = \{r \in R : sr \in \mathfrak{p} \text{ for some } s \in S\}.$$

Notice that whenever $sr \in \mathfrak{p}$ with $s \in S$, we must have $r \in \mathfrak{p}$. Therefore, we must have $\iota^{-1}\left(S^{-1}\mathfrak{p}\right) = \mathfrak{p}$.

Next, $S^{-1}\mathfrak{p}$ is proper since $\mathfrak{p} \cap S = \emptyset$. Finally, we will show that $S^{-1}\mathfrak{p}$ is prime.

Suppose

$$\frac{a}{s}\frac{b}{t} \in S^{-1}\mathfrak{p}$$

with $a, b \in R$ and $s, t \in S$. Then, we have $\frac{ab}{st} = \frac{p}{u}$ for some $p \in \mathfrak{p}$ and $u \in S$. There is thus $w \in S$ such that $w(uab - stp) = 0$, giving $(wu)(ab) = wstp \in \mathfrak{p}$. Since $wu \in S$ is disjoint from $\mathfrak{p}$, it follows that $wu \notin \mathfrak{p}$, and we have $ab \in \mathfrak{p}$. Therefore, $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$, so $\frac{a}{s} \in S^{-1}\mathfrak{p}$ or $\frac{b}{t} \in S^{-1}\mathfrak{p}$.

This gives the correspondence obtained by taking $\mathfrak{p} \mapsto S^{-1}\mathfrak{p}$ and $\mathfrak{q} \mapsto \iota^{-1}(\mathfrak{q})$. □

Example: Consider the special case of the multiplicative set $S = \left\{1, r, r^2, \dots\right\}$. We will show that $S^{-1}R \cong R[x]/(rx-1)$, which will emerge by showing that $R[x]/(rx-1)$ satisfies the universal property. That is, if $\varphi\colon R \to A$ is a ring homomorphism with $\varphi(1) = 1$ and $\varphi(S) \subseteq A^{\times}$, then there exists a unique ring homomorphism $\Phi\colon R[x]/(rx-1) \to A$ such that $\Phi \circ \varepsilon = \varphi$, where ε is the composition

$$R \xrightarrow{\iota} R[x] \xrightarrow{\pi} R[x]/(rx-1).$$

We start by setting $s = \varphi(r)^{-1}$, and extending φ to a map $\overline{\varphi}\colon R[x] \to A$ given by

$$\overline{\varphi}(a_0 + a_1 x + \cdots + a_n x^n) = \varphi(a_0) + \varphi(a_1)s + \cdots + \varphi(a_n)s^n.$$

To define Φ, it suffices to show that $(rx-1) \subseteq \ker(\overline{\varphi})$. This follows from the fact that

$$\begin{aligned} \overline{\varphi}(rx-1) &= \varphi(r)s - 1 \\ &= 0. \end{aligned}$$

Therefore, we have existence.

Now, suppose we have a homomorphism $\psi\colon R[x]/(rx-1) \to A$ such that $\psi \circ \varepsilon = \varphi$ on R. We will show that $\psi(x+(rx-1)) = \Phi(x+(rx-1))$, as we already know that $\psi(t+(rx-1)) = \varphi(t) = \Phi(t+(rx-1))$ for any $t \in R$. Observing that $rx \equiv 1 \mod (rx-1)$, we have

$$\begin{aligned}\psi(rx+(rx-1)) &= \psi(1+(rx-1))\\ &= 1,\end{aligned}$$

while

$$\begin{aligned}\psi(rx+(rx-1)) &= \psi(r+(rx-1))\psi(x+(rx-1))\\ &= \varphi(r)\psi(x+(rx-1)).\end{aligned}$$

Therefore, $\varphi(r)\psi(x+(rx-1)) = 1$, so that $\psi(x+(rx-1)) = \varphi(r)^{-1} = \Phi(x+(rx-1))$.

Unique Factorization Domains

Definition: An integral domain R is called a *unique factorization domain* if every element $r \in R$ that is nonzero and non-unit admits a factorization

$$r = up_1^{e_1}\cdots p_n^{e_n}$$

into irreducible elements of R. The factorization is unique up to associates, where $q \sim p$ are associates if there exists $v \in R^{\times}$ such that $q = vp$.

A general strategy with unique factorization domains consists of writing any nonzero non-units in terms of their irreducibles, and using divisibility to tease out properties. An example is discussed in the following.

Example (August '21, Problem 4)**:** Consider a nonzero non-unit $up_1^{d_1}\cdots p_m^{d_m} = f \in R$ in a unique factorization domain. Let $R_f = R\left[\frac{1}{f}\right]$. We will show that

$$R_f^{\times} = \left\{up_1^{e_1}\cdots p_m^{e_m} : u \in R^{\times}, e_i \in \mathbb{Z}\right\}.$$

To show one direction, observe that we may obtain any individual factor p_i of f by taking

$$p_i = \frac{f}{up_1^{d_1}\cdots p_i^{d_i-1}\cdots p_m^{d_m}},$$

so in the localization, we have all positive powers of $p_i^{d_i}$ are invertible in R_f, as well as all negative powers since

$$\frac{1}{p_i} = \frac{up_1^{d_1}\cdots p_i^{d_i-1}\cdots p_m^{d_m}}{f},$$

so that the latter set is contained in the former. Meanwhile, if $\frac{s}{f^{n_1}} \in R_f^{\times}$, then we use the fact that the factorization of f is unique to find that $\frac{s}{u}\left(p_1^{-d_1}\cdots p_m^{-d_m}\right) = su^{-1}\left(p_1^{-d_1}\cdots p_m^{-d_m}\right)$ is contained in the latter set.

This allows us to develop an isomorphism $R_f^{\times} \cong R^{\times}\times\mathbb{Z}^m$, where we take any unit in $R_f^{\times}$ to the tuple $(u, e_1, \dots, e_m)$.

Primality in the Gaussian Integers

We will discuss primality in the Gaussian integers. Recall that the Gaussian integers are given by

$$\mathbb{Z}[i] = \{a+bi \mid a, b \in \mathbb{Z}\}$$

$$\cong \frac{\mathbb{Z}[x]}{(x^2+1)}.$$

The Gaussian integers are equipped with a norm $N\colon \mathbb{Z}[i] \to \mathbb{Z}_{\geq 0}$ given by $N(a+bi) = (a+bi)(a-bi) = a^2+b^2$, which is a multiplicative norm satisfying $N(zw) = N(z)N(w)$. Furthermore, with this norm, the Gaussian integers form a Euclidean domain. The units of $\mathbb{Z}[i]$ are ± 1 and $\pm i$.

Lemma: Let $q \in \mathbb{Z}[i]$. Then, there is a prime integer $p \in \mathbb{Z}$ such that $N(q) = p$ or $N(q) = p^2$.

Proof. Since q is not a unit, $N(q) \neq 1$. Therefore, $N(q)$ is a nontrivial product of prime integers, and since q is prime in $\mathbb{Z}[i]$, q must divide one of the prime integer factors of $N(q)$. Let p be this prime integer.

Then, $q|p$ in $\mathbb{Z}[i]$, so $N(q)|N(p) = p^2$, so $N(q) = p$ or $N(q) = p^2$. □

Example: We observe that 3 is a prime element of $\mathbb{Z}[i]$. Since $\mathbb{Z}[i]$ is a Euclidean domain, it is a UFD, so irreducibles are prime, meaning we only need to show that 3 is irreducible in $\mathbb{Z}[i]$.

Since $N(3) = 9$, it follows that the norm of any factor of 3 must be a divisor of 9. Therefore, it follows that the norm of any divisor of 3 is equal to 1, 3, or 9. If the norm is 9, then the factor is an associate of 3, while if the norm is 1, then the factor is a unit, so it follows that any nontrivial factor of 3 would have norm equal to 3. Yet, there are no such elements.

Meanwhile, 5 is not a prime element of $\mathbb{Z}[i]$, since we may write $5 = (2+i)(2-i)$.

We will now focus on proving one of the most celebrated theorems in number theory: Fermat's Theorem on sums of squares. We will say a prime integer $p \in \mathbb{Z}$ *splits* in $\mathbb{Z}[i]$ if it is not a prime element of $\mathbb{Z}[i]$. For example, 5 splits in $\mathbb{Z}[i]$ while 3 does not.

Lemma: A positive prime integer $p \in \mathbb{Z}$ splits in $\mathbb{Z}[i]$ if and only if it is the sum of two squares in $\mathbb{Z}$.

Proof. Suppose $p = a^2 + b^2$ with $a, b \in \mathbb{Z}$. Then,

$$p = (a+bi)(a-bi)$$

in $\mathbb{Z}[i]$, so since $N(a \pm bi) = a^2 + b^2 = p \neq 1$, it follows that neither of the factors is a unit in $\mathbb{Z}[i]$, so that p is not irreducible, hence not prime.

Conversely, suppose p is not irreducible in $\mathbb{Z}[i]$. Then, p has an irreducible factor $q \in \mathbb{Z}[i]$ that is not an associate of p. Since $q|p$, multiplicativity of the norm gives that $N(q)|N(p) = p^2$, hence $N(q) = p$ as q and p are not associates and q is not a unit. If we have $q = a + bi$, then

$$\begin{aligned} p &= N(q) \\ &= a^2 + b^2. \end{aligned}$$

□

Next, we discuss a characterization of splitting via a single congruence.

Lemma: A positive odd prime integer p splits in $\mathbb{Z}[i]$ if and only if it is congruent to 1 modulo 4.

Proof. Our goal is to understand when $\mathbb{Z}[i]/(p)$ is an integral domain. Yet, we have the series of isomorphisms

$$\begin{aligned} \frac{\mathbb{Z}[i]}{(p)} &\cong \frac{\mathbb{Z}[x]/\left(x^2+1\right)}{(p)} \\ &\cong \frac{\mathbb{Z}[x]}{(p, x^2+1)} \\ &\cong \frac{\mathbb{Z}[x]/(p)}{(x^2+1)} \\ &\cong \frac{\mathbb{Z}/p\mathbb{Z}[x]}{(x^2+1)}. \end{aligned}$$

In particular, this means that p splits in $\mathbb{Z}[i]$ if and only if $x^2 + 1$ has a root in $\mathbb{Z}/p\mathbb{Z}$, or that there is some

n such that $n^2 \equiv -1$ modulo p. We will show that this condition is equivalent to p being congruent to 1 modulo 4.

Let $G = (\mathbb{Z}/p\mathbb{Z})^\times$, and fix a generator g. Observe that p is odd, so $p-1 = 2\ell$ for some integer ℓ. Let $\overline{n}$ denote the class of an integer n mod (p). For any $n \notin (p)$, it follows that there is some m such that $\overline{n} = g^m$.

Observe that the class $\overline{-1}$ generates the unique subgroup of order 2 of G, and since g^ℓ does the same, it follows that $g^\ell = -1$. In particular, we have that $n^2 = -1$ modulo p if and only if $g^{2m} = g^\ell$, or that $2m \equiv \ell$ mod 2ℓ, or that $(p-1) = 2\ell$ is a multiple of 4, or $p \equiv 1$ mod 4. □

This gives the following.

Theorem (Fermat's Theorem on Sums of Squares)**:** A positive odd prime $p \in \mathbb{Z}$ is a sum of two squares if and only if $p \equiv 1$ mod 4.

As a general technique for primality in a quadratic number ring $\mathbb{Z}\left[\sqrt{d}\right]$ for some square-free integer d, we use the fact that

$$\mathbb{Z}\left[\sqrt{d}\right] \cong \frac{\mathbb{Z}[x]}{(x^2-d)},$$

meaning that an element $p \in \mathbb{Z}$ is prime if and only if

$$\begin{aligned}\frac{\mathbb{Z}\left[\sqrt{d}\right]}{(p)} &\cong \frac{\mathbb{Z}[x]}{(p, x^2-d)} \\ &\cong \frac{(\mathbb{Z}/p\mathbb{Z})[x]}{(x^2-d)}\end{aligned}$$

is an integral domain; equivalently, we want to see whether or not x^2-d is irreducible mod p.

Module Theory

Useful Facts

Definition: A module M over R is called *cyclic* if there is a vector v such that $\langle v\rangle = Rv = M$.

Theorem: If a module M is cyclic with cyclic vector v, then

$$M \cong R/\operatorname{Ann}_R(v),$$

where $\operatorname{Ann}_R(v)$ is the *annihilator ideal*, defined by

$$\operatorname{Ann}_R(v) = \{r \in R : rv = 0\}.$$

What follows are the most important theorems in module theory. We'll discuss an application of it to linear algebra.

Definition: Let R be a PID. A matrix $A \in \operatorname{Mat}_{k,n}(R)$ is said to be in Smith Normal Form if there exists $m \leq \min(k,n)$ and nonzero elements $d_1,\dots,d_m$ such that $A = \operatorname{diag}_{k,n}(d_1,\dots,d_m)$ satisfying $(d_1) \supseteq (d_2) \supseteq \cdots \supseteq (d_m)$.

Theorem (Smith Normal Form Decomposition)**:** Let R be a PID, and let $A \in \operatorname{Mat}_{k,n}(R)$. Then, there are $U \in \operatorname{GL}_k(R)$ and $V \in \operatorname{GL}_n(R)$ such that UAV is in Smith Normal Form.

These matrices are found by a product of elementary row and column operations. Specifically, the following row operations can be interpreted by matrix multiplication, where $E_{ij}(\lambda)$ denotes the elementary matrix consisting of 1 along the diagonal and λ at the (i,j) position, while F_{ij} denotes the transposition matrix with row i and row j exchanged.

Elementary Row Operation	Interpretation as Matrix Multiplication
$R_i(A) \mapsto R_i(A) + \lambda R_j(A)$	$A \mapsto E_{ij}(\lambda)A$
$C_i(A) \mapsto C_i(A) + \lambda C_j(A)$	$A \mapsto AE_{ij}(\lambda)$
$R_i(A) \leftrightarrow R_j(A)$	$A \mapsto F_{ij}A$
$C_i(A) \leftrightarrow C_j(A)$	$A \mapsto AF_{ij}$

Theorem (Compatible Basis Theorem)**:** Let M be a rank n free module over a principal ideal domain R, and let N be a submodule of M. Then, N is a free module with rank $m \leq n$, and there is a basis $\{y_1,\dots,y_n\}$ for M and nonzero elements $d_1|d_2|\cdots|d_m$ such that $\{d_1y_1, d_2y_2,\dots,d_my_m\}$.

Moreover, the rank m of N is unique, and the $d_1,\dots,d_m$ are unique up to associates.

Proof. Select a basis $\{x_1,\dots,x_n\}$ for M, and a set of generators $\{u_1,\dots,u_k\}$ of N. Then, we may find a matrix $\left(a_{ij}\right)_{i,j} \in \mathrm{Mat}_{k,n}(R)$ such that

$$\begin{pmatrix} u_1 \\ \vdots \\ u_k \end{pmatrix} = \left(a_{ij}\right)_{i,j} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

We may put the matrix A into Smith normal form, yielding $A = PDQ$, where $D = \mathrm{diag}_{k,n}(d_1,\dots,d_m)$, $P \in \mathrm{GL}_k(R)$, and $Q \in \mathrm{GL}_n(R)$.

Writing

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = Q \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

$$\begin{pmatrix} v_1 \\ \vdots \\ v_k \end{pmatrix} = P^{-1} \begin{pmatrix} u_1 \\ \vdots \\ u_k \end{pmatrix},$$

we have that $\{y_1,\dots,y_n\}$ are a basis for M and $\{v_1,\dots,v_k\}$ generate N. We may write

$$\begin{pmatrix} v_1 \\ \vdots \\ v_k \end{pmatrix} = D \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix},$$

meaning that $k = m$ and $v_i = d_iy_i$. □

Theorem (Invariant Factors Decomposition)**:** Let M be a finitely generated module over a PID R. Then, there exists a unique k, unique ℓ, and unique up to associates $a_1|a_2|\cdots|a_\ell$ such that

$$M \cong R^k \oplus \left(\bigoplus_{i=1}^{\ell} R/(a_i)\right).$$

Proof Sketch. If $\{x_1,\dots,x_n\}$ forms a generating set for M, define a (surjective) R-module homomorphism $R^n = F \to M$ by taking $(r_1,\dots,r_n) \mapsto \sum_{i=1}^{n} r_ix_i$. Find a compatible basis for F and the kernel $N \leq F$ of this homomorphism. Then, we get that

$$\begin{aligned} M &\cong F/N \\ &= \frac{\bigoplus_{i=1}^{n} Ry_i}{\bigoplus_{i=1}^{m} (a_i)y_i} \end{aligned}$$

$$\cong R^{n-m} \oplus \left(\bigoplus_{i=1}^{m} R/(a_i)\right).$$

□

Theorem (Elementary Divisors Decomposition)**:** Let M be a finitely generated module over a PID R. Then, there are primes $p_1, \dots, p_t \in R$ not necessarily unique and $d_1, \dots, d_t \in \mathbb{N}$ such that

$$M \cong R^k \oplus \left(\bigoplus_{i=1}^{t} R/\left(p_i^{d_i}\right)\right).$$

Here, k is the same value as the k in the invariant factors decomposition.

Proof Sketch. Since R is a principal ideal domain, it is a unique factorization domain, so for each a_i in the invariant factors decomposition, we may write $a_i = p_{i,1}^{d_1} \cdots p_{i,t}^{d_t}$. The Chinese Remainder Theorem yields that

$$R/(a_i) \cong \bigoplus_{s=1}^{t} R/\left(p_{i,s}^{d_s}\right).$$

□

Definition: A module P over a (commutative, unital) ring R is called *projective* if it satisfies the following criterion: given a surjective R-module homomorphism $\varphi\colon M \to N$, any R-module homomorphism $\psi\colon P \to N$ extends to an R-module homomorphism $\widetilde{\psi}\colon P \to M$ satisfying $\varphi \circ \widetilde{\psi} = \psi$. The following diagram commutes.

$$\begin{tikzcd} & P \arrow[dl, dashed, "\exists\widetilde{\psi}"'] \arrow[d, "\psi"] & \\ M \arrow[r, "\varphi"'] & N \arrow[r] & 0 \end{tikzcd}$$

We now discuss a special decomposition. Suppose $e \in R$ is an idempotent with $e \neq 1$. Then, $f = 1 - e$ is idempotent as well, and satisfies $e(1-e) = 0$. Any R-module M then splits as

$$M = eM \oplus (1-e)M.$$

The sum is equal to all of M since $m = em + (1-e)m$ for each $m \in M$, and the sum is direct since for any $x \in eM \cap (1-e)M$, we have $x = em'$ and $x = (1-e)m''$, meaning $ex = x$ and $ex = e(1-e)m'' = 0$, meaning $x = 0$.

Inductively, we may define a complete set of orthogonal idempotents $e_1, \dots, e_n$ to be such that $e_i e_j = 0$ for any $i \neq j$ and $e_1 + \cdots + e_n = 1$. Then, we may take the module decomposition as

$$M = \bigoplus_{i=1}^{n} e_i M.$$

One of the most useful applications of this is if one is able to apply the Chinese Remainder Theorem to the underlying ring. Notice that if R is commutative, then the idempotents e_i are central, and since R is an R-module over itself, we have a decomposition

$$R \cong Re_1 \times \cdots \times Re_n$$

of the underlying ring.

Tensor Products

Definition: Let R be any ring. If M is a *right* R-module, N is a left R-module, and P is an abelian group, a map $\varphi\colon M\times N\to P$ is called *R-balanced* if

$$\begin{aligned}\varphi(m_1+m_2,n)&=\varphi(m_1,n)+\varphi(m_2,n)\\ \varphi(m,n_1+n_2)&=\varphi(m,n_1)+\varphi(m,n_2)\\ \varphi(mr,n)&=\varphi(m,rn)\end{aligned}$$

for all $m,m_1,m_2\in M$, $n,n_1,n_2\in N$, and $r\in R$.

If R is commutative, (i.e., both M and N can be considered as left R-modules) then the map φ is called *R-bilinear* if we also have $\varphi(rm,n)=r\varphi(m,n)=\varphi(m,rn)$.

Definition: Let R be any ring. If M and N are a right R-module and left R-module respectively, their *tensor product* is the unique abelian group satisfying the following universal property: for any R-balanced map $\phi\colon M\times N\to P$, there is a unique abelian group homomorphism $\varphi\colon M\otimes_R N\to P$ satisfying $\varphi(m\otimes n)=\phi(m,n)$.

If R is a commutative ring (i.e., left and right R-modules are the same), then $M\otimes_R N$ satisfies the following stronger universal property: for any R-bilinear map $\phi\colon M\times N\to P$ of left R-modules, there is a unique R-module homomorphism $\varphi\colon M\otimes_R N\to P$, where $M\otimes_R N$ is endowed with the R-module structure given by $r(m\otimes n)=(rm)\otimes n=m\otimes(rn)$.

A very useful case study is the *extension of scalars*. Given an inclusion of unital rings $R\to S$, there is a natural right R-module structure on S given by left multiplication; taking the left multiplication by S yields a (S,R)-bimodule structure on the ring S. If N is a left R-module, then $S\otimes_R N$ becomes a left S-module with a copy of N included via the map $\iota\colon N\to S\otimes_R N$ taking $n\mapsto 1\otimes n$.

If $f\colon R\to S$ is a unital ring homomorphism, then S receives the structure of a right R-module by taking $s\cdot r=sf(r)$, and S becomes a (S,R)-bimodule with respect to the standard left S-module structure and the described right R-module structure. The tensor product $S\otimes_R N$ of any left R-module with S yields a *change of base* from the ring R to the ring S.

Finally, if I is a two-sided ideal of a (not necessarily commutative) ring R, then R/I is a $(R/I,R)$-bimodule; given a left R-module N, then $R/I\otimes_R N$ becomes a left R/I-module (i.e., the change of base with respect to the quotient homomorphism). Defining

$$IN=\left\{\sum_{i=1}^{n}a_in_i : a_i\in I, n_i\in N\right\},$$

we obtain that IN is a left R-submodule of N.

The tensor product $(R/I)\otimes_R N$ is generated by the simple tensors $(r+I)\otimes n=r((1+I)\otimes n)$ for any $r\in R$ and $n\in N$. Therefore, the elements $(1+I)\otimes n$ generate $(R/I)\otimes_R N$ as an R/I-module. Define the map $N\to(R/I)\otimes_R N$ by taking $n\mapsto(1+I)\otimes n$. This is a surjective left R-module homomorphism with kernel containing IN, seeing as if $a\in I$ and $n\in N$, then $(1+I)\otimes an=(a+I)\otimes n=0$. We obtain a surjective R-module homomorphism $f\colon N/IN\to(R/I)\otimes_R N$ taking $n+IN\mapsto(1+I)\otimes n$. Define an inverse R-balanced map $R/I\times N\to N/IN$ taking $(r+I,n)\mapsto rn+IN$. Using the universal property yields the following standard isomorphism:

$$R/I\otimes_R N\cong N/IN.$$

Next, we discuss some categorical properties of tensor products. We will assume that our ring R is commutative (meaning that all left R-modules are right R-modules, and we work with R-bilinear maps).

If $\alpha\colon M_1\to M_2$ is an R-module homomorphism, and N is any R-module, then we may define an R-bilinear map $M_1\times N\to M_2\times N\to M_2\otimes N$ given by $(m_1,n)\mapsto\alpha(m_1)\otimes n$, which induces an R-module homomorphism $\alpha\otimes\mathrm{id}_N\colon M_1\otimes N\to M_2\otimes N$.

This map is functorial, since if $\beta\colon M_0 \to M_1$ is another homomorphism, then $(\alpha \otimes \mathrm{id}_N) \circ (\beta \otimes \mathrm{id}_N)$ agrees with $(\alpha \circ \beta) \otimes \mathrm{id}_N$ on simple tensors, so they must agree on all tensors.

Consider now the functor $P \mapsto \mathrm{Hom}_R(N, P)$, where any map $\varphi\colon P_1 \to P_2$ yields a map $\varphi_*\colon \mathrm{Hom}_R(N, P_1) \to \mathrm{Hom}_R(N, P_2)$ defined by $\varphi_*\theta(n) = \varphi(\theta(n))$ for any $\theta \in \mathrm{Hom}_R(N, P_1)$ and $n \in N$.

Recall that any R-linear map $M \otimes_R N \to P$ is equivalent to defining an R-bilinear map $M \times N \to P$. Any R-bilinear map $\varphi\colon M \times N \to P$ is defined by the fact that $\varphi(m, \cdot)$ and $\varphi(\cdot, n)$ are R-linear maps for all $m \in M$ and $n \in N$. That is, φ defines an R-module homomorphism $M \to \mathrm{Hom}_R(N, P)$. In particular, this means that an R-bilinear map is "equivalent" in a sense to an element of $\mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$.

Lemma: For all R-modules M, N, and P, we have an isomorphism of R-modules

$$\mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P)) \cong \mathrm{Hom}_R(M \otimes_R N, P).$$

Proof. We note that any $\alpha \in \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$ determines an R-bilinear map $\varphi\colon M \times N \to P$ defined by $(m, n) \mapsto \alpha(m)(n)$. By the universal property, φ factors through an R-linear map $\overline{\varphi}\colon M \otimes_R N \to P$, so α determines a well-defined $\overline{\varphi} \in \mathrm{Hom}_R(M \otimes_R N, P)$. The map $\alpha \mapsto \overline{\varphi}$ is R-linear and has inverse given by $\varphi \mapsto ((m, n) \mapsto \varphi(m \otimes n))$. $\square$

Corollary: For every R-module N, the functor $- \otimes_R N$ is left adjoint to the functor $\mathrm{Hom}_R(N, -)$

We can use the following general theorems to apply it to the special case of the functors $- \otimes_R N$ and $\mathrm{Hom}_R(N, -)$.

Theorem: If $\mathcal{F}$ is a functor with a right adjoint (i.e., a left-adjoint functor), then $\mathcal{F}$ commutes with categorical colimits (e.g., direct sums). Similarly, if $\mathcal{G}$ is a functor with a left adjoint (i.e., a right-adjoint functor), then $\mathcal{G}$ commutes with categorical limits (e.g., products).

Theorem: If $\mathcal{F}$ is a left-adjoint functor (on R-modules), then $\mathcal{F}$ preserves right-exact sequences. That is, given an exact sequence $A \to B \to C \to 0$,

$$\mathcal{F}(A) \to \mathcal{F}(B) \to \mathcal{F}(C) \to 0.$$

Similarly, if $\mathcal{G}$ is a right-adjoint functor on R-modules, then $\mathcal{G}$ preserves left-exact sequences. That is, given an exact sequence $0 \to A \to B \to C$,

$$0 \to \mathcal{G}(A) \to \mathcal{G}(B) \to \mathcal{G}(C).$$

Therefore, we obtain the following results.

Corollary: For all R-modules M_1, M_2, and N, we have

$$(M_1 \oplus M_2) \otimes_R N \cong (M_1 \otimes_R N) \oplus (M_2 \otimes_R N).$$

Corollary: If A and B are any sets, then

$$R^{\oplus A} \otimes_R R^{\oplus B} \cong R^{\oplus (A \times B)}.$$

Without having to go through the full process of proof, we obtain the following previously established result.

Corollary: For all R-modules N and all ideals I of R, we have

$$(R/I) \otimes_R N \cong N/IN.$$

Proof. From the exact sequence

$$0 \to I \to R \to R/I \to 0,$$

taking tensor products yields

$$I \otimes N \to R \otimes N \to (R/I) \otimes N \to 0$$

since tensor products preserve right exact sequences. Since $R \otimes N \cong N$ and $I \otimes N \cong IN$ in N, we thus get the identification $(R/I) \otimes N \cong N/IN$. $\square$

Corollary: For all ideals I and J of R,

$$(R/I) \otimes_R (R/J) \cong \frac{R}{I+J}.$$

Proof. We have

$$\begin{aligned}(R/I) \otimes_R (R/J) &\cong \frac{(R/J)}{I(R/J)}\\ &\cong \frac{(R/J)}{(I+J)/J}\\ &\cong \frac{R}{I+J},\end{aligned}$$

where the second line emerges from the correspondence in the fourth isomorphism theorem. □

A question we're interested in is what happens when the functor $-\otimes_R N$ preserves an entire short exact sequence (i.e., the functor is exact). This is not always the case. For instance, consider the following short exact sequence

$$0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0.$$

Taking the tensor product with $\mathbb{Z}/2\mathbb{Z}$ yields the operation $\mathbb{Z}/2\mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z}/2\mathbb{Z}$, which is the zero map (hence it is not injective).

Meanwhile, if the module $N \cong R^{\oplus A}$ is free, then $-\otimes_R N$ is exact, since any inclusion $M_1 \subseteq M_2$ yields a map $M_1 \otimes_R R^{\oplus A} \to M_2 \otimes_R R^{\oplus A}$, and since tensor products and direct sums commute, this yields an inclusion $M_1^{\oplus A} \to M_2^{\oplus A}$.

Definition: An R-module N is called *flat* if $-\otimes_R N$ is exact.

In other words, flatness is sort of the counterpart of projectivity, where we recall that an R-module N is projective if and only if the functor $\operatorname{Hom}_R(N, -)$ is exact.

Proposition: If R is a domain, then any flat R-module is torsion-free.

Proof. Let $x \in R$ be nonzero, and let M be a flat module over R. Observe that there is an exact sequence $0 \to R \xrightarrow{a \mapsto ax} R$, where the map $a \mapsto ax$ is injective since R has no zero divisors. Taking tensor products with M, we obtain that

$$0 \to R \otimes_R M \xrightarrow{(a \mapsto ax)\otimes \mathrm{id}_M} R \otimes_R M.$$

Notice that $R \otimes_R M \cong M$, so that $0 \to M \xrightarrow{(m \mapsto xm)} M$ is retained. In particular, this means that for all nonzero $m \in M$, $m \mapsto xm$ is injective, or that $xm \neq 0$ for any $x \in R$, so that M is torsion-free. □

Example: We consider a field extension L/K. Then, since L is a vector space over K, L is a free K-module. Given a polynomial $p(x) \in K[x]$, there is then a short exact sequence of K-modules given by

$$0 \to (p(x)) \to K[x] \to \frac{K[x]}{(p(x))} \to 0.$$

Taking tensor products with L over K then gives

$$0 \to L \otimes_K (p(x)) \to L \otimes_K K[x] \to L \otimes_K \frac{K[x]}{(p(x))} \to 0.$$

Notice that $L \otimes_K K[x] \cong L[x]$ under the identification $1 \otimes x^n \mapsto x^n$. Thus, we have

$$0 \to L \otimes_K (p(x)) \to L[x] \to L \otimes_K \frac{K[x]}{(p(x))} \to 0.$$

The injectivity of the map $L \otimes_K (p(x)) \to L[x]$ gives that $L \otimes_K (p(x))$ can be identified with the ideal $(p(x)) \subseteq L[x]$. Writing the exact sequence once more, we obtain

$$0 \to (p(x)) \to L[x] \to L \otimes_K \frac{K[x]}{(p(x))} \to 0.$$

Thus, this gives $L \otimes_K \frac{K[x]}{(p(x))} \cong \frac{L[x]}{(p(x))}$.

We now focus on showing that free and projective modules are flat.

Definition: Let R be a (commutative, unital) ring, and let M be an R-module. Let $\sum_{i=1}^{n} f_i x_i = 0$ be a relation in M. We say the relation is *trivial* if there exists $m \geq 0$, $y_j \in M$, and $a_{ij} \in R$ for each $i = 1, \dots, n$ and $j = 1, \dots, m$ such that

$$x_i = \sum_{j=1}^{n} a_{ij} y_j$$

and for each j,

$$0 = \sum_{i=1}^{n} f_i a_{ij}.$$

Lemma: An R-module M is flat if and only if every relation in M is trivial.

Proof. Suppose M is flat, and let $\sum_{i=1}^{n} f_i x_i = 0$ be a relation in M. Let I be the ideal $(f_1, \dots, f_n)$, and let $K \leq R^n$ be the kernel of the map $R^n \to I$ given by $(a_1, \dots, a_n) \mapsto \sum_{i=1}^{n} a_i f_i$. This yields the short exact sequence

$$0 \to K \to R^n \to I \to 0.$$

It follows that $\sum_{i=1}^{n} f_i \otimes x_i$ is an element of $I \otimes_R M$ that maps to 0 in $R \otimes_R M = M$, which follows from taking the tensor product with M of the exact sequence $0 \to I \to R$.

By the flatness of M, $\sum_{i=1}^{n} f_i \otimes x_i$ is zero in $I \otimes_R M$. By exactness, there exists an element of $K \otimes_R M$ mapping to $\sum_{i=1}^{n} e_i \otimes x_i \in R^n \otimes_R M$, where e_i denotes the ith element of the standard basis. Writing this element as $\sum_{j=1}^{m} k_j \otimes y_j$, we may write the image of k_j in R^n as $\sum_{i=1}^{n} a_{ij} e_i$ to obtain our desired result.

Now, assume every relation in M is trivial. Let I be a finitely generated ideal, and let $x = \sum_{i=1}^{n} f_i \otimes x_i$ be an element of $I \otimes_R M$ that maps to zero in $R \otimes_R M \cong M$. This means that $\sum_{i=1}^{n} f_i x_i$ is a relation in M. Since this relation is trivial, we have

$$\begin{aligned} x &= \sum_{i=1}^{n} f_i \otimes x_i \\ &= \sum_{i=1}^{n} \left(\sum_{j=1}^{m} f_i a_{ij} \right) \otimes y_j \\ &= 0. \end{aligned}$$

$\square$

Equivalently, we have the following characterization of flatness that follows from jiu-jitsu with the above lemma.

Theorem: An R-module M is flat if and only if for every R-module homomorphism $f \colon R^n \to M$ from a finitely generated free R-module to M, and for all submodules $K \leq \ker(f)$, there exists a free R-module $G \cong R^m$ and maps $g \colon F \to G$ and $h \colon G \to M$ satisfying $h \circ g = f$ and $g(K) = 0$.

$$\begin{tikzcd} K \leq \ker(f) \arrow[r, "\iota"] & R^n \arrow[r, "f"] \arrow[d, dashed, "g"'] & M \\ & G \cong R^m \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

The following is a condensation of the previous result. Recall that an R-module M is called finitely presented if there is an exact sequence $R^m \to R^n \to M \to 0$ (i.e., the module M is defined by a finite generating set and finite generating set for its module of relations).

Corollary: Let M be an R-module. Then, M is flat if and only if for any finitely-presented R-module P and R-module homomorphism $f\colon P \to M$, there is a free finitely-generated R-module F and maps $h\colon P \to F$ and $g\colon F \to M$ such that $f = g \circ h$.

$$\begin{tikzcd} P \arrow[r, "f"] \arrow[d, dashed, "g"'] & M \\ F \cong R^n \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

Localization of Modules

Similar to how, given a multiplicative subset $S \subseteq R$ of a unital commutative ring, we may construct the ring of fractions $S^{-1}R$, it is possible to construct a localization of any R-module $S^{-1}M$ by a similar method. That is, define an equivalence relation on $M \times S$ by $(m,s) \sim (n,t)$ if and only if there exists $x \in S$ such that $x(tm - sn) = 0$, and write equivalence classes as $[(m,s)] = \frac{m}{s}$. This yields the module of fractions of M with respect to S, and is a $S^{-1}R$-module.

In fact, the localization $S^{-1}M$ is an R-module (with each r acting as $r/1$), and there is an R-module homomorphism $M \to S^{-1}M$ taking $m \mapsto \frac{m}{1}$. Notice that $\ker(\iota) = \{m \in M : sm = 0 \text{ for some } s \in S\}$.

Proposition: Let S be a multiplicative subset of R, and let M be an R-module. There is an isomorphism $S^{-1}M \cong S^{-1}R \otimes_R M$ as $S^{-1}R$-modules.

Proof. The map from $S^{-1}R \times M$ to $S^{-1}M$ taking $(r/s, m) \mapsto \frac{rm}{d}$ is well-defined and R-balanced, so we have a homomorphism $S^{-1}R \otimes_R M \to S^{-1}M$.

We obtain a map $\frac{m}{d} \mapsto \frac{1}{d} \otimes m$ yields a well-defined inverse homomorphism, since if $\frac{m}{d} = \frac{m'}{d'}$, then we may write $t(d'm - dm') = 0$, whence $\frac{1}{d} \otimes m$ can be written as $\frac{1}{td'd} \otimes td'm = \frac{1}{xd'd} \otimes xdm' = \frac{1}{d'} \otimes m'$.

Therefore, $S^{-1}M$ is isomorphic to $S^{-1}R \otimes_R M$ as $S^{-1}R$-modules since these inverse isomorphism are also $D^{-1}R$-module homomorphisms. ☐

Proposition: Let R be a commutative ring with 1, and let $S^{-1}R$ be the localization.

(i) If I and J are ideals of R, then $S^{-1}(I+J) = S^{-1}I + S^{-1}J$ and $S^{-1}(I \cap J) = S^{-1}I \cap S^{-1}J$. Additionally, localization commutes with quotients: $S^{-1}R/S^{-1}I \cong S^{-1}(R/I)$, where the localization on the right is with respect to the image of S in the quotient R/I.

(ii) If N is the nilradical of R, then $S^{-1}N$ is the nilradical of $S^{-1}R$. Similarly, if

$$\sqrt{I} = \{r \in R : r^n \in I \text{ for some } n \in \mathbb{N}\}$$

denotes the radical of I, then $\sqrt{S^{-1}I} = S^{-1}\sqrt{I}$.

(iii) If Q is a P-primary ideal (i.e., $\sqrt{Q} = P$ and $xy \in Q$ if and only if $x \in Q$ or $y^n \in Q$ for some $n > 0$), then if $S \cap P \neq \emptyset$ we have $S^{-1}Q = S^{-1}R$, and if $S \cap P = \emptyset$, $S^{-1}P$ is a prime ideal and the extension $S^{-1}Q$ of Q is a $S^{-1}P$-primary ideal in $S^{-1}R$. The contraction to R of $S^{-1}Q$ is Q.

(iv) If L and N are submodules of the R-module M, then

- $S^{-1}(L+N) = S^{-1}L + S^{-1}N$;

- $S^{-1}(L \cap N) = S^{-1}L \cap S^{-1}N$;
- $S^{-1}N$ is a submodule of $S^{-1}M$;
- $S^{-1}M/S^{-1}N = S^{-1}(M/N)$.

(v) If M and N are R-modules, then $S^{-1}(M \oplus N) \cong S^{-1}M \oplus S^{-1}N$.

(vi) The localization $S^{-1}R$ is a flat R-module. That is, if

$$0 \to L \to M \to N \to 0$$

is a short exact sequence of R-modules, then

$$0 \to S^{-1}L \to S^{-1}M \to S^{-1}N \to 0$$

is a short exact sequence of $S^{-1}R$-modules.

Proof.

(i) Notice that $\frac{i+j}{s} = \frac{i}{s} + \frac{j}{s}$ for any $i \in I$, $j \in J$, and $s \in S$, giving $S^{-1}(I+J) \subseteq S^{-1}I + S^{-1}J$. Meanwhile, we observe that $\frac{i}{s_1} + \frac{j}{s_2} = \frac{s_2 i + s_1 j}{s_1 s_2}$, giving the reverse direction.

Similarly, we have $S^{-1}(I \cap J) \subseteq S^{-1}I \cap S^{-1}J$. If $\frac{a}{s} \in S^{-1}I \cap S^{-1}J$, then $s_1 a \in I$ and $s_2 a \in J$ for some $s_1, s_2 \in S$. Then, $s_1 s_2 a \in I \cap J$, and since $\frac{a}{s} = \frac{s_1 s_2 a}{s_1 s_2 s}$, we obtain the reverse direction $S^{-1}I \cap S^{-1}J \subseteq S^{-1}(I \cap J)$.

Finally, we apply the result in (vi) to the exact sequence

$$0 \to I \to R \to R/I \to 0$$

to show that localization commutes with quotients.

(ii) Suppose $a \in \sqrt{I}$, so that $a^n \in I$ for some $n \geq 1$. Then, $(a/s)^n = \frac{a^n}{s^n} \in S^{-1}I$, so $S^{-1}\sqrt{I} \subseteq \sqrt{S^{-1}I}$. Conversely, if $\frac{a}{s} \in \sqrt{S^{-1}I}$, then $(a/s)^n \in S^{-1}I$ for some $n \geq 1$, meaning there is s_1 such that $s_1 a^n \in I$. Therefore, $(s_1 a)^n = s_1^{n-1}(s_1 a^n) \in I$, so that $s_1 a \in \sqrt{i}$. Thus, $\frac{a}{s} = \frac{s_1 a}{s_1 s} \in S^{-1}\sqrt{I}$, so that $\sqrt{S^{-1}I} \subseteq S^{-1}\sqrt{I}$.

The statement with regard to the nilradical emerges from the fact that the nilradical is the radical of the ideal generated by 0.

(iii) We start by observing that $S \cap P = \emptyset$ if and only if $S \cap Q = \emptyset$. One direction emerges from the fact that $Q \subseteq \sqrt{Q} = P$, while the other emerges from the fact that $s \in S \cap P$ implies $s^n \in S \cap Q$ for some n.

Next, if $S \cap P = \emptyset$, then $S^{-1}P$ is a prime ideal. We will show that $S^{-1}Q$ is a primary ideal in $S^{-1}R$. Let $\frac{a}{s_1}\frac{b}{s_2} \in S^{-1}Q$ with $\frac{a}{s_1} \notin S^{-1}Q$. There is some element $s \in S$ such that $sab \in Q$. Since $a \notin Q$ and Q is primary, we have $(sb)^n \in Q$ for some $n \geq 1$. We have $\left(\frac{b}{d_2}\right)^n = \frac{d^n b^n}{d^n d_2^n} \in S^{-1}Q$, so that $S^{-1}Q$ is primary. Furthermore, the radical of $S^{-1}Q$ is $S^{-1}P$ by (ii).

Finally, the contraction of $S^{-1}Q$ is the saturation of Q, consisting of the elements $r \in R$ such that $sr \in Q$ for some $s \in S$. Since Q is P-primary, the definition of primary implies that if $sr \in Q$ and $s \notin P$, then $r \in Q$, whence the contraction of $S^{-1}Q$ is Q.

(iv) Since ideals I and J are submodules of R, we may apply the same method for proving I and J satisfy (i) to the submodules L and N.

(v) From (vi), we have that if the exact sequence

$$0 \to L \to M \to N \to 0$$

of R-modules splits, then the exact sequence

$$0 \to S^{-1}L \to S^{-1}M \to S^{-1}N \to 0$$

of $S^{-1}R$-modules splits (as tensor products and direct sums commute).

(vi) Suppose we have a short exact sequence

$$0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$$

of R-modules. Every element of $S^{-1}N$ is of the form $\frac{n}{s}$ for some $n \in N$ and $s \in S$. Since φ is surjective, $n = \varphi(m)$ for some $m \in M$, so the extension of φ yields $\varphi'\left(\frac{m}{s}\right) = \frac{\varphi(m)}{s} = \frac{n}{s}$, meaning that φ' is surjective.

If $\frac{m}{s} \in \ker(\varphi)$, then $s_1\varphi(m) = 0$ for some $s_1 \in S$. So, we have $\varphi(s_1 m) = 0$, so $s_1 m = \psi(\ell)$ for some $\ell \in L$ by exactness of the original sequence at M. Therefore,

$$\begin{aligned}\frac{m}{s} &= \frac{s_1 m}{s_1 s}\\ &= \frac{\psi(\ell)}{s_1 s}\\ &= \psi'\left(\frac{\ell}{s_1 s}\right),\end{aligned}$$

meaning that $\ker(\varphi') \subseteq \operatorname{im}(\psi')$. If $\frac{\psi(\ell)}{s} \in \operatorname{im}(\psi')$, then

$$\begin{aligned}\varphi'\left(\frac{\psi(\ell)}{s}\right) &= \frac{\varphi(\psi(\ell))}{d}\\ &= 0,\end{aligned}$$

so that $\operatorname{im}(\psi') \subseteq \ker(\varphi')$. This yields exactness of the induced sequence at $S^{-1}M$.

Suppose $\psi'\left(\frac{\ell}{d}\right) = 0$. Then, $s_2\psi(\ell) = 0$ for some $s_2 \in S$, meaning that $\psi(s_2\ell) = 0$. By the injectivity of ψ, $s_2\ell = 0$, so that

$$\begin{aligned}\frac{\ell}{s} &= \frac{s_2\ell}{s_2 s}\\ &= 0.\end{aligned}$$

Thus, the sequence

$$0 \to S^{-1}L \to S^{-1}M \to S^{-1}N \to 0$$

is exact.

□

Nakayama's Lemma

Definition: The *Jacobson radical* $\mathfrak{j}$ of a commutative ring A is the intersection of all the maximal ideals.

Lemma: Let $x \in A$. The following are equivalent:

(i) $x \in \mathfrak{j}$;

(ii) $1 + xy \in A^\times$ for all $y \in R$.

Proof. Suppose $x \in \mathfrak{j}$. If it were the case that $1+x \notin A^\times$, then $(1+x) \neq A$, so it is contained in some maximal ideal $\mathfrak{m}$ of A. Yet, that implies that 1 is in the maximal ideal $\mathfrak{m}$ since $x \in \mathfrak{m}$, implying that $\mathfrak{m} = A$.

Suppose $x \notin \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$ of A. Then, $\mathfrak{m} + (x) = A$, so there is some $y \in A$ such that $1 + xy \in \mathfrak{m}$, implying that $1 + xy \notin A^\times$. □

Lemma: Let $\mathfrak{a}$ be an ideal of a commutative ring A that is contained in the Jacobson radical $\mathfrak{j}$. Let E be a finitely generated A-module. Suppose $\mathfrak{a}E = E$. Then, $E = \{0\}$.

Proof. We induct on the number of generators of E. Let $x_1, \dots, x_s$ be generators of E. Then, there exist $a_1, \dots, a_s \in \mathfrak{a}$ such that

$$x_s = a_1 x_2 + \cdots + a_s x_s,$$

meaning there is some $a \in \mathfrak{a}$ such that $(1 + a)x_s \in \langle x_1, \dots, x_{s-1} \rangle$. Notice that $1 + a$ is a unit in A since $\mathfrak{a} \subseteq \mathfrak{j}$. Therefore, $x_s \in \langle x_1, \dots, x_{s-1} \rangle$, so induction gives the desired result. □

Corollary: Let A be a ring, M an A-module, and let N and N' be submodules of M with $\mathfrak{a}$ an ideal of A. Suppose $M = N + \mathfrak{a}N'$.

(a) If $\mathfrak{a}$ is nilpotent, then $M = N$.

(b) If $\mathfrak{a}$ is contained in the Jacobson radical and N' is finitely generated, then $M = N$.

Proof.

(a) We observe that

$$\begin{aligned} M/N &= \mathfrak{a}\overline{N'} \\ &\cong \mathfrak{a}\frac{N' + N}{N} \\ &\subseteq \frac{N' + N}{N} \\ &= M/N, \end{aligned}$$

meaning that in particular, $\mathfrak{a}(M/N) = \mathfrak{a}\overline{N'} = \overline{N'} = M/N$. Therefore, since $\mathfrak{a}$ is nilpotent, we have

$$\begin{aligned} M/N &= \mathfrak{a}(M/N) \\ &= \mathfrak{a}^2(M/N) \\ &\vdots \\ &= 0. \end{aligned}$$

(b) By Nakayama's Lemma, if $\mathfrak{a}$ is contained in the Jacobson radical, then since $\mathfrak{a}(M/N) = M/N$, it follows that $M/N = 0$.

□

Jordan Canonical Form

If we are given a finite-dimensional vector space V over a field F and a linear transformation $T\colon V \to V$, then V attains the structure of an $F[x]$-module by defining the action of x by $x \cdot v = Tv$. In particular, this means that we may apply the invariant factors and elementary divisors decomposition(s) to V to obtain different canonical forms of T. The application of the invariant factors decomposition yields the rational canonical form. It can be shown that the rational canonical form of $A \in \mathrm{Mat}_n(F)$ is computed by taking the Smith Normal Form of the matrix $xI - A$.

Here, we will focus on applying the elementary divisors decomposition (in the special case of an algebraically closed field) to obtain the Jordan Canonical Form. Then, we will discuss some applications of this. By the definition of the $F[x]$-module structure on V, the element T acts on V by x acting by multiplication on each of the cyclic summands $F[x]/(x - \lambda)^k$.

Define an ordered basis

$$\left\{ (\overline{x} - \lambda)^{k-1}, (\overline{x} - \lambda)^{k-2}, \dots, \overline{x} - \lambda, 1 \right\}$$

for the quotient $F[x]/(x-\lambda)^k$. Viewing $(\overline{x}-\lambda)^k = 0$ in the quotient, we have that x acts as

$$\begin{aligned} x(\overline{x}-\lambda)^{k-1} &= \lambda(\overline{x}-\lambda)^{k-1} + (\overline{x}-\lambda)^k \\ &= \lambda(\overline{x}-\lambda)^{k-1} \\ x(\overline{x}-\lambda)^{k-2} &= \lambda(\overline{x}-\lambda)^{k-2} + \left(\overline{x}-\lambda^{k-1}\right) \\ &\vdots \\ \overline{x}-\lambda &= \lambda(\overline{x}-\lambda) + (\overline{x}-\lambda)^2 \\ x &= \lambda(1) + (\overline{x}-\lambda). \end{aligned}$$

This yields the matrix

$$\begin{pmatrix} \lambda & 1 & & & \\ & \lambda & \ddots & & \\ & & \ddots & 1 & \\ & & & \lambda & 1 \\ & & & & \lambda \end{pmatrix}.$$

Each of the cyclic summands $F[x]/(x-\lambda)^k$ correspond to the *generalized eigenspaces* of T corresponding to the eigenvalue λ.

Notice that the minimal polynomial of a Jordan block $J_d(\lambda)$ is given by $(x-\lambda)^d$. In particular, if $\lambda = 0$, we have that $F[x]/x^d$ admits a minimal polynomial of $\mu(x) = x^d$.

> **Theorem:** The total number of Jordan blocks with eigenvalue λ is given by the geometric multiplicity of λ. That is, the number of Jordan blocks with eigenvalue λ is given by $\dim\ker(A-\lambda I)$.

Assume $\lambda \neq 0$. Considering the transformation corresponding to $T^2 = J_n(\lambda)^2$, we observe that we may write

$$x^2 - \lambda^2 = (x-\lambda)(x+\lambda).$$

In particular, this means that the minimal polynomial $\mu_{T^2}(y)$ divides $(y-\lambda)^n(y+\lambda)^n$. Since the greatest common divisor of $(x-\lambda)$ and $(x+\lambda)$ is 1, we have that $(\overline{x}+\lambda)^d$ is invertible in $F[x]/(x-\lambda)^n$ for any exponent d; the minimal polynomial $\mu_{T^2}(y)$ of T^2 will thus divide $\left(y-\lambda^2\right)^n$. Since $\left(\overline{x}^2-\lambda^2\right)^d = (\overline{x}-\lambda)^d\theta$, where θ is some invertible element of $F[x]/(x-\lambda)^n$, it follows that $\left(\overline{x}^2-\lambda^2\right)^d \neq 0$ for any $0 \leq d < n$, meaning that the minimal polynomial μ_{T^2} of T^2 is equal to $\left(x-\lambda^2\right)^n$.

Representation Theory

We assume that the group G is finite and that all representations are complex unless stated otherwise.

- **Schur's Lemma:** If M is a simple R-module, then $\operatorname{End}_R(M)$ is a division ring. Equivalently, if M is a simple $\mathbb{C}$-module, then any element $T \in \operatorname{End}_{\mathbb{C}}(M)$ is a diagonal operator $T = \lambda I$.
- **Maschke's Theorem:** Suppose we are given a representation $\rho\colon G \to \mathrm{GL}_n(F)$ over an arbitrary field. If $\operatorname{char}(F)$ does not divide the order of G (or $\operatorname{char}(F) = 0$), then the representation ρ is completely reducible. Equivalently, the $F[G]$-module structure on V decomposes as a direct sum

  $$V \cong \bigoplus_{i=1}^{r} V_i$$

  of irreducible $F[G]$-submodules V_i.

- **Direct Sums:** If $\rho_1 \colon G \to GL(V_1)$ and $\rho_2 \colon G \to GL(V_2)$ are any two representations over the same field, then the direct sum $\rho_1 \oplus \rho_2 \colon G \to GL(V_1 \oplus V_2)$ is given by $(\rho_1 \oplus \rho_2)(g)(v_1, v_2) = (\rho_1(g)v_1, \rho_2(g)v_2)$.

  The character of the direct sum representation is given by $\chi_{\rho_1 \oplus \rho_2} = \chi_{\rho_1} + \chi_{\rho_2}$.

- **Tensor Products:** If $\rho_1 \colon G \to GL(V_1)$ and $\rho_2 = G \to GL(V_2)$ are any two representations over the same field, then the tensor product $\rho_1 \otimes \rho_2 \colon G \to GL(V_1 \otimes V_2)$ is defined by $(\rho_1 \otimes \rho_2)(g)(v_1 \otimes v_2) = \rho_1(g)v_1 \otimes \rho_2(g)v_2$.

  The character of the tensor product representation is given by $\chi_{\rho_1 \otimes \rho_2} = \chi_{\rho_1}\chi_{\rho_2}$.

- **Fixed-Point Formula:** Given an action $G \curvearrowright X$, the permutation representation $G \to \operatorname{Sym}(X)$ yields a representation $G \to GL(V)$, where $\dim(V) = |X|$ and we consider the elements of $GL(V)$ as the corresponding permutation matrices in $\operatorname{Mat}_{|X|}(\mathbb{C})$. Since the fixed points of any permutation are precisely equal to the trace of the permutation matrix, it follows that the character $\chi(g)$ has value equal to the number of fixed points of g.

- **Number of Characters:** The characters $\chi_\rho(\cdot) = \operatorname{Tr}(\rho(\cdot))$ of any representation $\rho \colon G \to GL_k(\mathbb{C})$ are constant on conjugacy classes. The number of (equivalence classes of) *irreducible* complex characters is equal to the number of conjugacy classes.

- **Finding the Center:** A conjugacy class K in G is an element of $Z(G)$ if and only if the size of the conjugacy class equals 1.

- **Abelianization:** The number of one-dimensional irreducible characters is equal to the order $|G/[G,G]|$ of the abelianization. Furthermore, a conjugacy class K of G is contained in $[G,G]$ if and only if $\chi_s(K) = 1$ for all 1-dimensional characters χ_s.

  The one-dimensional characters $\chi \colon G \to \mathbb{C}^\times$ descend to a map on the abelianization $\chi \colon G/[G,G] \to \mathbb{C}^\times$. The group of one-dimensional characters with pointwise multiplication is isomorphic to the abelianization $G/[G,G]$.

- **First Orthogonality Relation:**

$$\begin{aligned} \left\langle \chi_j, \chi_k \right\rangle &= \frac{1}{|G|}\sum_{g \in G} \overline{\chi_j(g)}\chi_k(g) \\ &= \begin{cases} 1 & j = k \\ 0 & \text{else.} \end{cases} \end{aligned}$$

- **Second Orthogonality Relation:**

$$\sum_{j=1}^{r} \chi_j(g)\overline{\chi_j(h)} = \begin{cases} 0 & g \text{ and } h \text{ are not conjugate} \\ \frac{|G|}{|K_G(g)|} & g \text{ and } h \text{ are conjugate.} \end{cases}$$

  Equivalently, for any $g \in G$, we have

$$|C_G(g)| = \sum_{i=1}^{r} |\chi_i(g)|^2.$$

- **Irreducibility Criterion:** a representation ρ with character χ_ρ is irreducible if and only if $1 = \left\langle \chi_\rho, \chi_\rho \right\rangle$.

- **Order of the Group:** If $\chi_1, \dots, \chi_r$ are the irreducible complex characters, then

$$|G| = \sum_{j=1}^{r} \chi_j(e)^2.$$

- **Irreducibility and Quotients:** Suppose N is a normal subgroup of a group G. If $\rho\colon G/N \to \mathrm{GL}(V)$ is an irreducible representation, then $\widetilde{\rho}\colon G \to \mathrm{GL}(V)$, defined by $\widetilde{\rho}(g) = \rho(gN)$, is irreducible, and has $\widetilde{\rho}(n) = \mathrm{id}_V$ for all $n \in N$.
- **Normal Subgroups:** A conjugacy class K_i is a member of some normal subgroup of G if $\chi(g) = \chi(e)$ for some irreducible character χ and $g \in K_i$.
- **Decomposition of Regular Representation:** If G is a finite group, then the regular representation $\lambda\colon G \to \mathbb{C}[G]$ decomposes into a direct sum of irreducible representations

$$\begin{aligned}\mathbb{C}[G] &\cong \bigoplus_{i=1}^{r} V_i^{\oplus n_i}\\ &\cong \bigoplus_{i=1}^{r} \mathrm{Mat}_{n_i}(\mathbb{C})\end{aligned}$$

  where each n_i is equal to the dimension of the irreducible representation V_i.
- **Counting Multiplicity:** Given any complex representation V and its decomposition into irreducibles

$$V \cong \bigoplus_{i=1}^{n} V_i^{\oplus d_i},$$

  the multiplicity d_i can be found by taking

$$\begin{aligned}d_i &= \left\langle \chi_V, \chi_{V_i}\right\rangle\\ &= \frac{1}{|G|}\sum_{g\in G}\overline{\chi_V(g)}\chi_{V_i}(g).\end{aligned}$$

The Projection Formula and its Applications

If $\rho\colon G \to \mathrm{GL}(V)$ is a representation of G, then the fixed points V^G form a subrepresentation. Notice that given an endomorphism $g\colon V \to V$, this is not an $F[G]$-module homomorphism, but the average

$$\varphi = \frac{1}{|G|}\sum_{g\in G}\rho(g)$$

is a $F[G]$-module homomorphism. Furthermore, $\varphi\colon V \to V$ is a projection onto V^G, and

$$\begin{aligned}\dim\left(V^G\right) &= \mathrm{Tr}(\varphi)\\ &= \frac{1}{|G|}\sum_{g\in G}\mathrm{Tr}(\rho(g))\\ &= \frac{1}{|G|}\sum_{g\in G}\chi_V(g).\end{aligned}$$

If V and W are representations of G, then $\mathrm{Hom}(V,W)$ is a representation of G, where we identify $\mathrm{Hom}(V,W) \cong V^* \otimes W$. Furthermore, the set of $F[G]$-module homomorphisms $\mathrm{Hom}_G(V,W)$ has dimension given by

$$\begin{aligned}\dim(\mathrm{Hom}_G(V,W)) &= \dim(\mathrm{Hom}_G(F, V^*\otimes W))\\ &= \mathrm{Tr}\left(\frac{1}{|G|}\sum_{g\in G}\rho_{V^*\otimes W}(g)\right)\\ &= \frac{1}{|G|}\sum_{g\in G}\chi_{V^*\otimes W}(g)\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{|G|}\sum_{g\in G}\chi_{V^*}(g)\chi_W(g)\\
&= \frac{1}{|G|}\sum_{g\in G}\overline{\chi_V(g)}\chi_W(g)\\
&= \langle \chi_V, \chi_W\rangle.
\end{aligned}$$

If V is irreducible, then the space $\operatorname{Hom}_G(V,W)$ of $F[G]$-module homomorphism $V \to W$ has dimension equal to the multiplicity of V in W, and similarly, if W is irreducible, then $\operatorname{Hom}_G(V,W)$ has dimension equal to the multiplicity of W in V. If V and W are both irreducible, then Schur's Lemma gives

$$\dim(\operatorname{Hom}_G(V,W)) = \begin{cases} 1 & V \cong W \\ 0 & V \ncong W. \end{cases}$$

Since the character $\chi_{\operatorname{Hom}(V,W)}$ is equal to $\chi_V\overline{\chi_W}$ by the identification $\operatorname{Hom}(V,W) \cong V \otimes W^*$. This yields the First Orthogonality Relation: if V and W are irreducible representations, then

$$\begin{aligned}
\langle \chi_V, \chi_W\rangle &= \frac{1}{|G|}\sum_{g\in G}\overline{\chi_V(g)}\chi_W(g)\\
&= \begin{cases} 1 & V \cong W \\ 0 & V \ncong W. \end{cases}
\end{aligned}$$

Furthermore, notice that this gives that the characters χ_{V_i} of the irreducible representations V_i are linearly independent. If V is any representation of G, then by Maschke's Theorem there is a decomposition into irreducible representations

$$V \cong \bigoplus_{i=1}^{n} V_i^{\oplus d_i},$$

so that

$$\chi_V = \sum_{i=1}^{n} d_i \chi_{V_i}.$$

We may then compute the multiplicity d_i by $\left\langle \chi_V, \chi_{V_i}\right\rangle$.

Example (Character Table of S_4)**:** We wish to find the character table of S_4. We start by observing that the partitions of 4 yield cycle types given by:

- the identity (1 element);
- transpositions (6 elements);
- 3-cycles (8 elements);
- 4-cycles (6 elements);
- cycles of type $2+2$ (3 elements);

There are three representations that we can already establish:

- the trivial representation;
- the sign representation $\operatorname{sgn}\colon S_4 \to \mathbb{C}$ given by $\tau \mapsto \operatorname{sgn}(\tau)$;
- the standard representation ρ_{std} emerging in the decomposition $\rho_{\text{def}} = \rho_{\text{triv}} \oplus \rho_{\text{std}}$, where $\rho_{\text{def}}\colon S_4 \to \operatorname{GL}_4(\mathbb{C})$ is given by taking a permutation to the corresponding permutation matrix.

These have character table rows given by

- $\chi_{\text{triv}} = (1,1,1,1,1)$;
- $\chi_{\text{sgn}} = (1,-1,1,-1,1)$;
- $\chi_{\text{std}} = \chi_{\text{def}} - \chi_{\text{triv}} = (4,2,1,0,0) - (1,1,1,1,1) = (3,1,0,-1,-1)$.

Verifying the irreducibility of χ_{std} can be determined by taking the inner product with itself. We determine that there are two other representations with dimensions 2 and 3. First, observe that taking the tensor product $\rho_{\text{sgn}} \otimes \rho_{\text{std}}$ yields another representation with dimension 3 with character

$$\chi_{\text{twist}} = (3,-1,0,1,-1).$$

Observe that $\langle \chi_{\text{twist}}, \chi_{\text{twist}} \rangle = 1$, so this is another irreducible representation.

Finally, using the orthogonality relations we may fill out the final character to be $\chi_{\text{quot}} = (2,0,-1,0,2)$. This yields the following character table.

	id	$(1,2)$	$(1,2,3)$	$(1,2,3,4)$	$(1,2)(3,4)$
ρ_{triv}	1	1	1	1	1
ρ_{sgn}	1	-1	1	-1	1
ρ_{quot}	2	0	-1	0	2
ρ_{std}	3	1	0	-1	-1
ρ_{twist}	3	-1	0	1	-1

To determine what χ_{quot} is, we observe that any of the cycles of type $2+2$ are a trace 2 involution. In particular, this means that any cycle of type $2+2$ must map to the identity. Therefore, ρ_{quot} is a representation of the quotient group $S_4/V \cong S_3$, where $V \trianglelefteq S_4$ is a copy of the Klein 4-group with elements

$$V = \{\text{id}, (1,2)(3,4), (1,3)(2,4), (1,4)(2,3)\}.$$

The Dimension of Irreducible Representations

Definition: An algebraic integer (resp. algebraic number) is a root α of a monic polynomial $p(x) \in \mathbb{Z}[x]$ (resp. $p(x) \in \mathbb{Q}[x]$). Equivalently, taking the companion matrix, α is an eigenvalue of a matrix $P \in \text{Mat}_n(\mathbb{Z})$ (resp. $P \in \text{Mat}_n(\mathbb{Q})$) for some n.

The set of algebraic integers is denoted $\overline{\mathbb{Z}}$ and the set of algebraic numbers is denoted $\overline{\mathbb{Q}}$.

It can be shown that the algebraic numbers are a field and the algebraic integers are a ring.

Proposition: The only rational algebraic integers are integers. That is, $\overline{\mathbb{Z}} \cap \mathbb{Q} = \mathbb{Z}$.

Proof. Let α be a root of

$$p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0,$$

and suppose we have

$$\alpha = \frac{p}{q}$$

with $p, q \in \mathbb{Z}$ and $\gcd(p,q) = 1$. Then, the leading term in $p(\alpha)$ has q^n in the denominator, while all other terms have a lower power of q. If $q \neq \pm 1$, then $p(\alpha) \notin \mathbb{Z}$, which is a contradiction. Therefore, $\alpha \in \overline{\mathbb{Z}} \cap \mathbb{Q}$ implies $\alpha \in \mathbb{Z}$. The reverse direction emerges from the fact that $\alpha \in \mathbb{Z}$ is a root of the monic polynomial $p(x) = x - \alpha$. $\square$

Proposition: Let V be an irreducible representation of a finite group G over $\mathbb{C}$. Let $C_1, \dots, C_r$ be the conjugacy classes of G. Let g_i be representatives of C_i, and set

$$\lambda_i = \chi_V(g_i)\frac{|C_i|}{\dim(V)}.$$

Then, λ_i is an algebraic integer for each i.

Proof. Let C be a conjugacy class in G. Set $P = \sum_{h\in C} h$. Then, P is a central element of $\mathbb{Z}[G]$, so by Schur's Lemma, it acts via a scalar λ. This scalar is an algebraic integer since $\mathbb{Z}[G]$ is a finitely generated $\mathbb{Z}$-module, so that every element of $\mathbb{Z}[G]$ is integral over $\mathbb{C}$. Taking the trace of P in V yields

$$\begin{aligned}\lambda \dim(V) &= \operatorname{tr}(\lambda \operatorname{id}_V)\\ &= \operatorname{tr}(P)\\ &= \operatorname{tr}\left(\sum_{h\in C} h\right)\\ &= |C|\chi_V(g)\end{aligned}$$

for the representative $g \in C$. Therefore, by the definition of λ, we have

$$\lambda = \frac{|C|\chi_V(g)}{\dim(V)}.$$

Since λ was defined to be an algebraic integer, this gives the desired result. □

Theorem: Let G be a finite group, and let V be an irreducible representation of G over $\mathbb{C}$. Then, $\dim(V)$ divides $|G|$.

Proof. Letting C_i and g_i be the conjugacy classes and representatives of G as above, we have that the value $\sum_{i=1}^{r} \lambda_i \overline{\chi_V(g_i)}$ is an algebraic integer. This follows from the fact that the λ_i are algebraic integers by above, $\chi_V(g_i)$ are a sum of roots of unity — they are the sum of eigenvalues of a finite-order linear operator given by $\rho(g_i)$ — and $\overline{\mathbb{Z}}$ is a ring.

By the definition of the λ_i and the fact that the χ_V are class functions, we have

$$\begin{aligned}\sum_{i=1}^{r} \lambda_i \overline{\chi_V(g_i)} &= \sum_{i=1}^{r} \frac{|C_i|\chi_V(g_i)\overline{\chi_V(g_i)}}{\dim(V)}\\ &= \frac{1}{\dim(V)}\sum_{g\in G}\chi_V(g)\overline{\chi_V(g)}\\ &= \frac{|G|\langle \chi_V, \chi_V\rangle}{\dim(V)}\\ &= \frac{|G|}{\dim(V)},\end{aligned}$$

where the final line emerges from the fact that $\langle \chi_V, \chi_V\rangle = 1$. Therefore, since $\frac{|G|}{\dim(V)}$ is a rational number that is an algebraic integer, it is an integer. □

Field Theory and Galois Theory

Normal and Separable Extensions

Theorem: Suppose E/F is a finite extension, and let K/F be any extension. If $\mathcal{B}$ is a (finite) basis for E/F, then $\mathcal{B}$ spans EK over K. In particular,

$$[EK : K] \leq [E : F].$$

Proof. Let $\mathcal{B} = \{\beta_1, \dots, \beta_n\}$ be a basis for E over F. We may lift the basis to $EK = K(\beta_1, \dots, \beta_n)$, where each β_i is algebraic over K. Since EK is then the set of polynomials over K in $\beta_1, \dots, \beta_n$, and each monomial in the β_i is a linear combination of $\beta_1, \dots, \beta_n$ over F, it follows that EK is the set of linear combinations of the β_i over K. Therefore, $\mathcal{B}$ spans EK over K. □

Definition: Let $\sigma\colon F \to L$ be an embedding (i.e., injective ring/field homomorphism) of the field F into L, and let E/F be an extension. An embedding $\overline{\sigma}\colon E \to L$ for which $\overline{\sigma}|_F = \sigma$ is an extension of σ to E. If $\overline{\sigma}$ extends the identity map $\iota\colon F \to F$, then we call it an F-embedding.

The set of all F-embedding $\sigma\colon E \to L$ is denoted $\operatorname{Hom}_F(E, L)$.

Definition: If K/F is an algebraic extension for which every F-embedding $\sigma\colon K \to \overline{F}$ takes $\sigma(K) = K$, then we call this extension *normal*. Some texts write $F \triangleleft K$.

Proposition: Let K/F be an algebraic extension. The following are equivalent:

(i) K is normal;

(ii) if $p(x) \in F[x]$ is any irreducible polynomial, then if $p(x)$ has a root in K, then $p(x)$ splits over K;

(iii) K is a splitting field for the family $\{\mu_{\alpha,F}(x) : \alpha \in K\}$.

Proof. Suppose K is normal. If $p(x)$ is any irreducible polynomial over F that has a root α in K, then if β is a root of $p(x)$ in $\overline{F}$, there is an embedding $\sigma\colon \operatorname{Hom}_F\left(K, \overline{F}\right)$ that takes $\sigma(\alpha) = \beta$. Since K is invariant under σ, it follows that $\beta \in K$. This also gives that K/F is the splitting field for the set of all minimal polynomials $\mu_{\alpha,F}(x)$ with $\alpha \in K$. This gives (i) implies (ii) and (ii) implies (iii).

Now, suppose K is a splitting field of a family $\mathcal{F}$ of polynomials over F. Then, $K = F(R)$, where R is the set of all roots of the polynomials in $\mathcal{F}$. Any embedding $\sigma\colon K \to \overline{F}$ must send roots to roots, so we have

$$\begin{aligned}\sigma(K) &= \sigma(F(R))\\ &= F(\sigma(R))\\ &= F(R)\\ &= K,\end{aligned}$$

where we use the fact that σ is an F-embedding that takes roots to roots. Since σ is an embedding of K to itself over F, and K/F is algebraic, it follows that σ is an F-automorphism of K. □

Theorem (Inheritance Properties of Normal Extensions)**:**

(i) If $E/K/F$ is a tower of extensions, and $F \triangleleft E$ is normal, then $K \triangleleft E$ is normal.

(ii) If $F \triangleleft E$ is normal, and K/F is any extension, then $K \triangleleft EK$ is normal.

(iii) If $\{E_i\}_{i\in I}$ is a family of fields, and $F \triangleleft E_i$ is normal for each i, then

$$\begin{aligned}F &\triangleleft \bigvee_{i\in I} E_i\\ F &\triangleleft \bigwedge_{i\in I} E_i,\end{aligned}$$

where $\bigvee_{i\in I} E_i$ denotes the composite of all the fields E_i and $\bigwedge_{i\in I} E_i$ denotes the infimum (/intersection) of all the fields E_i.

Proof.

(i) A splitting field for a family of polynomials over F is also a splitting field for the same family of polynomials over K.

(ii) Let E be a splitting field for a family $\mathcal{F}$ of polynomials over F, and let R be the set of roots in E of all the polynomials in $\mathcal{F}$. Then, $E = F(R)$. Therefore, $EK = K(R)$, so EK is a splitting field for the family $\mathcal{F}$ considered as polynomials over K. Therefore, $K \triangleleft EK$.

(iii) Let $\sigma\colon \bigvee_{i\in I} E_i \to \overline{F}$ be an embedding. Then, $\sigma|_{E_i}\colon E_i \to \overline{F}$ is an embedding, so since each E_i is normal over F, $\sigma(E_i) = E_i$, whence

$$\begin{aligned}\sigma\left(\bigvee_{i\in I} E_i\right) &= \bigvee_{i\in I} \sigma(E_i)\\ &= \bigvee_{i\in I} E_i,\end{aligned}$$

we obtain that σ is an automorphism of $\bigvee_{i\in I} E_i$. Similarly, $\sigma\colon \bigwedge_{i\in I} E_i \to \overline{F}$ has

$$\begin{aligned}\sigma\left(\bigwedge_{i\in I} E_i\right) &= \bigwedge_{i\in I} \sigma(E_i)\\ &= \bigcap_{i\in I} E_i,\end{aligned}$$

giving the desired result.

□

Definition: If E/F is not a normal extension, then there is a smallest extension N of E in an algebraic closure $\overline{F}$ for which N is normal over F. This is denoted the *normal closure* of E over F, and we write $\langle E/F\rangle$.

Proposition: Let E/F be an algebraic extension. Then,

$$\langle E/F\rangle = \bigvee_{\sigma\in \mathrm{Hom}_F(E,\overline{F})} \sigma(E).$$

Definition: An algebraic extension K/F is called separable if for any $\alpha \in K$, $\mu_{\alpha,F}(x)$ is a separable polynomial (i.e., the polynomial has distinct roots).

We note that, given a field embedding $\sigma\colon K \to L$, it is possible to define a ring homomorphism $\sigma_*\colon K[x] \to L[x]$ by taking

$$\sigma_*\left(\sum_{i=0}^{n} a_i x^i\right) = \sum_{i=0}^{n} \sigma(a_i)x^i.$$

Theorem (Simple Extension Lemma): Suppose we have a field embedding $\sigma\colon M \to L$, extensions M'/M and L'/L, and algebraic elements $\alpha \in M'$ and $\beta \in L'$. The following are equivalent:

(i) there exists a field embedding $\sigma_1 : M(\alpha) \to L(\beta)$ with $\sigma_1|_M = \sigma$ and $\sigma_1(\alpha) = \beta$;

(ii) $\sigma_*(\mu_{\alpha,M}(x)) = \mu_{\beta,L}(x)$.

Proof. Suppose such a field embedding exists. Then, pushing forward σ_1 to $(\sigma_1)_*\colon \frac{M[x]}{(\mu_{\alpha,M}(x))} \to \frac{L[x]}{(\mu_{\beta,L}(x))}$, we have that, since $\sigma_1(\alpha) = \beta$, it follows that $(\sigma_1)_*(\mu_{\alpha,M}(x)) = \mu_{\beta,L}(x)$.

In the reverse direction, since α is algebraic over x, we have that any element of $M(\alpha)$ is given by $p(\alpha)$ for some $p \in M[x]$. Defining

$$\sigma_1(p(\alpha)) = (\sigma_*(p))(\beta),$$

we have a well-defined extension of σ since if $p(\alpha) = q(\alpha)$, then $(p-q)(\alpha) = 0$, whence $\mu_{\alpha,M}(x)|(p-q)(x)$. Since σ_* is a ring homomorphism, it preserves divisibility, so that $\sigma_*(\mu_{\alpha,M}(x))|\sigma_*(p)-\sigma_*(q)$, whence $(\sigma_*(p))(\beta) = (\sigma_*(q))(\beta)$. □

Definition: Let F be a field, and let $\alpha \in \overline{F}$. We define the separable degree, $\mathrm{sdeg}_F(\alpha)$, to be the number of distinct $\overline{F}$-roots of $\mu_{\alpha,F}(x)$.

Proposition: Let K/F be an algebraic extension. Then, the following are equivalent:

(i) K/F is separable;

(ii) $K = F(\alpha_1, \dots, \alpha_n)$ where each $\alpha_i \in K$ is separable over F;

(iii) there exists a chain of subfields

$$F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n = K$$

and elements $\alpha_i \in K_i$ such that $K_i = K_{i-1}(\alpha)$ and each α_i is separable over K_{i-1};

(iv) $\left|\mathrm{Hom}_F\left(K, \overline{F}\right)\right| = [K : F]$.

Proof. Assume (i). Since K/F is separable, every element of K is separable. If $\alpha_1, \dots, \alpha_n$ are a basis for K/F, then $K = F(\alpha_1, \dots, \alpha_n)$ where each α_i are separable. This gives (ii).

Assuming (ii), we observe that $K_i = F(\alpha_1, \dots, \alpha_i)$ has $K_i = K_{i-1}(\alpha_i)$, and for any $\alpha \in K_i$, $\mu_{\alpha,K_{i-1}} | \mu_{\alpha,F}$, meaning that $\mu_{\alpha,K_{i-1}}$ is separable, whence α_i is separable over K_{i-1}. This gives (iii).

Now, suppose we have K_i and α_i defined as in (iii). We will construct an F-embedding of $K = K_n$ into $\overline{F}$ as follows. Start with the inclusion $\sigma_0 \colon F \to \overline{F}$, and extend via the simple extension lemma to $\sigma_1 \colon K_1 \to \overline{F}$, $\sigma_2 \colon K_2 \to \overline{F}$, etc. up to $\sigma_n \colon K_n \to \overline{F}$.

Counting degrees, we see that if $\sigma_{i-1} \colon K_{i-1} \to \overline{F}$ is chosen, then the extension to σ_i is uniquely determined by its value at α_i, and the valid choices are the roots of $(\sigma_{i-1})_*\left(\mu_{\alpha_i,K_{i-1}}\right)$. The number of these roots is equal to the separable degree of α_i over K_{i-1}. Yet, since the separable degree of the α_i equals the degree of the α_i, it follows that

$$\begin{aligned}
\left|\mathrm{Hom}_F\left(K, \overline{F}\right)\right| &= \prod_{i=1}^{n} \mathrm{sdeg}_{K_{i-1}}(\alpha_i) \\
&= \prod_{i=1}^{n} \deg\left(\mu_{\alpha_i,K_{i-1}}\right) \\
&= \prod_{i=1}^{n} [K_i : K_{i-1}] \\
&= [K : F].
\end{aligned}$$

Finally, assume that $\mathrm{sdeg}_{K_{i-1}}(\alpha_i) = \deg\left(\mu_{\alpha_i,K_{i-1}}\right)$ for any choices of α_i and K_i. We may take the case of one arbitrary element α_1 that is algebraic over F, hence separable by assumption. □

Theorem (Primitive Element Theorem)**:** If K/F is a finite separable extension, then there is $\gamma \in K$ such that $K = F(\gamma)$.

Proof. If F is finite, then K is finite, so $K^\times$ is cyclic, so there exists $\gamma \in K^\times$ such that $K^\times = \langle \gamma \rangle$, whence $F(\gamma) \supseteq \{0\} \cup \langle \gamma \rangle = K$.

If F is infinite, then $K = F(\alpha_1, \dots, \alpha_n)$ by (ii) above. It thus suffices to prove for the case of $n = 2$.

Suppose we have $K = F(\alpha, \beta)$, and let $m = [K : F]$. Since K/F is separable, $\left|\mathrm{Hom}_F\left(K, \overline{F}\right)\right| = m$. Call these embeddings $\sigma_1, \dots, \sigma_m$.

We claim that there exists some $c \in F$ such that $\{\sigma_i(\alpha + c\beta) : 1 \leq i \leq m\}$ are all distinct. It suffices to show that the set of all $c \in F$ such that there exists $i \neq j$ with $\sigma_i(\alpha + c\beta) = \sigma_j(\alpha + c\beta)$ is finite.

Notice that we have $\sigma_i(\alpha) + c\sigma_i(\beta) = \sigma_j(\alpha) + c\sigma_j(\beta)$. If it were the case that $\sigma_i(\beta) = \sigma_j(\beta)$, then we would have $\sigma_i(\alpha) = \sigma_j(\beta)$, implying that $\sigma_i = \sigma_j$ since they're defined as elements of $\mathrm{Hom}_F\left(K, \overline{F}\right)$ solely by where they map α and β. If it were the case that $\sigma_i(\beta) \neq \sigma_j(\beta)$, then we solve for c by taking

$$c = \frac{\sigma_j(\alpha) - \sigma_i(\alpha)}{\sigma_i(\beta) - \sigma_j(\beta)},$$

which has only finitely many options.

We will obtain the primitive element $\gamma = \alpha + c\beta$ as in the claim. Then, for all $1 \leq i \leq m$, we have $\sigma_i|_{F(\gamma)}$ is an element of $\text{Hom}_F\left(F(\gamma), \overline{F}\right)$, so that

$$\begin{aligned}[F(\gamma) : F] &\geq \left|\text{Hom}_F\left(F(\gamma), \overline{F}\right)\right| \\ &\geq m \\ &= [K : F] \\ &\geq [F(\gamma) : F].\end{aligned}$$

□

Irreducibility of Cyclotomic Polynomials over Finite Fields

We will let Φ_n denote the nth cyclotomic polynomial, defined by

$$\Phi_n(x) = \prod_{\gcd(d,n)=1} \left(x - \zeta_n^d\right).$$

Recall that the cyclotomic polynomials satisfy

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

Theorem: If n does not divide p, then the image of $\text{Gal}\left(\mathbb{F}_p(\zeta_n)/\mathbb{F}_p\right)$ in $(\mathbb{Z}/n\mathbb{Z})^\times$ under the standard embedding is the subgroup $\langle [p]_n \rangle$.

Proof. The polynomial $x^n - 1$ is separable in $\mathbb{F}_p[x]$, and the Galois group $\text{Gal}\left(\mathbb{F}_p(\zeta_n)/\mathbb{F}_p\right)$ is generated by the Frobenius map $\text{fr}\colon x \mapsto x^p$. The image of fr under the standard embedding of $\text{Gal}\left(\mathbb{F}_p(\zeta_n)/\mathbb{F}_p\right)$ into $(\mathbb{Z}/n\mathbb{Z})^\times$ is given by the congruence class $[a]_n$, where we have $\text{fr}(\theta) = \theta^a$ for all primitive roots of unity θ. In particular, this means that $\theta^p = \theta^a \bmod n$, meaning that the embedding maps fr to $[p]_n$. □

Theorem: If p does not divide n, then the irreducible factors of $\overline{\Phi_n(x)} \in \mathbb{F}_p[x]$ are distinct and each has degree equal to the order of p modulo n.

Proof. We observe that $\Phi_n(x) | (x^n - 1)$ in $\mathbb{Z}[x]$, so the divisibility relation is preserved when reducing modulo p, meaning that $\overline{\Phi_n(x)}$ is separable in $\mathbb{F}_p[x]$ as $x^n - \overline{1}$ is separable in $\mathbb{F}_p[x]$ by the assumption that $\gcd(p,n) = 1$.

Let α be a root of $\overline{\Phi_n(x)}$ in an extension of $\mathbb{F}_p$. We will show that α is a primitive nth root of unity.

Since $\overline{\Phi_n(x)} \left| \left(x^n - \overline{1}\right)\right.$, it follows that we have $\alpha^n = 1$. If α were not of order n, then it has some order m that properly divides n. In particular, this means that α is a root of some $\Phi_d(x)$ with $d|n$ properly. In particular, since $d|n$, it follows that $\overline{\Phi_n(x)\Phi_d(x)}$ divides $x^n - \overline{1}$, meaning that α is a double root of $x^n - \overline{1}$, contradicting the fact that $x^n - \overline{1}$ does not have any repeated roots. Therefore, α is a primitive root of unity.

If $\pi(x)$ is an irreducible factor of $\overline{\Phi_n(x)}$ in $\mathbb{F}_p[x]$, and α is a root of $\pi(x)$, then α is a primitive nth root of unity, so the degree of π, which is equal to $\left[\mathbb{F}_p(\alpha) : \mathbb{F}_p\right]$, is the order of p modulo n. □

The Nullstellensatz

We know that if K is an algebraically closed field, then the only irreducible polynomials are linear, so that the maximal ideals are of the form $(x - c)$ for some $c \in K$.

It turns out this extends to polynomial rings over several indeterminates. That is, if K is algebraically closed, then every maximal ideal over $K[x_1, \ldots, x_n]$ is of the form $(x_1 - c_1, \ldots, x_n - c_n)$. This result is part of the *Nullstellensatz* (or zero locus theorem) originally proved by Hilbert.

We will not prove the general case (which holds for any algebraically closed field), but we will prove it for the special case of uncountable fields.

Definition: We say an R-algebra S is *finitely generated* (or of "finite type") if there is a surjective R-algebra homomorphism $R[x_1,\dots,x_n] \to S$ from the free algebra on some finite number of indeterminates to S. That is, a finite-type algebra is some quotient of a polynomial ring.

Theorem (Nullstellensatz, Special Case)**:** Let K/F be a field extension. Assume that K is a finite-type F-algebra. Then, K/F is an algebraic extension.

Proof. Assume F is uncountable. Let K/F be a field extension, and assume K is a finitely generated F-algebra. Then, it is finitely generated as a field extension. Let $\alpha \in K \setminus F$. We will show that α is the root of some nonzero $f(x) \in F[x]$.

By the hypothesis, we have a surjective homomorphism $F[x_1,\dots,x_n] \to K$, meaning that there is a countable basis for K as a vector space over F, since there is a countable basis for $F[x_1,\dots,x_n]$ as a vector space over F.

Since the set

$$\left\{\frac{1}{\alpha - c} : c \in F\right\}$$

is uncountable, it is linearly dependent over F, so there exist distinct $c_1,\dots,c_m \in F$ and nonzero $\lambda_1,\dots,\lambda_m \in F$ such that

$$0 = \frac{\lambda_1}{\alpha - c_1} + \cdots + \frac{\lambda_m}{\alpha - c_m}.$$

Taking a common denominator yields

$$0 = \frac{f(\alpha)}{g(\alpha)}$$

for some $f(x) \neq 0 \in F[x]$ and $g(\alpha) \neq 0$. Therefore, $f(\alpha) = 0$ for some nonzero $f(x) \in F[x]$. □

Useful Galois Theory Facts

Theorem: If $f(x) \in K[x]$ is a separable polynomial of degree n, then f is irreducible if and only if the Galois group $\operatorname{Gal}(\operatorname{spl}_K(f(x))/K)$ is a transitive subgroup of S_n.

Proof. If $f(x)$ is irreducible, then we know that the Galois group acts transitively on the roots.

Meanwhile, if $f(x)$ is reducible, then $f(x)$ is a product of distinct irreducibles since it is separable. Let r_i and r_j be roots of different irreducible factors of $f(x)$. Then, for each $\sigma \in \operatorname{Gal}(\operatorname{spl}_K(f(x))/K)$, we have $\sigma(r_i)$ has the same minimal polynomial over K as r_i, so we cannot have $\sigma(r_i) = r_j$, whence the Galois group is not transitive. □

Theorem (Artin's Lemma)**:** Let K be a field, G a finite subgroup of $\operatorname{Aut}(K)$, and $F = K^G$ the corresponding fixed field. Then,

(i) K/F is Galois;

(ii) K/F is a finite;

(iii) $\operatorname{Gal}(K/F) = G$ as a subgroup of $\operatorname{Aut}(K)$.

- An extension K/F is Galois if and only if K is the splitting field of a separable polynomial $f(x) \in F[x]$.
- Correspondence: if K/F is a finite Galois extension, then the correspondence between subfields of K/F (that is, subfields of K that contain F) and subgroups of $\operatorname{Gal}(K/F)$ is given by $K \supseteq L \mapsto \operatorname{Gal}(K/L) \subseteq \operatorname{Gal}(K/F)$ and $\operatorname{Gal}(K/F) \geq H \mapsto K^H \subseteq K$.

- Suppose $K_1, K_2 \subseteq K$ are subfields of a Galois extension K/F corresponding to subgroups H_1 and H_2. Then, the composite K_1K_2 corresponds to $H_1 \cap H_2$, while the intersection $K_1 \cap K_2$ corresponds to the group $\langle H_1, H_2\rangle$ generated by H_1 and H_2.
- If K/F is a finite Galois extension, L is a subfield of K/F, then L/F is a Galois extension if and only if $\operatorname{Gal}(K/L) \trianglelefteq \operatorname{Gal}(K/F)$ is normal. Furthermore, $\operatorname{Gal}(L/F) \cong \operatorname{Gal}(K/F)/\operatorname{Gal}(K/L)$.
- If K/F is a Galois extension, and F'/F is any extension, then the composite KF'/F' is Galois, and $\operatorname{Gal}(KF'/F') \cong \operatorname{Gal}(K/K \cap F')$.
- If K_1 and K_2 are Galois extensions over F, then $K_1 \cap K_2$ is Galois over F and K_1K_2 is Galois over F, with $\operatorname{Gal}(K_1K_2/F)$ isomorphic to the subgroup of $\operatorname{Gal}(K_1/F) \times \operatorname{Gal}(K_2/F)$ consisting of all (σ, τ) with $\sigma|_{K_1 \cap K_2} = \tau|_{K_1 \cap K_2}$.

  If K/F is Galois and $\operatorname{Gal}(K/F) \cong G_1 \times G_2$ is the direct product of two subgroups, then K is the composite of two Galois extensions K_1 and K_2 with $K_1 \cap K_2 = F$.
- If $\mathbb{F}_p$ denotes the field of order p, then any finite field $\mathbb{F}_{p^m}$ over $\mathbb{F}_p$ has cyclic Galois group with order m generated by $\operatorname{fr}\colon \overline{\mathbb{F}_p} \to \overline{\mathbb{F}_p}$, given by $\operatorname{fr}(x) = x^p$.

  Furthermore, if $\mathbb{F}_{q^k}/\mathbb{F}_q$ is a finite extension of a finite field of order $q = p^\ell$, then the Galois group is generated by fr^ℓ and is of order k/ℓ.
- Given a separable irreducible polynomial $f(x) \in F[x]$ and its splitting field K/F, the minimal polynomial of any element $\theta \in K$ is given by

$$\mu_{\theta,F}(x) = \prod_{i=1}^{r} (x - \theta_i),$$

  where $\{\theta_1, \dots, \theta_r\}$ is the $\operatorname{Gal}(K/F)$-orbit of θ.

We now discuss the special case of the discriminant, and a simple decision process to compute Galois group for cubics and quartics

Define the discriminant of a polynomial $p(x) \in F[x]$ by

$$\Delta = \prod_{i<j} \left(\alpha_i - \alpha_j\right)^2,$$

where $\{\alpha_1, \dots, \alpha_n\}$ denotes the roots of the polynomial $p(x)$, with possible multiplicity. If $p(x)$ is separable and irreducible with splitting field K, observe that $\operatorname{Gal}(K/F)$ preserves Δ (i.e., $\Delta \in F$), and it preserves $\sqrt{\Delta}$ if and only if $\operatorname{Gal}(K/F) \leq A_n$.

We now focus on the special cases of cubics and quartics.

- Assume $p(x)$ is cubic (and irreducible and separable). A root of $p(x)$ then generates a degree 3 extension over F, so the degree of the splitting field K/F is divisible by 3.
  - If Δ is a square in F, then $\operatorname{Gal}(K/F) \cong A_3 \cong \mathbb{Z}/3\mathbb{Z}$, and the splitting field is given by $K = F(\theta)$ for any root θ of $p(x)$.
  - If Δ is *not* a square in F, then $\operatorname{Gal}(K/F) \cong S_3$, and the splitting field is given by $K = F\left(\sqrt{\Delta}, \theta\right)$ for any root θ of $p(x)$. The subgroup is then generated by automorphisms σ and τ, where σ shuffles the root θ to the other roots of f while fixing $\sqrt{\Delta}$ and τ takes $\sqrt{\Delta} \mapsto -\sqrt{\Delta}$ and fixes θ.
- Assume $p(x)$ is quartic (and irreducible and separable). Define the resolvent cubic to be the polynomial in $g(x) \in F[x]$ with roots

$$\begin{aligned}\theta_1 &= (\alpha_1 + \alpha_2)(\alpha_3 + \alpha_4)\\ \theta_2 &= (\alpha_1 + \alpha_3)(\alpha_2 + \alpha_4)\end{aligned}$$

$$\theta_3 = (\alpha_1 + \alpha_4)(\alpha_2 + \alpha_3).$$

Notice that $g(x)$ is indeed in $F[x]$ since the Galois group $\mathrm{Gal}(K/F)$ simply shuffles the θ_i. Furthermore, it can be shown that the discriminant of $g(x)$ is equal to the discriminant of $p(x)$.

Furthermore, observe that the stabilizer of the element θ_1 is a copy of the dihedral group D_4 sitting inside S_4, while the stabilizer of all three of the θ_i is equal to the intersection of these three subgroups, which is the copy of the Klein 4-group V sitting inside S_4.

- If the resolvent cubic $g(x)$ is irreducible, then we consider the discriminant Δ.
  - If Δ is not a square, then $\mathrm{Gal}(K/F)$ is not contained in A_4 and the Galois group of the resolvent cubic is equal to S_3. Therefore, the degree of the splitting field for $p(x)$ is divisible by 6, meaning $\mathrm{Gal}(K/F) \cong S_4$.
  - If Δ is a square, then $\mathrm{Gal}(K/F)$ is contained in A_4, and the Galois group of the resolvent cubic is A_3. Hence, 3 divides the order of the Galois group (and 4 divides the order of the Galois group by the assumption that $p(x)$ is irreducible), so $\mathrm{Gal}(K/F) \cong A_4$.
- If the resolvent cubic is reducible, then we consider the following cases.
  - If the resolvent cubic splits completely into linear factors, then all of θ_1, θ_2, and θ_3 are in F, so they are all fixed by the Galois group. Therefore, $\mathrm{Gal}(K/F) \cong V$.
  - If the resolvent cubic splits into a linear factor and an irreducible quadratic factor, then $\mathrm{Gal}(K/F) \subseteq D_4$ and $\mathrm{Gal}(K/F) \not\subseteq V$.

    Consider the field $F\left(\sqrt{\Delta}\right)$, which is the fixed field of $\mathrm{Gal}(K/F) \cap A_4$. Since we assume that $p(x)$ is irreducible, it follows that we must have $\mathrm{Gal}(K/F) = D_4$ or $\mathrm{Gal}(K/F) = C$, where $C = \langle (1,2,3,4) \rangle$. Notice that $D_4 \cap A_4 = V$ and $C \cap A_4 = \{\mathrm{id}, (1,3)(2,4)\}$, the latter of which is not transitive.
    - If $p(x)$ is irreducible over $F\left(\sqrt{\Delta}\right)$, then the Galois group $\mathrm{Gal}\left(K/F\left(\sqrt{\Delta}\right)\right) = \mathrm{Gal}(K/F) \cap A_4$ is transitive. Thus, $\mathrm{Gal}\left(K/F\left(\sqrt{\Delta}\right)\right) = V$, so $\mathrm{Gal}(K/F) \cong D_4$.
    - If $p(x)$ is reducible over $F\left(\sqrt{\Delta}\right)$, then the Galois group $\mathrm{Gal}\left(K/F\left(\sqrt{\Delta}\right)\right) = \mathrm{Gal}(K/F) \cap A_4$ is not transitive. Therefore, $\mathrm{Gal}\left(K/F\left(\sqrt{\Delta}\right)\right) = \{\mathrm{id}, (1,3)(2,4)\}$, so $\mathrm{Gal}(K/F) \cong C$.

Linear Independence of Characters

Definition: A (linear) character χ of a group G with values in a field L is a homomorphism from G to the multiplicative group of L, $\chi\colon G \to L^\times$.

Theorem (Linear Independence of Characters)**:** If $\chi_1, \dots, \chi_n$ are distinct characters of G with values in L, then they are linearly independent over L.

Proof. Find the minimal dependence relation with m terms not all equal to 0 given by

$$a_1\chi_1 + a_2\chi_2 + \cdots + a_m\chi_m = 0.$$

For any g, we have

$$a_1\chi_1(g) + a_2\chi_2(g) + \cdots + a_m\chi_m(g) = 0. \tag{$*$}$$

There is g_0 such that $\chi_1(g_0) \neq \chi_m(g_0)$. Therefore, we have

$$a_1\chi_1(g_0 g) + a_2\chi_2(g_0 g) + \cdots + a_m\chi_m(g_0 g) = 0.$$

Multiplying $(*)$ by $\chi_m(g_0)$, using the fact that the χ_i are homomorphisms, and subtracting, we obtain

$$(\chi_m(g_0) - \chi_1(g_0))a_1\chi_1(g) + (\chi_m(g_0) - \chi_2(g_0)a_2\chi_2(g)) + \cdots + (\chi_m(g_0) - \chi_{m-1}(g_0))a_{m-1}\chi_{m-1}(g) = 0.$$

Since this holds for all $g \in G$, and the first coefficient is nonzero, this contradicts the minimality of the linear dependence relation. □

Corollary: If $\sigma_1, \dots, \sigma_n \in \operatorname{Gal}(K/F)$, then $\sigma_1, \dots, \sigma_n$ are linearly independent.

Theorem (Normal Basis Theorem)**:** Let K/F be a finite Galois extension of degree n. Let $\sigma_1, \dots, \sigma_n$ be elements of the Galois group G. Then, there exists an element $w \in K$ such that $\{\sigma_1(w), \dots, \sigma_n(w)\}$ forms a basis of K over F.

Proof. We will prove for the case that F is infinite. For any $\sigma, \tau \in G$, let X_σ be an indeterminate and let $t_{\sigma,\tau} = X_{\sigma^{-1}\tau}$. If $X_i = X_{\sigma_i}$, then we let

$$f(X_1, \dots, X_n) = \det\left(t_{\sigma_i, \sigma_j}\right).$$

We observe that f is not identically zero; since F is infinite, f is reduced as a polynomial, so f is not equal to zero for all $x \in K$ substituting $\sigma_i(x)$ for X_i in the expression of f. In particular, this means that $\det\left(\sigma_i^{-1}\sigma_j(w)\right) \neq 0$ for some $w \in K$.

If $a_1, \dots, a_n \in F$ are such that

$$a_1\sigma_1(w) + \cdots + a_n\sigma_n(w) = 0,$$

then applying σ_i^{-1} to this relation for each $i = 1, \dots, n$, we find that we obtain a system of linear equations regarding the a_j as unknowns. Since the determinant $\det\left(\sigma_i^{-1}\sigma_j(w)\right)$ is nonzero, it follows that we must have $a_j = 0$ for each $j = 1, \dots, n$, so that w is the desired element. ☐

Galois Groups with Nested Square Roots

We will let $F = \mathbb{Q}$ for each of the following.

(a) Let α be the nested square root $\sqrt{4+\sqrt{5}}$. Let $\alpha' = \sqrt{4-\sqrt{5}}$, so that the polynomial

$$\begin{aligned} f(x) &= (x-\alpha)(x+\alpha)(x-\alpha')(x+\alpha') \\ &= x^4 - 8x^2 + 11 \end{aligned}$$

is irreducible. If K is the splitting field for $f(x)$ over F, then we have $F(\alpha, \alpha') = K$; since f is irreducible, $[F(\alpha) : F] = 4$, and since $\sqrt{5}$ is in $F(\alpha)$, we have α' has degree at most 2 over $F(\alpha)$, so we hope to determine whether K/F is degree 4 or 8.

Writing

$$\begin{aligned} \alpha_1 &= \alpha \\ \alpha_2 &= \alpha' \\ \alpha_3 &= -\alpha \\ \alpha_4 &= -\alpha', \end{aligned}$$

we observe that any automorphism that takes $\alpha_1 \mapsto \alpha_i$ takes $\alpha_3 \mapsto -\alpha_i$. Permutations of this form yield the dihedral group D_4 generated by $\sigma = (1,2,3,4)$ and $\tau = (2,4)$. Therefore, we must determine whether this group is isomorphic to $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (the unique copy of V in S_4 that is transitive), or D_4.

We observe that the permutation $(1,3)(2,4)$ is contained in all three subgroups of D_4, so it extends to $\rho \in \operatorname{Gal}(K/F)$; we hope to determine the subgroup N of $\operatorname{Gal}(K/F)$ with order 2. Notice that

$$\begin{aligned} \alpha^2 &= 4 + \sqrt{5} \\ \alpha\alpha' &= \sqrt{11}, \end{aligned}$$

meaning that K^N contains $L = F\left(\sqrt{5}, \sqrt{11}\right)$. The chain of fields $F \subseteq L \subseteq K^N \subseteq K$ yields that $[K : F] \leq 8$, $[L : F] = 4$, and $\left[K : K^N\right] = 2$. Therefore, $L = K^N$, and $[K : F] = 8$, so $\operatorname{Gal}(K/F) \cong D_4$.

(b) Let $\alpha = \sqrt{2+\sqrt{2}}$. Then, $f(x) = x^4 - 4x^2 + 2$ is the minimal polynomial for α over F, with roots given by α, $\alpha' = \sqrt{2-\sqrt{2}}$, $-\alpha$, and $-\alpha'$. Notice that $\alpha\alpha' = \sqrt{2}$, and $\sqrt{2} \in F(\alpha)$, so $\alpha' \in F(\alpha)$. Thus, $[K : F] = 4$, meaning that $\operatorname{Gal}(K/F) \cong \mathbb{Z}/4\mathbb{Z}$ or $\operatorname{Gal}(K/F) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Since $f(x)$ is irreducible, the operation of $G = \operatorname{Gal}(K/F)$ on the roots of f is transitive. There is an element σ' of G that takes $\alpha \mapsto \alpha'$, so since $\alpha^2 = 2+\sqrt{2}$ and $\alpha'^2 = 2-\sqrt{2}$, we have σ' takes $\sqrt{2} \mapsto -\sqrt{2}$ and $\alpha\alpha' \mapsto -\alpha\alpha'$. In particular, this means that $\alpha' \mapsto -\alpha$, so that $\sigma' = \sigma$, where σ is the permutation from part (a), so that $\operatorname{Gal}(K/F) \cong \mathbb{Z}/4\mathbb{Z}$.

(c) Let $\alpha = \sqrt{4+\sqrt{7}}$. The corresponding minimal polynomial over F is $f(x) = x^4 - 8x^2 + 9$. Then, $\alpha\alpha' = 3$, and α' is in the field $F(\alpha)$, so that $[K : F] = 4$. If σ' takes $\alpha \mapsto \alpha'$, then since $\alpha\alpha' = 3$, we must have $\alpha' = \alpha$, so that the Galois group is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Galois Groups of Polynomials

Definition: Let $x_1, \dots, x_n$ be indeterminates. The *elementary symmetric functions* $s_1, \dots, s_n$ are defined by

$$\begin{aligned} s_1 &= x_1 + x_2 + \cdots + x_n \\ s_2 &= \prod_{i<j} x_i x_j \\ &\vdots \\ s_n &= x_1 x_2 \cdots x_n. \end{aligned}$$

The *general polynomial* of degree n is the polynomial

$$g(x) = (x - x_1)(x - x_2)\cdots(x - x_n)$$

with roots $x_1, \dots, x_n$.

Observe that the general polynomial can be written as

$$(x - x_1)(x - x_2)\cdots(x - x_n) = x^n - s_1 x^{n-1} + \cdots + (-1)^n s_n.$$

This means that if F is some field, then the extension $F(x_1, \dots, x_n)$ is a Galois extension of the field $F(s_1, \dots, s_n)$ since it is the splitting field of the general polynomial of degree n. Furthermore, $F(s_1, \dots, s_n)$ is contained in the fixed field of the action of S_n on $F(x_1, \dots, x_n)$, where $\sigma \cdot x_i = x_{\sigma(i)}$ for any $\sigma \in S_n$. Since the fixed field of S_n has index $n!$ in $F(x_1, \dots, x_n)$ by the fundamental theorem of Galois theory, it follows that we have

$$[F(x_1, \dots, x_n) : F(s_1, \dots, s_n)] \leq n!,$$

whence $F(s_1, \dots, s_n)$ is the full fixed field of $F(x_1, \dots, x_n)$.

Theorem: The general polynomial

$$x^n - s_1 x^{n-1} + s_2 x^{n-2} + \cdots + (-1)^n s_n$$

over the field $F(s_1, \dots, s_n)$ is separable with Galois group S_n.

Now, we will discuss the discriminant, which is an essential tool in understanding the Galois groups of polynomials, especially those of degrees two, three, and four.

Definition: The *discriminant* of $x_1, \dots, x_n$ is given by

$$\Delta = \prod_{i<j} \left(x_i - x_j\right)^2.$$

The discriminant of a polynomial is the discriminant of the roots of the polynomial.

We start by observing that the discriminant is a symmetric function in $x_1,\dots,x_n$, so it is an element of the field $K = F(s_1,\dots,s_n)$.

Recalling the definition of the alternating group A_n, we have that $\sigma \in S_n$ is in A_n if and only if σ fixes

$$\sqrt{\Delta} = \prod_{i<j}\left(x_i - x_j\right)$$

in $\mathbb{Z}[x_1,\dots,x_n]$. Notice then that if F has characteristic not equal to 2, then $\sqrt{\Delta}$ generates the fixed field of A_n, which is a quadratic extension of K.

If $f(x)$ is a monic degree n polynomial with roots $\alpha_1,\dots,\alpha_n$, then the discriminant is given by

$$\Delta = \prod_{i<j}\left(\alpha_i - \alpha_j\right)^2.$$

Observe that $\Delta \in F$ since Δ is fixed by the Galois group. Furthermore, $\Delta = 0$ if and only if $f(x)$ is separable, so if F is perfect (i.e., characteristic zero or finite), then $f(x)$ is reducible if and only if $\Delta = 0$ since every irreducible polynomial over a perfect field is separable. Additionally, observe that $\sqrt{\Delta} \in F$ if and only if the Galois group of $f(x)$ is contained in A_n.

Now, we start our investigation with the polynomials of degree 2. The discriminant of a polynomial $x^2 + ax + b$ is given by $\Delta = (\alpha - \beta)^2$, which can be written as

$$\begin{aligned}\Delta &= s_1^2 - 4s_2\\ &= a^2 - 4b.\end{aligned}$$

The polynomial is separable if and only if $a^2 - 4b \neq 0$, and the Galois group is a subgroup of $S_2 = \mathbb{Z}/2\mathbb{Z}$. This means that the Galois group is trivial if and only if $a^2 - 4b$ is a rational square.

Next, we consider the cubic polynomials. If we have

$$f(x) = x^3 + ax^2 + bx + c,$$

which upon the substitution $x = y - a/3$, yields a reduced cubic

$$g(y) = y^3 + py + q,$$

where

$$\begin{aligned}p &= \frac{1}{3}\left(3b - a^2\right)\\ q &= \frac{1}{27}\left(2a^3 - 9ab + 27c\right).\end{aligned}$$

Since the roots of the polynomials f and g differ by the constant $a/3$, and the formula for the discriminant involves the difference of these roots, it follows that the polynomials have the same discriminant.

Letting α,β,γ be the roots of the polynomial $g(y)$, then we have

$$\begin{aligned}g'(y) &= (y-\alpha)(y-\beta) + (y-\alpha)(y-\gamma) + (y-\beta)(y-\gamma)\\ g'(\alpha) &= (\alpha-\beta)(\alpha-\gamma)\\ g'(\beta) &= (\beta-\alpha)(\beta-\gamma)\\ g'(\gamma) &= (\gamma-\alpha)(\gamma-\beta).\end{aligned}$$

Therefore, we have

$$\Delta = ((\alpha-\beta)(\alpha-\gamma)(\beta-\gamma))^2$$

$$= -g'(\alpha)g'(\beta)g'(\gamma).$$

Since $g'(y) = 3y^2 + p$, it follows that

$$\begin{aligned}
-\Delta &= \left(3\alpha^2 + p\right)\left(3\beta^2 + p\right)\left(3\gamma^2 + p\right) \\
&= 27\alpha^2\beta^2\gamma^2 + 9p\left(\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2\right) + 3p^2\left(\alpha^2 + \beta^2 + \gamma^2\right) + p^3.
\end{aligned}$$

These expressions are elementary symmetric functions of the roots, where for g, $s_1 = 0$, $s_2 = p$, and $s_3 = -q$, whence

$$\Delta = -4p^3 - 27q^2.$$

Therefore, expressing Δ in terms of a, b, c, we have

$$\Delta = a^2b^2 - 4b^3 - 4a^3c - 27c^2 + 18abc.$$

This allows us to compute the Galois group as follows.

(i) If $f(x)$ is reducible, then either $f(x)$ splits into three linear factors or into a linear factor and an irreducible quadratic. In the first case, the Galois group is trivial and in the second case, the Galois group is of order 2.

(ii) If $f(x)$ is irreducible, then a root of $f(x)$ generates a degree 3 extension over F. Therefore, the degree of the splitting field over F is divisible by 3.

Since the Galois group is a subgroup of S_3, it follows that there are only two options, either A_3 or S_3. In particular, the Galois group is A_3 if and only if the discriminant Δ is a square.

If Δ is a square, then the splitting field of the irreducible cubic $f(x)$ is obtained by adjoining any root of $f(x)$ to F. If Δ is not the square of an element of F, then the splitting field of $f(x)$ is of degree 6 over F, given by $F\left(\theta, \sqrt{\Delta}\right)$ where θ is some root of $f(x)$. This extension is Galois over F generated by the automorphisms σ and τ, where σ takes θ to any other root of $f(x)$ fixing $\sqrt{\Delta}$ and τ takes $\sqrt{\Delta} \mapsto -\sqrt{\Delta}$ and fixing θ.

Finally, we discuss the Galois group of a quartic polynomial

$$f(x) = x^4 + ax^3 + bx^2 + cx + d.$$

Then, by substituting with $x = y - a/4$, this yields the quartic

$$g(y) = y^4 + py^2 + qy + r$$

with

$$\begin{aligned}
p &= \frac{1}{8}\left(-3a^2 + 8b\right) \\
q &= \frac{1}{8}\left(a^3 - 4ab - 8c\right) \\
r &= \frac{1}{256}\left(-3a^4 + 16a^2b - 64ac + 256d\right).
\end{aligned}$$

Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be the roots of $g(y)$ and G the Galois group for the splitting field of $g(y)$.

If $g(y)$ is reducible, then the following cases hold.

(i) If $g(y)$ splits into a linear and a cubic, then G is the Galois group of the cubic.

(ii) If $g(y)$ splits into two irreducible quadratics, then the splitting field is $F\left(\sqrt{\Delta_1}, \sqrt{\Delta_2}\right)$, where Δ_1 and Δ_2 are the discriminants of the quadratics.

If Δ_1 and Δ_2 do not differ by a square, then G is isomorphic to a Klein 4-subgroup of S_4.

If Δ_1 is a square multiplied by Δ_2, then $F\left(\sqrt{\Delta_1}, \sqrt{\Delta_2}\right) = F\left(\sqrt{\Delta_1}\right)$ is a quadratic extension with G isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

Now, we have the case of $g(y)$ irreducible.Then, the Galois group is transitive on its roots. There are five transitive subgroups (up to conjugation) of S_4, given by copies of S_4, A_4, D_4, V, C, where V denotes the Klein 4-group and C denotes the copy of $\mathbb{Z}/4\mathbb{Z}$.

We consider the elements

$$\begin{aligned}\theta_1 &= (\alpha_1+\alpha_2)(\alpha_3+\alpha_4)\\ \theta_2 &= (\alpha_1+\alpha_3)(\alpha_2+\alpha_4)\\ \theta_3 &= (\alpha_1+\alpha_4)(\alpha_2+\alpha_3)\end{aligned}$$

inside the splitting field for $g(y)$. Notice that the elements are permuted among themselves by the permutations of S_4. The stabilizer of θ_1 is the dihedral group D_4, while the stabilizer of all three of θ_1, θ_2, and θ_3 is the copy of V.

The elementary symmetric functions in the θ_i are fixed by S_4, so they are in F. These elementary symmetric functions evaluated on $\theta_1, \dots, \theta_3$ are given by $2p$, p^2-4r, and $-q^2$. This gives that θ_1, θ_2, and θ_3 are the roots of

$$h(x) = x^3 - 2px^2 + \left(p^2-4r\right)x + q^2.$$

This is known as the *resolvent cubic* for the quartic $g(y)$. It can be shown (see DF p.614) that the discriminant Δ of the resolvent cubic is equal to the discriminant of $g(y)$. In particular, this gives

$$\Delta = 16p^4r - 4p^3q^2 - 128p^2r^2 + 144pq^2r - 27q^4 + 256r^3.$$

Remark: In the special case where $g(y) = y^4 + py^2 + r$ (which emerges during analysis of quartics whose roots are nested square roots), we have

$$\Delta = 16p^4r - 128p^2r^2 + 256r^3.$$

In particular, since the splitting field for the resolvent cubic is a subfield of the splitting field of the original quartic, the Galois group for $h(x)$ is a quotient of G, so knowing the action of the Galois group of $h(x)$ on the roots of $h(x)$ gives information about the Galois group of $g(y)$.

(i) If the resolvent cubic is irreducible, and Δ is not a square, then G is not contained in A_4 with the Galois group of the resolvent cubic equal to S_3. In particular, this means that the degree of the splitting field for $g(y)$ is divisible by 6, so that $G = S_4$.

(ii) If the resolvent cubic is irreducible, and Δ is a square, then G is a subgroup of A_4 and 3 divides the order of G since the Galois group of the resolvent cubic is A_3, so that $G = A_4$.

(iii) If the resolvent cubic is reducible, and $h(x)$ splits completely, then every element of G fixes the elements θ_1, θ_2, and θ_3, so $G = V$.

(iv) If the resolvent cubic splits into a linear factor and a quadratic, then G stabilizes θ_1 but not θ_2 or θ_3, so $G \subseteq D_4$ and $G \not\subseteq V$, so $G = D_4$ or $G = C$.

    Now, notice that $F\left(\sqrt{\Delta}\right)$ is the fixed field of the elements of $G \cap A_4$. We have $D_4 \cap A_4 = V$ and $C \cap A_4 = \{e, (1,3)(2,4)\}$, the latter of which is not transitive. Therefore, we have the case $D_4 \cap A_4 = V$ if and only if $g(y)$ is irreducible over $F\left(\sqrt{\Delta}\right)$.

Solvability by Radicals

Definition: An extension K/F is called *cyclic* if it is Galois with cyclic Galois group.

Proposition: Suppose F is a field of characteristic not dividing n that contains the nth roots of unity. Then, the extension $F\left(\sqrt[n]{a}\right)$, for some $a \in F$, is cyclic over F with degree dividing n.

Proof. The extension $K = F\left(\sqrt[n]{a}\right)$ is Galois over F if F contains the nth roots of unity, since it is the splitting field for the polynomial $x^n - a$. For any $\sigma \in \mathrm{Gal}(K/F)$, we have $\sigma\left(\sqrt[n]{a}\right)$ is another root of this polynomial, so $\sigma\left(\sqrt[n]{a}\right) = \zeta_\sigma \sqrt[n]{a}$ for some nth root of unity ζ_σ.

We thus get a map $\mathrm{Gal}(K/F) \to \mu_n$ taking $\sigma \mapsto \zeta_\sigma$, where μ_n denotes the group of nth roots of unity. Since F contains μ_n, it follows that every nth root of unity is fixed by every element of $\mathrm{Gal}(K/F)$, so that $\zeta_{\sigma\tau} = \zeta_\sigma\zeta_\tau$. In particular, this means that the map is a homomorphism, so there is an injection of $\mathrm{Gal}(K/F)$ into the cyclic group μ_n with order n. ☐

Now, assume that K is a cyclic extension of degree n over a field F with characteristic not dividing n that contains the nth roots of unity. Let σ be a generator for the cyclic group $\mathrm{Gal}(K/F)$.

Definition: Let $\alpha \in K$ and ζ an nth root of unity. Define the *Lagrange resolvent* (α, ζ) by

$$(\alpha,\zeta) = \alpha + \zeta\sigma(\alpha) + \zeta^2\sigma^2(\alpha) + \cdots + \zeta^{n-1}\sigma^{n-1}(\alpha).$$

Note that by applying σ to (α,ζ), we get

$$\sigma(\alpha,\zeta) = \zeta^{-1}(\alpha,\zeta).$$

In particular, this means that $(\alpha,\zeta)^n$ is fixed by $\mathrm{Gal}(K/F)$, so it is an element of F for any $\alpha \in K$.

If ζ is a primitive nth root of unity, then since the automorphisms $\mathrm{id}, \sigma, \dots, \sigma^{n-1}$ are linearly independent, there is an element $\alpha \in K$ with $(\alpha,\zeta) \neq 0$. Therefore, we have

$$\sigma^i(\alpha,\zeta) = \zeta^{-i}(\alpha,\zeta),$$

so it follows that σ^i does not fix (α,ζ) for $1 \leq i < n$. Therefore, the element cannot lie in any proper subfield of K, whence $K = F((\alpha,\zeta))$. Since $(\alpha,\zeta)^n = a \in F$, it follows that $F\left(\sqrt[n]{a}\right) = F((\alpha,\zeta)) = K$. Therefore, we have the following.

Proposition: Any cyclic extension of degree n over a field F of characteristic not dividing n which contains the nth roots of unity is of the form $F\left(\sqrt[n]{a}\right)$ for some $a \in F$.

A group G is said to have *exponent n* if $g^n = 1$ for each $g \in G$. If F is a field of characteristic not dividing n that contains the nth roots of unity, then if $a_1, \dots, a_k \in F^\times$, we observe that the extension

$$K = F\left(\sqrt[n]{a_1}, \dots, \sqrt[n]{a_k}\right)$$

is an abelian extension of F whose Galois group is of exponent n. Furthermore, any abelian extension of exponent n is of this form. The Galois group is isomorphic to the group generated by the elements $a_1, \dots, a_k$ in the group $F^\times/(F^\times)^n$, and any two such extensions are equal if and only if their associated groups are equal.

From now on, we will consider base fields of characteristic 0. The results are valid over fields whose characteristics do not divide the orders of the roots taken.

Definition: An element α that is algebraic over F can be *solved in terms of radicals* if α is an element of a field K that can be obtained by a succession of simple radical extensions

$$F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_{i+1} \subseteq K_i \subseteq K_s = K,$$

where $K_{i+1} = K_i\left(\sqrt[n_i]{a_i}\right)$ for some $a_i \in K_i$, where $\sqrt[n_i]{a_i}$ denotes some root of the polynomial $x^{n_i} - a_i$. Such a field K is called a *root extension* of F.

A polynomial $f(x) \in F[x]$ is said to be solved by radicals if all its roots can be solved for in terms of radicals.

Lemma: Suppose α is contained in a root extension K of F. Then, α is contained in a root extension which is Galois over F where each extension K_{i+1}/K_i is cyclic.

Proof. Let L be the Galois closure of K over F. For any $\sigma \in \mathrm{Gal}(L/F)$, we have the chain of subfields

$$F = \sigma K_0 \subseteq \sigma K_1 \subseteq \cdots \subseteq \sigma K_i \subseteq \sigma K_{i+1} \subseteq \cdots \subseteq \sigma K_s = \sigma K,$$

where $\sigma K_{i+1}/\sigma K_i$ is a simple radical extension since it is generated by the element $\sigma\left(\sqrt[n_i]{a_i}\right)$, which is a root of the equation $x^{n_i} - \sigma(a_i)$ over $\sigma(K_i)$.

The composite of two root extensions is again a root extension, since if K' is a root extension with subfields K_i', we may take the composite of K_1' with the fields $K_0, K_1, \dots, K_s$, then the composite with K_2', etc., so that each extension is a simple radical extension.

It follows that the composite of all the conjugate fields $\sigma(K)$ for all $\sigma \in \mathrm{Gal}(L/F)$ is again a root extension. Yet, since this field is L, it follows that α is contained in a Galois root extension.

Now, adjoin to F the n_ith roots of unity for all the roots $\sqrt[n_i]{a_i}$ of the simple radical extensions of the Galois root extension K/F. This gives the field F'. Form the composite of F' with the root extension

$$F \subseteq F' = F'K_0 \subseteq F'K_1 \subseteq \cdots \subseteq F'K_i \subseteq F'K_{i+1} \subseteq \cdots \subseteq F'K_s = F'K.$$

The field $F'K$ is a Galois extension of F since it is the composite of two Galois extensions. The extension from F to $F' = F'K_0$ can be given as a chain of subfields with cyclic extensions (since this holds for any abelian extension). Each extension $F'K_{i+1}/F'K_i$ is a simple radical extension, and since there are roots of unity in the base fields, it follows that each individual extension from F' to $F'K$ is a cyclic extension, so $F'K/F$ is a root extension which is Galois over F with cyclic intermediate extensions. □

Recall that a finite group G is called *solvable* if there is a subgroup series

$$1 = G_s \trianglelefteq G_{s-1} \trianglelefteq \cdots \trianglelefteq G_{i+1} \trianglelefteq G_i \trianglelefteq G_0 = G$$

where G_i/G_{i+1} is cyclic.[III] Note that subgroups and quotients of solvable groups are solvable, and that if $H \trianglelefteq G$ and G/H are both solvable, then G is solvable.

Theorem: The polynomial $f(x)$ can be solved by radicals if and only if its Galois group is a solvable group.

Proof. Suppose $f(x)$ can be solved by radicals. Then, each root of $f(x)$ is contained in a root extension, and the composite L of such extensions is another root extension. Let G_i be the subgroups of the Galois group corresponding to the subfields K_i where $i = 0, 1, \dots, s-1$. Since

$$\mathrm{Gal}(K_{i+1}/K_i) = G_i/G_{i+1}$$

by the Galois correspondence. Each such group is cyclic since K_{i+1}/K_i is a root extension. It follows that the Galois group $G = \mathrm{Gal}(L/F)$ is a solvable group, and since the Galois group of $f(x)$ is a quotient of the solvable group G, it follows that the Galois group is solvable.

Now, suppose that the Galois group G of $f(x)$ is solvable, and let K be the splitting field of $f(x)$. Taking fixed fields of the subgroups in the series, we have a chain

$$F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_i \subseteq K_{i+1} \subseteq \cdots \subseteq K_s = K,$$

where K_{i+1}/K_i is a cyclic extension of degree n_i. If F' is the cyclotomic field of all roots of unity with order n_i, we get a sequence of extensions $K_i' = F'K_i$ given by

$$F \subseteq F' = F'K_0 \subseteq K_1' \subseteq \cdots \subseteq K_i' \subseteq K_{i+1}' \subseteq \cdots \subseteq K_s' = F'K.$$

The extension $F'K_{i+1}/F'K_i$ is cyclic of degree n_i. Since there are the appropriate roots of unity in the base fields, it follows that each of these cyclic extensions is a simple radical extension, so each of the roots of $f(x)$ is contained in the root extension $F'K$, so that $f(x)$ is solvable by radicals. □

Theorem: The general quintic is not solvable by radicals.

Proof. The general quintic has Galois group S_5, and S_5 is not solvable. □

[III] Equivalently, the derived series $D^{(i)}(G)$ terminates at the identity. See the group theory notes for an explanation of the equivalence.

Galois Groups of Cyclotomic Polynomials

If $\mathbb{Q}(\zeta_n)$ denotes the extension of $\mathbb{Q}$ by the nth cyclotomic polynomial, then we observe that any $\mathbb{Q}$-automorphism is determined by its action on the primitive nth root of unity ζ_n. That is, $\sigma_a\colon \zeta_n \mapsto \zeta_n^a$ for some $a \in (\mathbb{Z}/n\mathbb{Z})^\times$. In fact, the collection of all such σ_a defines the Galois group $\operatorname{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$.

If $n = p$ is an odd prime, we observe that a basis for $\mathbb{Q}(\zeta_p)$ over $\mathbb{Q}$ is given by

$$\left\{\zeta_p, \zeta_p^2, \dots, \zeta_p^{p-1}\right\}.$$

Any $\sigma \in \operatorname{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ permutes the basis elements as these are the primitive pth roots of unity.

If $H \leq (\mathbb{Z}/p\mathbb{Z})^\times$ is a subgroup, then

$$\theta = \sum_{\sigma\in H} \sigma(\zeta_p)$$

is an element of the fixed field for H. If it were the case that α were fixed by some $\tau \notin H$, then $\tau(\alpha)$ implies that $\tau(\zeta_p) = \sigma(\zeta_p)$ for at least one $\sigma \in H$ as the set of all ζ_p forms a basis, but this means that $\tau\sigma^{-1} = \mathrm{id}$ as $\tau\sigma^{-1}(\zeta_p) = \zeta_p$. Therefore, $\tau = \sigma \in H$, which is a contradiction. In particular, this means that $\mathbb{Q}(\alpha)$ is the fixed field for H (and necessarily any such fixed field is determined in this fashion).

In the special case where H is the unique index 2 subgroup of $(\mathbb{Z}/p\mathbb{Z})^\times$, it can be shown that $\mathbb{Q}(\zeta_p)^H = \mathbb{Q}(\sqrt{\pm p})$, where the sign is positive if and only if $p \equiv 1 \bmod 4$.

Exam Solutions

January 2018

Problem (Problem 1)**:** Classify up to isomorphism all finite groups of order $2p$ where p is a prime number.

Solution: If $p = 2$, then the groups of order 4 are $\mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

If p is odd, then there is a p-Sylow subgroup isomorphic to $\mathbb{Z}/p\mathbb{Z}$ with index 2 (hence normal), and there is a 2-Sylow subgroup isomorphic to $\mathbb{Z}/2\mathbb{Z}$. Since $\operatorname{Aut}(\mathbb{Z}/p\mathbb{Z}) \cong (\mathbb{Z}/(p-1)\mathbb{Z})^\times$, it follows that the only homomorphisms $\mathbb{Z}/2\mathbb{Z} \to \operatorname{Aut}(\mathbb{Z}/p\mathbb{Z})$ are $1 \mapsto 1$ and $1 \mapsto -1$ as the latter is the unique order 1 subgroup of $\operatorname{Aut}(\mathbb{Z}/p\mathbb{Z})$ and the latter generates the unique order 2 subgroup. These yield the two semidirect products $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/p\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$.

Problem (Problem 3)**:** Determine whether the given polynomial is irreducible. Include arguments.

(a) $x^2 - 2i$ in $\mathbb{Z}[i][x]$;

(b) $x^3 - 49x^2 + (3 + \sqrt{2})x + 7$ in $\mathbb{Z}[\sqrt{2}][x]$;

(c) $x^2 + xy + y^2$ in $\mathbb{C}[x, y]$.

Solution:

(a) Observe that $(1 + i)^2 = 2i$, so that $x^2 - 2i = (x - (1 + i))(x + (1 + i))$ is a factorization over $\mathbb{Z}[i][x]$.

(b) We start by claiming that the ideal $(3 + \sqrt{2}) \subseteq \mathbb{Z}[\sqrt{2}]$ is prime. For this, observe that $\mathbb{Z}[\sqrt{2}] = \frac{\mathbb{Z}[x]}{(x^2-2)}$, yielding (through various applications of the third isomorphism theorem)

$$\begin{aligned}\frac{\mathbb{Z}[\sqrt{2}]}{(3+\sqrt{2})} &\cong \frac{\frac{\mathbb{Z}[x]}{(x^2-2)}}{(\overline{x}+3)}\\ &\cong \frac{\mathbb{Z}[x]}{(x+3, x^2-2)}\end{aligned}$$

$$\cong \frac{\frac{\mathbb{Z}[x]}{(x+3)}}{\frac{(x+3,x^2-2)}{(x+3)}}$$
$$\cong \frac{\mathbb{Z}/3\mathbb{Z}[x]}{(x^2-2)}.$$

Now, we observe that 2 is not a square mod 3, meaning that x^2-2 is not reducible mod 3, whence this is quotient is an integral domain. By the Eisenstein criterion applied with the ideal $\left(3+\sqrt{2}\right)$, we find that this polynomial is irreducible since $7 \notin \left(11+6\sqrt{2}\right)$.

(c) Let ω denote a primitive cube root of unity; then, $x^2+xy+y^2 = (x-\omega y)\left(x-\omega^{-1}y\right)$.

Problem (Problem 4): Let A be a finite abelian group of order n, p a prime divisor of n, and $n = p^k m$ with $k, m \in \mathbb{N}$ such that $\gcd(p,m) = 1$. Denote by A_p the p-Sylow subgroup of A.

(a) Show that the abelian groups A_p and $\mathbb{Z}/p^k\mathbb{Z} \otimes_{\mathbb{Z}} A$ are isomorphic.

(b) Describe $\mathbb{Z}/p\mathbb{Z} \otimes_{\mathbb{Z}} A$ as an abelian group without using tensor products.

Solution:

(a) Let

$$A = A_p \oplus (\mathbb{Z}/p_1^{d_1}\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/p_\ell^{d_\ell}\mathbb{Z})$$

be the elementary divisors decomposition for A, where $p_1, \dots, p_\ell \neq p$. Since tensor products commute with direct sums, we have

$$\mathbb{Z}/p^k\mathbb{Z} \otimes_{\mathbb{Z}} A = \left(\mathbb{Z}/p^k\mathbb{Z} \otimes_{\mathbb{Z}} A_p\right) \oplus \bigoplus_{i=1}^{\ell} \mathbb{Z}/p^k\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/p_i^{d_i}\mathbb{Z}.$$

We start by observing that in all of the latter factors, we have $\gcd\left(p^k, p_i^{d_i}\right) = 1$, meaning that given an elementary tensor $[m] \otimes [n]$ in a summand, we may write $[n] = \left[p^k\right]\left[np^{-k}\right]$ as p^k is invertible in $\mathbb{Z}/p_i^{d_i}\mathbb{Z}$. Since the tensor products are $\mathbb{Z}$-balanced, it follows that elementary tensor in $\mathbb{Z}/p^k\mathbb{Z} \otimes \mathbb{Z}/p_i^{d_i}\mathbb{Z}$ is zero, so that these respective tensor products are zero.

Now, we may write

$$\mathbb{Z}/p^k\mathbb{Z} \otimes_{\mathbb{Z}} A = \bigoplus_{j=1}^{s} \mathbb{Z}/p^k\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/p^{e_j}\mathbb{Z}$$

for some multiset $\left\{e_j\right\}_{j=1}^{s}$ of indices, again by the elementary divisors decomposition (this time applied to A_p). Observe that each of the e_j are less than or equal to k. Furthermore, we have the exact sequence

$$\mathbb{Z} \xrightarrow{\cdot p^k} \mathbb{Z} \to \mathbb{Z}/p^k\mathbb{Z} \to 0,$$

and since tensor products preserve surjections, we find taking the tensor product with $\mathbb{Z}/p^{e_j}\mathbb{Z}$, we get

$$\mathbb{Z}/p^{e_j}\mathbb{Z} \xrightarrow{\cdot p^k} \mathbb{Z}/p^{e_j}\mathbb{Z} \to \mathbb{Z}/p^{e_j}\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/p^k\mathbb{Z} \to 0.$$

We observe then that

$$\mathbb{Z}/p^{e_j}\mathbb{Z} \otimes \mathbb{Z}/p^k\mathbb{Z} \cong \frac{\mathbb{Z}/p^{e_j}\mathbb{Z}}{p^k(\mathbb{Z}/p^{e_j}\mathbb{Z})}.$$

Observe that $p^k(\mathbb{Z}/p^{e_j}\mathbb{Z})$ is zero whenever $k \geq e_j$, so that $\mathbb{Z}/p^{e_j}\mathbb{Z} \otimes \mathbb{Z}/p^k\mathbb{Z} \cong \mathbb{Z}/p^{e_j}\mathbb{Z}$ for each summand in the tensor product. This gives our desired result that $\mathbb{Z}/p^k\mathbb{Z} \otimes A \cong A_p$.

(b) Observe that in the exact sequence and quotient identification we exhibited above, if we substitute p for p^k, then we have

$$\mathbb{Z}/p^{e_j}\mathbb{Z} \otimes \mathbb{Z}/p\mathbb{Z} \cong \frac{\mathbb{Z}/p^{e_j}\mathbb{Z}}{p(\mathbb{Z}/p^{e_j}\mathbb{Z})}.$$

If $e_j > 1$, then this quotient is isomorphic to $\mathbb{Z}/p\mathbb{Z}$ as it emerges from the surjection $[m]_{p^{e_j}} \mapsto [m]_p$, while if $e_j = 1$, then we have that $p(\mathbb{Z}/p^{e_j}\mathbb{Z}) = 0$, so $\mathbb{Z}/p\mathbb{Z} \otimes_{\mathbb{Z}} A$ is given by $(\mathbb{Z}/p\mathbb{Z})^r$, where r denotes the number of summands corresponding to the prime p in the elementary divisors decomposition.

Problem (Problem 5)**:** Let $M = \text{Mat}_3(\mathbb{Q})$, let 0 denote the zero matrix, and 1 the identity matrix of M.

(a) Prove or disprove: if $A \in M$ satisfies $A^6 = 0$, then $A^3 = 0$.

(b) Classify, up to similarity, all matrices in M satisfying $A^6 = 0$. Exhibit one representative for each similarity class.

(c) Prove or disprove: if $A \in M$ satisfies $A^6 = 1$, then $A^3 = 1$.

(d) Classify, up to similarity, all matrices in M satisfying $A^6 = 1$. Exhibit one representative for each such similarity class.

Solution:

(a) If $A^6 = 0$, then it follows that $\mu_A(x) | x^6$. Since $\mu_A(x) | \chi_A(x)$, and $\chi_A(x)$ has degree 3, it follows that $\mu_A(x) | x^3$, meaning that $A^3 = 0$.

(b) We use the rational canonical form and the possibilities for $\mu_A(x)$ given that $\mu_A(x) | x^3$. If $\mu_A(x) = x$, then $A = 0$. If $\mu_A(x) = x^2$, then the invariant factors of A are x and x^2, so that the rational canonical form is given by

$$A_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Finally, if $\mu_A(x) = x^3$, then $\mu_A(x) = \chi_A(x)$, and there is one invariant factor given by x^3, which has rational canonical form given by

$$A_2 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Since rational canonical form classifies matrices up to similarity, it follows that these are representatives of the similarity classes of matrices satisfying $A^6 = 0$.

(c) Observe that if $A = -1$, then $A^3 = (-1)^3 = -1 \neq 1$, while $A^6 = 1$. Therefore, this assertion is false.

(d) We observe that $\mu_A(x) | x^6 - 1$. Factorizing $x^6 - 1 = (x-1)\left(x^2+x+1\right)(x+1)\left(x^2-x+1\right)$, so we consider all the possible minimal polynomials and find their rational canonical forms.

- If $\mu_A(x) = x - 1$, then $A_0 = 1$.
- If $\mu_A(x) = x + 1$, then $A_1 = -1$.
- If $\mu_A(x) = x^2 - 1$, then there are two invariant factor decompositions, given by $(x-1) \supseteq \left(x^2-1\right)$ and $(x+1) \supseteq \left(x^2-1\right)$, yielding rational canonical forms

$$A_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$A_4 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

- If $\mu_A(x) = x^3 - 1$, then $\mu_A(x) = \chi_A(x)$, and A is the companion matrix

$$A_5 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

- If $\mu_A(x) = x^3 + 1$, then $\mu_A(x) = \chi_A(x)$, and A is the companion matrix

$$A_6 = \begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Since $x^2 \pm x + 1$ is irreducible, it follows that $A \in \mathrm{Mat}_3(\mathbb{Q})$ cannot have invariant factors that divide μ_A unless those invariant factors are equal to μ_A, implying that A would be a $2k \times 2k$ matrix.

Problem (Problem 7)**:**

(a) Using cyclotomic fields, construct a Galois extension K of $\mathbb{Q}$ of degree 3.

(b) Find a polynomial $f(x) \in \mathbb{Q}[x]$ such that the field K is the splitting field of $f(x)$ over $\mathbb{Q}$.

Solution:

(a) If ω is a primitive 7th root of unity, then $\mathbb{Q}(\omega)/\mathbb{Q}$ is a cyclotomic field of degree $\varphi(7) = 6$ with Galois group $(\mathbb{Z}/7\mathbb{Z})^\times$. Elements of the Galois group are given by $\zeta_7 \mapsto \zeta_7^a$ for each $a \in (\mathbb{Z}/7\mathbb{Z})^\times$.

There is a unique index 3 subgroup of $(\mathbb{Z}/7\mathbb{Z})^\times$ given by inversion, and we see that

$$\theta = \omega + \omega^{-1}$$

is invariant under this subgroup. In particular, this means that $[\mathbb{Q}(\theta) : \mathbb{Q}] \leq 3$. Furthermore, observe that ω is a root of $f(x) = x^2 - \theta x + 1$, meaning that $[\mathbb{Q}(\omega) : \mathbb{Q}(\theta)] \leq 2$, meaning that $[\mathbb{Q}(\theta) : \mathbb{Q}] \geq 3$, or that $[\mathbb{Q}(\theta) : \mathbb{Q}] = 3$. Setting $K = \mathbb{Q}(\theta)$, we observe that K is the fixed field for a subgroup of the abelian group $(\mathbb{Z}/7\mathbb{Z})^\times$, so $K/\mathbb{Q}$ is a Galois extension.

(b) We will find a polynomial whose roots are the Galois conjugates of θ, which are given by

$$\begin{aligned} \theta &= \omega + \omega^{-1} \\ \sigma_2(\theta) &= \omega^2 + \omega^{-2} \\ \sigma_3(\theta) &= \omega^3 + \omega^{-3}. \end{aligned}$$

A polynomial with these roots is given by

$$\begin{aligned} f(x) &= \left(x - \left(\omega + \omega^{-1}\right)\right)\left(x - \left(\omega^2 + \omega^{-2}\right)\right)\left(x - \left(\omega^3 + \omega^{-3}\right)\right) \\ &= x^3 + x^2 - 2x - 1. \end{aligned}$$

Since the Galois group $\mathrm{Gal}(K/\mathbb{Q})$ is given by the quotient $\mathrm{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})/\mathrm{Gal}(\mathbb{Q}(\omega)/\mathbb{Q}(\theta))$, it follows that this Galois group acts transitively on the roots of $f(x)$, so f is irreducible of degree 3 with its roots contained in K, whence K is the splitting field for f over $\mathbb{Q}$.

August 2018

Problem (Problem 2)**:** Let G be a non-cyclic group of order 57. Determine the number of elements of all orders of G.

Solution: Notice that $57 = 3\cdot 19$, so there is a (necessarily normal) 19-Sylow subgroup. Since $\text{Aut}(\mathbb{Z}/19\mathbb{Z}) \cong \mathbb{Z}/18\mathbb{Z}$, it follows that there is a nonzero homomorphism $\mathbb{Z}/3\mathbb{Z} \to \text{Aut}(\mathbb{Z}/19\mathbb{Z})$; by structural properties for groups of order pq it follows that the resulting semidirect product $\mathbb{Z}/19\mathbb{Z} \rtimes \mathbb{Z}/3\mathbb{Z}$ is unique. Since G is a non-cyclic group of order 57, there are no elements of order 57; we thus observe that elements of the form (a, b) where $b \in \mathbb{Z}/3\mathbb{Z}$ and $a \in \mathbb{Z}/19\mathbb{Z}\setminus\{0\}$ are of order 19, while elements of the form $(0, b)$ with $b \in \mathbb{Z}/3\mathbb{Z}\setminus\{0\}$ are of order 3. The last one we are left with is the identity $(0, 0)$. Therefore, combined, we find there are 54 elements of order 19, 2 elements of order 3, and 1 element of order 1.

Problem (Problem 3)**:** Let R be a commutative ring such that all prime ideals are finitely generated. Prove that R is Noetherian by using the following steps.

(a) Let $\mathcal{X}$ be the set of non-finitely generated ideals of R. Prove that if R is not Noetherian, then $\mathcal{X}$ has a maximal element, which we call I.

(b) Prove that there exist elements $x, y \in R$ such that $x, y \notin I$ and $xy \in I$.

(c) Prove that $I_x = I + (x)$ and $J_x = \{r \in R : rx \in I\}$ are finitely generated ideals. Here, x is the same element as in the previous part.

(d) Conclude that I is finitely generated, and derive a contradiction.

Solution:

(a) Suppose R is not Noetherian. Then, there is an ideal that is not finitely generated, so the set $\mathcal{X}$ is nonempty. If $\mathcal{C}$ is a chain in X, then we claim that $\bigcup\mathcal{C}$ is not finitely generated. If it were, then taking a generating set $\{x_1, \dots, x_n\}$ for $\bigcup\mathcal{C}$, we may find an ideal J in $\mathcal{C}$ containing all elements of the generating set, so it would follow that $\bigcup\mathcal{C} = J$, but J was assumed to not be finitely generated since $\mathcal{C}$ is a chain in $\mathcal{X}$. Therefore, by Zorn's Lemma, there is a maximal element I of $\mathcal{X}$.

(b) Since I is not finitely generated, I cannot be prime, so there must exist $x, y \in R$ such that $xy \in I$ but $x, y \notin I$.

(c) Since $x \notin I$, it follows that $I \subsetneq I_x$; if I_x were not finitely generated, then this would contradict maximality. Similarly, since $y \notin I$, we must have that $I \subsetneq J_x$, again contradicting maximality.

(d) Denote by $\{f_1, \dots, f_n\}$ a finite generating set for I_x, and write

$$f_i = s_i + r_i x.$$

Similarly, let $\{t_1, \dots, t_m\}$ be a finite generating set for J_x. Observe then that $\{s_1, \dots, s_n, xt_1, \dots, xt_n\} \subseteq I$ by the definitions of I_x and J_x. Our task is to now show that this is a generating set for I.

Let $a \in I$. Then, $a \in I_x$ and $a \in J_x$, so we may find $u_1, \dots, u_n$ in R such that

$$\begin{aligned} a &= \sum_{i=1}^{n} u_i f_i \\ &= \sum_{i=1}^{n} u_i s_i + u_i r_i x \end{aligned}$$

Observe then that

$$\left(\sum_{i=1}^{n} u_i r_i\right) x = a - \sum_{i=1}^{n} u_i s_i,$$

meaning that the left hand side is contained in I, so that $\sum_{i=1}^{n} u_i r_i \in J_x$. There are then $v_1, \dots, v_m$ in

J_x such that

$$\left(\sum_{j=1}^{m} v_j t_j\right) x = a - \sum_{i=1}^{n} u_i s_i,$$

whence

$$a = \sum_{i=1}^{n} u_i s_i + \sum_{j=1}^{m} \left(v_j x\right) t_j,$$

so that $I \subseteq (s_1, \dots, s_n, t_1, \dots, t_m)$. Thus, we get that I is finitely generated, yielding our final contradiction that, in fact, the set $\mathcal{X}$ was empty, and so every ideal of R is finitely generated.

Problem (Problem 4): Let R be an integral domain, and let M be a nontrivial torsion R-module.

(a) If M is finitely generated, then the annihilator of M is nontrivial.

(b) Find an integral domain R and a torsion module M over R whose annihilator is the zero ideal.

Solution:

(a) Let $\{x_1, \dots, x_n\}$ be a generating set for M, and let $0 \neq r_i$ be such that $r_i x_i = 0$ for each $i = 1, \dots, n$. Since each $r_i \neq 0$, it follows that $r = r_1 \cdots r_n$ is nonzero, and since R is commutative, $r x_i \neq 0$ for each $i = 1, \dots, n$, so $r \in \operatorname{Ann}(M)$ as r annihilates all generators of M. Therefore, $\operatorname{Ann}(M)$ is nontrivial.

(b) Let $M = \mathbb{Q}/\mathbb{Z}$ as a $\mathbb{Z}$-module. Observe that $\left[\frac{a}{b}\right]$ is torsion as $b\left[\frac{a}{b}\right] = [a] = [0]$. Yet, for all $b > 1$, we have $\left[\frac{1}{b}\right] \in \mathbb{Q}/\mathbb{Z}$, meaning that any ideal that annihilates every element of $\mathbb{Q}/\mathbb{Z}$ annihilates $\left[\frac{1}{b}\right]$, giving that

$$\begin{aligned} \operatorname{Ann}(M) &\subseteq \bigcap_{b>1} (b) \\ &= (0). \end{aligned}$$

Problem (Problem 5): Let F be a field, and let n be a natural number.

(a) Let $F = \mathbb{R}$. Classify the matrices $a \in \operatorname{Mat}_n(\mathbb{R})$ satisfying $A^3 = A$, up to similarity.

(b) For an appropriate F and n, find a matrix $a \in \operatorname{Mat}_n(F)$ which is not diagonalizable and satisfies $A^3 = A$.

Solution:

(a) Notice that if $A^3 = A$, then $\mu_A(x) | x^3 - x$, so we may classify rational canonical forms using invariant factors as follows.

- If $\mu_A(x) = x$, then $A = 0$.
- If $\mu_A(x) = x - 1$, then $A = I$.
- If $\mu_A(x) = x + 1$, then $A = -I$.
- If $\mu_A(x) = x^2 - 1$, then there are two possible invariant factor decompositions.
  - If we have invariant factors given by $(x-1) \supseteq \left(x^2 - 1\right)$, then we have the similarity class with representative given by

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

    in rational canonical form.

- If we have invariant factors given by $(x+1) \supseteq \left(x^2-1\right)$, then we have similarity class with representative given by

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

in rational canonical form.

Since these are the only possible invariant factor decompositions of the polynomial $x^3 - x$, these are the similarity classes.

(b) If $F = \mathbb{F}_2$, then then we have $\mu_A(x) | x(x-1)(x+1) = x(x-1)^2$. If $\mu_A(x) = (x-1)^2$, then A is not diagonalizable, and we have invariant factors given by $(x-1) \supseteq (x-1)^2$, and a similarity class representative

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Problem (Problem 6)**:** Let $p(x)$ be a polynomial defined over $\mathbb{R}$, and suppose that p takes on rational values at rational numbers. Prove that the coefficients of p are rational.

Solution: Write

$$p(x) = a_n x^n + \cdots + a_1 x + a_0$$

with $a_0, \dots, a_n \in \mathbb{R}$. By Lagrange interpolation, a selection of $(n+1)$ distinct elements of $\mathbb{Q}$ with evaluations uniquely determines a polynomial of degree n in $\mathbb{Q}[x]$. Therefore, if we let $x_0, \dots, x_n$ be $(n+1)$ distinct points of $\mathbb{Q}$ with values given by $p(x_0), \dots, p(x_n)$, then there is a unique polynomial $q(x)$ of degree n in $\mathbb{Q}[x]$ such that $q(x_i) = p(x_i)$.

This means that $q(x) = p(x)$ for all $x \in \mathbb{Q}$; using facts from analysis, since polynomials are continuous over $\mathbb{R}$, it follows that $p(x) = q(x)$ for all $x \in \mathbb{R}$, so that $p(x)$ must have rational coefficients.

This is not the case for integer coefficients, as we may consider the polynomial $p(x) = \frac{1}{2}(x)(x+1)$, which is integral at all integers since any integer is either even or adjacent to an even integer.

Problem (Problem 7)**:** What is the Galois group of $x^5 - 1$ over a field with 7 elements?

Solution: Factoring $x^5 - 1 = (x-1)\Phi_5(x)$, we start by observing that $7 \nmid 5$, so by the irreducibility criterion for cyclotomic polynomials over finite fields, we have that Φ_5 splits into irreducible factors with degree equal to the multiplicative order of 7 mod 5. Since $7 \equiv 2 \bmod 5$, and 2 has multiplicative order 4 mod 5, it follows that $\Phi_5(x)$ is irreducible, so we have that the Galois group of $x^5 - 1$ is given by the Galois group of $\mathbb{F}_{7^4}$ over $\mathbb{F}_7$. Since Galois groups of finite extensions of finite fields are given by cyclic groups of order equal to the degree of the extension, it follows that the Galois group of x^5-1 over a field with 7 elements is given by $\mathbb{Z}/4\mathbb{Z}$.

Problem (Problem 8)**:** Describe the splitting field of $x^4 + x^2 + 1$ over $\mathbb{Q}$. What is its degree?

Solution: Observe that we may write

$$x^6 - 1 = (x-1)(x+1)\left(x^4 + x^2 + 1\right),$$

where we may write $x^4 + x^2 + 1 = \Phi_3(x)\Phi_6(x)$. Observe that if θ is a primitive sixth root of unity, then θ^2 is a primitive cube root of unity, so that the splitting field of x^4+x^2+1 is given by adjoining a primitive sixth root of unity over $\mathbb{Q}$. Since $\deg(\Phi_6(x)) = 2$, it follows that this is the degree of the splitting field over $\mathbb{Q}$.

January 2019

Problem (Problem 2): Let G be a finite group of order n.

(a) Prove there exists an injective homomorphism $\varphi\colon G \to S_n$ such that for all $g \in G$, the permutation $\varphi(g)$ is a product of n/k disjoint cycles of length k.

(b) Suppose n is even, and a Sylow 2-subgroup of G is cyclic. Use (a) to prove that G has a subgroup of index 2.

Solution:

(a) Label the elements of G by $\{g_1, \dots, g_n\}$, and let G act on itself by left regular permutations. By Cayley's Theorem, this yields an injective homomorphism $\varphi\colon G \to S_n$ as G acts on itself freely and faithfully. Note that this means that $\varphi(e) = \mathrm{id}$.

Now, let $g \in G$ with $o(g) = k$, where $k|n$ and $k \neq 1$. Observe that orders are preserved since φ is an isomorphism onto its image, it follows that φ preserves orders. In particular, this means that the cycle decomposition of $\varphi(g)$ consists exclusively of k-cycles. Furthermore, since we assume $g \neq e$, it follows that g, and hence $\varphi(g)$, does not have any fixed points (as G acts on itself freely), so that there must be n/k such k-cycles that permute the n elements of $\{g_1, \dots, g_n\}$ without fixed points.

(b) If $|G| = 2^k$, then G is cyclic, so there is a subgroup of G with index 2.

Else, let $|G| = 2^k m$, where m is odd with odd prime p dividing m. Let x denote the generator of the assumed cyclic Sylow 2-subgroup. Then, we observe that $\varphi(x)$ has order 2^k, and that $\varphi(x)$ consists of m cycles of length 2^k. Notice that this means $\varphi(x) \notin A_n$ as even-length cycles are odd permutations.

Since there is a p-Sylow subgroup (where p is the aforementioned odd prime dividing m), this p-Sylow subgroup admits an element y of order p by Cauchy's theorem. The cycle decomposition of $\varphi(y)$ has $2^k m/p$ cycles of length p. Since p is odd, $\varphi(y) \in A_n$ since odd-length cycles are even permutations. Therefore, we have that $\varphi(G) \cap A_n \leq G$ is normal (by the second isomorphism theorem) and has index at least 2, with the second isomorphism theorem giving

$$\begin{aligned} 2 &\leq [\varphi(G) : \varphi(G) \cap A_n] \\ &= [\varphi(G)A_n : A_n] \\ &\leq [S_n : A_n] \\ &= 2, \end{aligned}$$

so that the inverse image of $\varphi(G) \cap A_n$ under φ is a subgroup of index 2 in G.

Problem (Problem 3): Let $R = \mathbb{Z}[x]$. Find the number of maximal ideals of R which contain $x^2 + 1$ and 15, and find explicit generators for each such ideal.

Solution: By the fourth isomorphism theorem, the maximal ideals in $\mathbb{Z}[x]$ containing $x^2 + 1$ and 15 correspond to the maximal ideals in the ring

$$\begin{aligned} \frac{\mathbb{Z}[x]}{(x^2+1, 15)} &\cong \frac{\frac{\mathbb{Z}[x]}{(x^2+1)}}{\frac{(x^2+1,15)}{(x^2+1)}} \\ &\cong \frac{\mathbb{Z}[i]}{(15)}. \end{aligned}$$

In particular, this means we want to find the maximal ideals in $\mathbb{Z}[i]$ containing 15; we consider $\overline{x} = i$ in $\mathbb{Z}[x]$ to correspond to i in $\mathbb{Z}[x]/\left(x^2 + 1\right)$. Since $\mathbb{Z}[i]$ is a Euclidean domain, maximal ideals are prime, and primes are irreducible as $\mathbb{Z}[i]$ is a unique factorization domain, so we only need to find the primes that divide 15. Using the classification of primes in $\mathbb{Z}[i]$, it follows that there are three such maximal ideals given by $\{(3), (2-i), (2+i)\}$. Lifting to $\mathbb{Z}[x]$ after application of the homomorphism $\mathbb{Z}[x] \to \mathbb{Z}[i]$, we observe that these ideals correspond to $\left\{\left(x^2+1, 3\right), \left(x^2+1, 2-x\right), \left(x^2+1, 2+x\right)\right\}$ respectively.

Problem (Problem 5)**:** Let F be an algebraically closed field, $n \in \mathbb{N}$, and $A \in \mathrm{GL}_n(F)$ an invertible $n \times n$ matrix over F.

(a) Assume that $\mathrm{char}(F) \neq 2$. Prove that if A^2 is diagonalizable, then A is also diagonalizable over F.

(b) Give an example where $\mathrm{char}(F) = 2$, A^2 is diagonalizable, and A is not diagonalizable.

Solution:

(a) Since A^2 is diagonalizable and invertible, it follows that $\mu_{A^2}(x)$ splits into distinct linear factors

$$\mu_{A^2}(x) = \prod_{i=1}^{r} (x - \lambda_i)$$

with each $\lambda_i \neq 0$. We observe then that, by definition, we have $\mu_A(x) | \prod_{i=1}^{r} \left(x^2 - \lambda_i\right)$. Since F is algebraically closed and not of characteristic 2, it follows there are $\mu_i \neq 0$ such that $\left(x^2 - \lambda_i\right) = (x - \mu_i)(x + \mu_i)$. Therefore, we have that $\mu_A(x)$ consists of distinct linear factors, so A is diagonalizable.

(b) If over $\mathbb{F}_2$, we have

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

then A is in Jordan canonical form, so A is not diagonalizable, but

$$A^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is the identity matrix, which is already diagonal.

Problem (Problem 6)**:** Let R be a commutative ring with 1, M an R-module, and N a submodule of M.

(a) Prove that if both N and M/N are finitely generated, then M is finitely generated.

(b) Give an example where M is finitely generated and N is not.

Solution:

(a) Let $m \in M$. Take $\{x_1 + N, \dots, x_m + N\}$ to be a finite generating set for M/N, and $\{y_1, \dots, y_n\}$ a finite generating set for N.

Then, we may write

$$m + N = \left(\sum_{j=1}^{m} r_j x_j\right) + N,$$

so there exists $n \in N$ such that

$$m - \left(\sum_{j=1}^{m} r_j x_j\right) = n,$$

meaning we may write

$$n = \sum_{i=1}^{n} s_i y_i,$$

and thus

$$m = \sum_{j=1}^{m} r_j x_j + \sum_{i=1}^{n} s_i y_i,$$

so that $\{x_1,\dots,x_m,y_1,\dots,y_n\}$ is a finite generating set for M.

(b) If $M = \mathbb{Z}[x_1,x_2,\dots]$ is the polynomial ring in $\mathbb{Z}$ with infinitely many indeterminates, viewed as a ring over itself, then the ideal $N = (x_1,x_2,\dots)$ is infinitely generated, but M is finitely generated since it is the free module over the ring $\mathbb{Z}[x_1,x_2,\dots]$ with rank 1.

Problem (Problem 8)**:** Let $p > 2$ be a prime, $\mathbb{F}_p$ the field of order p, and $\overline{\mathbb{F}_p}$ an algebraic closure. Let $f(x) = x^m + 1$ for some $m \in \mathbb{N}$. Assume that f is irreducible, and let α be a root of f in $\overline{\mathbb{F}_p}$.

(a) Prove that the multiplicative order of α is equal to $2m$.

(b) Prove that $2m$ divides $p^m - 1$, and $2m$ does not divide $p^k - 1$ for any $0 < k < m$.

(c) Prove that $m \neq 4$.

Solution:

(a) Since f is irreducible, we have that $\mathbb{F}_p(\alpha) = \mathbb{F}_{p^m}$, and α is not contained in any proper subfield of $\mathbb{F}_p(\alpha)$. In particular, this means that $\alpha^k \notin \mathbb{F}_p$ for any $1 \leq k < m$, so since we have $\alpha^m = -1 \neq 1$ (where the latter emerges from the fact that $p \neq 2$), it follows that $\alpha^{2m} = 1$, so that the multiplicative order of α is $2m$.

(b) We have that $\alpha \neq 0$, so $\alpha \in \left(\mathbb{F}_{p^m}\right)^\times$, meaning that $o(\alpha) = 2m \Big| \left|\mathbb{F}_{p^m}^\times\right| = p^m - 1$, and since $\alpha \notin \mathbb{F}_{p^k}$ for any $1 \leq k < m$, it follows that $o(\alpha) \nmid p^k - 1$.

(c) We may show that x^4+1 is reducible mod p by observing that, since p is odd, $p^2 \equiv 1$ modulo 8. Therefore, we have that $x^8 - 1 | x^{p^2-1} - 1$ over $\mathbb{F}_p$, or that $x^4 + 1 | x^{p^2} - x$. In particular, this means that $x^4 + 1$ splits over $\mathbb{F}_{p^2}$. Yet, since any element of $\mathbb{F}_{p^2}$ has a minimal polynomial over $\mathbb{F}_p$ with degree at most 2, it follows that $x^4 + 1$ is necessarily reducible over $\mathbb{F}_p$.

August 2019

Problem (Problem 2)**:** Let $n \geq 3$ be an integer, and S_n the symmetric group on $\{1,2,\dots,n\}$. Let H be a subgroup of S_n with $[S_n : H] = n$. Prove that $H \cong S_{n-1}$.

Solution: Let S_n act on the coset space S_n/H by left multiplication; that is, $g \cdot H = gH$. This defines a permutation representation $\rho\colon S_n \to \mathrm{Sym}(S_n/H)$. Observe that this action is transitive, and that H is the stabilizer of this action; in particular, this means that $K = \ker(\rho)$ of this action is contained in H.

Since K is normal in S_n, there are three cases for K if K is non-trivial.

- If $K = S_n$, then by orbit–stabilizer theorem, we have $n = [S_n : H] \leq [S_n : K] = 1$, but $n \geq 3$, which is a contradiction.
- If $K = A_n$, then this implies that $|K| = \frac{n!}{2}$, but since $|H| = (n-1)!$ and $|H| \geq |K|$, this implies that $(n-1)! \geq \frac{n!}{2}$, which is not the case for any $n \geq 3$.
- If $n = 4$ and $K = V$, the Klein 4-group, then since $|H| = 6$, Lagrange's theorem implies that $4|6$, which is certainly false.

Therefore, it must be the case that $K = \{e\}$ is injective. Since ρ is an injective map between groups of the same order, it is an isomorphism, so that the action of S_n on $\{1,\dots,n\}$ is equivalent to the action of S_n on S_n/H. In particular, this means that, up to relabeling, H is the stabilizer of $1 \in \{1,\dots,n\}$, so that $H \cong S_{n-1}$.

Problem (Problem 3)**:** Let m and n be positive integers with $m|n$. Let $f : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z}$ be the natural projection. Prove that the associated map of the group of units $f : (\mathbb{Z}/n\mathbb{Z})^\times \to (\mathbb{Z}/m\mathbb{Z})^\times$ is surjective.

Solution: By Chinese Remainder Theorem and divisibility of n by m, we may write

$$\mathbb{Z}/n\mathbb{Z} \cong \frac{\mathbb{Z}}{p_1^{d_1}\mathbb{Z}} \oplus \cdots \oplus \frac{\mathbb{Z}}{p_r^{d_r}\mathbb{Z}}$$

$$\begin{aligned}
\mathbb{Z}/m\mathbb{Z} &\cong \frac{\mathbb{Z}}{p_1^{e_1}\mathbb{Z}} \oplus \cdots \oplus \frac{\mathbb{Z}}{p_r^{e_r}\mathbb{Z}}\\
(\mathbb{Z}/n\mathbb{Z})^{\times} &\cong \left(\frac{\mathbb{Z}}{p_1^{d_1}\mathbb{Z}}\right)^{\times} \times \cdots \times \left(\frac{\mathbb{Z}}{p_r^{d_r}\mathbb{Z}}\right)^{\times}\\
(\mathbb{Z}/m\mathbb{Z})^{\times} &\cong \left(\frac{\mathbb{Z}}{p_1^{e_1}\mathbb{Z}}\right)^{\times} \times \cdots \times \left(\frac{\mathbb{Z}}{p_r^{e_r}\mathbb{Z}}\right)^{\times},
\end{aligned}$$

where $d_i \geq 1$ and $e_i \leq d_i$ for each i. Therefore, we reduce the question to the case of a single factor $\mathbb{Z}/p^d\mathbb{Z} \to \mathbb{Z}/p^e\mathbb{Z}$; if $e = 0$, then there is no map of units since the latter is the zero ring, so we will assume $d \geq e \geq 1$, and we will show that the map $\left(\mathbb{Z}/p^d\mathbb{Z}\right)^{\times} \to (\mathbb{Z}/p^e\mathbb{Z})^{\times}$ is surjective.

Now, if $a \in (\mathbb{Z}/p^e\mathbb{Z})^{\times}$ with $1 \leq a < p^e$, then $\gcd(a, p^e) = 1$. In particular, this means that the prime factorization of a does not contain any factors of p, so that $\gcd\left(a, p^d\right) = 1$, whence the class of a in $\mathbb{Z}/p^d\mathbb{Z}$ is a unit, and $a \mapsto a$ gives a surjection of the units of $\mathbb{Z}/p^d\mathbb{Z}$ and $\mathbb{Z}/p^e\mathbb{Z}$.

Problem (Problem 4)**:** Let R be a commutative ring with 1, S a subring of R with 1, and let M, N be R-modules.

(a) Prove that there exists a surjective S-module homomorphism $\varphi\colon M \otimes_S N \to M \otimes_R N$ such that $\varphi(m \otimes_S m) = m \otimes_R n$.

(b) Assume that R and S are fields, $R \neq S$, and M and N are nonzero. Prove that φ is not injective.

(c) Suppose we only know that R is a field. Does the conclusion of (b) remain true?

Solution:

(a) Define a S-balanced map $\phi\colon M \times N \to M \otimes_R N$ by $\varphi(m, n) = m \otimes_R n$. Use the universal property to define a unique S-module homomorphism $\varphi\colon M \otimes_S N \to M \otimes_R N$ that is necessarily surjective since any sum of elementary tensors in $M \otimes_R N$ is given by the sum of elementary tensors in $M \otimes_S N$.

(b) Given $m \in M$ and $n \in N$ nonzero, with $r \in R \setminus S$, consider the vector

$$u = m \otimes (rn) - (rm) \otimes n.$$

Then, we observe that $u \in \ker(\varphi)$ since elementary tensors in $M \otimes_R N$ are R-bilinear.

We will show that $u \neq 0$ by determining an S-linear map $\theta\colon M \otimes_S N \to S$ such that $\theta(u) \neq 0$.

Since R and S are fields, M and N are nonzero free R-modules that are also free S-modules; since $S \neq R$, we have that m and rm are S-linearly independent. Extending $\{m, rm\}$ to an S-basis $\mathcal{B}$ for M, we may define $f\colon M \to S$ by taking $f(m) = 1$ and $f(m') = 0$ for any $m' \neq m$ in $\mathcal{B}$, and extending linearly by using the universal property. Similarly, n and rn are S-linearly independent, we may extend $\{n, rn\}$ to an S-basis $\mathcal{C}$ of N, and taking $g\colon N \to S$ by taking $g(n) = 1$ for all $n \in \mathcal{C}$ and extending linearly.

Defining $\beta\colon M \times N \to S$ by $(m, n) \mapsto f(m)g(n)$, we observe that β is S-bilinear, so it extends to an S-linear map $\theta\colon M \otimes_S N \to S$ satisfying $\theta(m \otimes n) = f(m)g(n)$. Therefore, we have

$$\begin{aligned}
\theta(u) &= f(m)g(rn) - f(rm)g(n)\\
&= 1\\
&\neq 0.
\end{aligned}$$

Therefore, $u \neq 0$.

(c) Consider $R = \mathbb{Q}$ and $S = \mathbb{Z}$. Observe that any simple tensor in $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ can be written as

$$\frac{a}{b} \otimes \frac{c}{d} = \frac{a}{b} \otimes b\frac{c}{bd}$$

$$\begin{aligned} &= b\frac{a}{b}\otimes\frac{c}{bd}\\ &= a\otimes\frac{c}{bd}\\ &= 1\otimes\frac{ac}{bd}, \end{aligned}$$

meaning that element in $\mathbb{Q}\otimes_{\mathbb{Z}}\mathbb{Q}$ is written $1\otimes q$, while under φ, we have that this element is given by $1\otimes q$ in $\mathbb{Q}\otimes_{\mathbb{Q}}\mathbb{Q}$; since any element in $\mathbb{Q}\otimes_{\mathbb{Q}}\mathbb{Q}$ is given by $1\otimes q$ by a similar process, it follows that φ only maps $1\otimes q$ to 0 when $q=0$, meaning that $\ker(\varphi)=\{0\}$.

Problem (Problem 5)**:** Let F be a field, V a vector space over F of finite dimension n, and let $T\colon V\to V$ an F-linear map.

(a) Assume that F is algebraically closed. Prove that V has at least $n+1$ T-invariant subspaces including 0 and V.

(b) Give an example showing that if F is not algebraically closed, then (a) may be false.

(c) Assume that T is diagonalizable over F and has n distinct eigenvalues. Prove that the number of T-invariant subspaces depends only on n, and find that number.

Solution:

(a) Since F is algebraically closed, V admits a basis $\mathcal{B}=\{v_1,v_2,\dots,v_n\}$ such that $[T]_{\mathcal{B}}$ is in Jordan canonical form; in particular, $[T]_{\mathcal{B}}$ is upper triangular. Taking the chain

$$0\subseteq \operatorname{span}\{v_1\}\subseteq\operatorname{span}\{v_1,v_2\}\subseteq\cdots\subseteq\operatorname{span}\{v_1,v_2,\dots,v_n\},$$

we have a chain of contained subspaces of length $n+1$, and observe that $Tv_i\in\operatorname{span}\{v_1,v_2,\dots,v_i\}$ by the fact that $[T]_{\mathcal{B}}$ is upper triangular, so that each of these subspaces is T-invariant.

(b) Considering the matrix

$$T=\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}$$

in $\operatorname{Mat}_2(\mathbb{R})$, we observe that this is a rotation by $\pi/2$, so it has the only invariant subspaces given by 0 and $\mathbb{R}^2$.

(c) If we consider V as an $F[x]$-module with action $x\cdot v=Tv$, then we observe that a submodule is precisely a T-invariant subspace.

Since T is diagonalizable, we may write

$$V\cong\bigoplus_{i=1}^{n}\frac{F[x]}{(x-\lambda_i)}$$

where each of the $\lambda_1,\dots,\lambda_n$ are distinct by assumption. Since each of the $(x-\lambda_i)\subseteq F[x]$ are maximal ideals, the modules $\frac{F[x]}{(x-\lambda_i)}$ are irreducible as $F[x]$-modules. In particular, this means there are no submodules in any summand, and the summands are pairwise non-isomorphic by the fact that $\lambda_1,\dots,\lambda_n$ are distinct; counting submodules is thus given by counting the total number of decompositions of V into these summands, giving 2^n such submodules.

Problem (Problem 8)**:** Let q be a prime power, $\mathbb{F}_q$ a field of order q, and let $a\in\mathbb{F}_q$ be a nonzero element. Consider the polynomial

$$f(x)=x^{q+1}-a.$$

Let L be the splitting field of $f(x)$ over $K=\mathbb{F}_q$. Prove that $[L:K]=2$.

Solution: Write $\mathbb{F}_q = \mathbb{F}_{p^n}$. Observe that $f'(x) = x$, so that $\gcd(f, f') = 1$, so in particular, $f(x)$ is separable. Furthermore, since $a \in \mathbb{F}_q \setminus \{0\}$, we have that $a^{p^n - 1} = 1$, so we observe that if α is a root of $f(x)$, then

$$\begin{aligned} \alpha^{(p^n+1)(p^n-1)} &= \alpha^{p^{2n}-1} \\ &= 1, \end{aligned}$$

meaning that $\alpha \in \mathbb{F}_{p^{2n}} = \mathbb{F}_{q^2}$.

Since $f(x)$ is separable, it follows that $f(x)$ admits $p^n + 1$ distinct roots in $\overline{\mathbb{F}_p}$, so by the pigeonhole principle, at least one of the roots cannot be contained in $\mathbb{F}_q$; in particular, if α is this root, then $\alpha \in \mathbb{F}_{q^2}$, whence $L = \mathbb{F}_{q^2}$ has degree 2 over $\mathbb{F}_q$.

January 2020

Problem (Problem 1)**:**

(a) Let C be a cyclic group of order $n \geq 2$. Explain why $\operatorname{Aut}(C) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, and why $\operatorname{Aut}(C) \cong \mathbb{Z}/(n-1)\mathbb{Z}$ if n is prime.

(b) Let G be a finite group, p the smallest prime dividing G, and suppose G contains a normal subgroup C of order p. Prove that C lies in the center of G.

Solution:

(a) An automorphism $\mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ is determined by where it sends the generator 1, and it must send 1 to another generator of $\mathbb{Z}/n\mathbb{Z}$, namely an element of $(\mathbb{Z}/n\mathbb{Z})^\times$. If n is prime, then $\mathbb{Z}/n\mathbb{Z}$ is a field, so $(\mathbb{Z}/n\mathbb{Z})^\times$ is cyclic of order $n-1$.

(b) Let x be the generator for C. Then, observe that, since C is normal, $gxg^{-1} = x^k$ for some $1 \leq k < p$. If it were the case that $k \neq 1$, then this gives rise to a nontrivial homomorphism $G \to \operatorname{Aut}(C) \cong \mathbb{Z}/(n-1)\mathbb{Z}$. This implies that $p-1$ divides the order of G, but p is the smallest prime that divides the order of G, which is a contradiction.

Problem (Problem 4)**:** Let R be a commutative ring with 1, and let $x \in R$.

(a) Prove that $x \notin R^\times$ if and only if $x \in M$ for some maximal ideal M of R.

(b) Prove that the following are equivalent:

  (i) $x \in M$ for every maximal ideal M of R;

  (ii) $1 + xy \in R^\times$ for every $y \in R$.

Solution:

(a) If $x \in M \cap R^\times$, then we have that $x^{-1} \in M \cap R^\times$, meaning that $1 \in M$, so that $M = R$, but M cannot be equal to R by the definition of a maximal ideal. Meanwhile, if $x \notin M$ for all maximal ideals M, we have that $(x) = R$, meaning that $1 \in (x)$, so there exists some $y \in R$ such that $xy = 1 = yx$.

(b) If $x \in M$ for all maximal ideals M of R, then $(x) \subseteq \bigcap_{M \text{ maximal}} M$. Given $y \in R$, $1+M = (1+xy)+M$ for any $y \in R$ since $xy \in M$ for all maximal ideals M. Therefore, we have that $1 + xy \notin M$ for all maximal ideals M, so that $1 + xy \in R^\times$ by (a).

Suppose that $x \notin M$ for some maximal ideal M. Then, since M is maximal, $M + (x) = R$, so there is some $y \in R$ such that $1 + xy \in M$, so $1 + xy \notin R^\times$.

Problem (Problem 5)**:** Let F be a field, and let $A, B \in \operatorname{Mat}_n(F)$ for some $n \in \mathbb{N}$. Suppose $A^2 = B^2 = I$, and $\operatorname{rk}(A - I) = \operatorname{rk}(B - I)$. Prove that A and B are similar.

Solution: We start by assuming that F is not characteristic 2. Then, $\mu_A(x) | (x-1)(x+1)$, so that the eigenvalues of A (and those of B) are drawn from $\{1, -1\}$, meaning that both A and B are diagonalizable. Since $\operatorname{rk}(A - I) = \operatorname{rk}(B - I)$, it follows that the eigenspaces $E_1(A)$ and $E_1(B)$ are isomorphic, so in the diagonalization of A and B, there are an equal number of 1s in the decomposition. Furthermore, since the remaining eigenvalues are equal to -1, it follows that $E_{-1}(A) \cong E_{-1}(B)$, so A and B have identical eigenvalues with identical geometric multiplicity, meaning that A and B have equal diagonalizations. Since Jordan canonical forms define a unique similarity class, and A and B have identical Jordan canonical forms, it follows that A and B are similar.

August 2020

Problem (Problem 1)**:** Let A be a finitely generated abelian group. Prove that the following are equivalent:

(a) $\dim_{\mathbb{Q}}(A \otimes_{\mathbb{Z}} \mathbb{Q}) \leq 1$;

(b) $\operatorname{Aut}(A)$ is finite.

Solution: Using the classification theorem, we may write

$$A \cong \mathbb{Z}^d \oplus T,$$

where T is a finite abelian group.

Suppose $A \otimes_{\mathbb{Z}} \mathbb{Q}$ has $\mathbb{Q}$-dimension less than or equal to 1. Noting that $T \otimes_{\mathbb{Z}} \mathbb{Q} \cong \{0\}$, it follows that $A \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q}^d$, meaning that there are two cases.

(i) If $d = 0$, then A is a finite group, meaning that $\operatorname{Aut}(A)$ is finite as $\operatorname{Aut}(A)$ is a subgroup of the finite group $\operatorname{Sym}(A)$.

(ii) If $d = 1$, then we consider all the possible automorphisms of A. First, observe that any element of the form $(0, s)$, where $s \in T$, must be mapped to an element of the form $(0, s)$, since orders are preserved under automorphisms and s is necessarily finite-order.

If $(1, 0)$ denotes the generator of the $\mathbb{Z}$ summand in the expression of A given by the classification, then we observe that if $\alpha \in \operatorname{Aut}(A)$ is any element, then $\alpha(1, 0) = (n, s)$ and $\alpha^{-1}(1, 0) = (n', s')$. Since any automorphism must preserve T, it follows that the automorphism α descends to an automorphism of the quotient $A/T \cong \mathbb{Z}$; since automorphisms preserve generators, it follows that $n = \pm 1$ and similarly, $n' = \pm 1$. In particular, this means that any automorphism of A is a composition of an automorphism of $\mathbb{Z}$, a shearing term s, and an automorphism of T; since each of these sets are finite, it follows that $\operatorname{Aut}(A)$ is finite.

Now, suppose $A \otimes_{\mathbb{Z}} \mathbb{Q}$ has $\mathbb{Q}$-dimension at least 2. Then, $d \geq 2$, meaning that we may write $A \cong \mathbb{Z}^2 \oplus S$ for some finitely generated abelian group S. In particular, any automorphism of the form $U \oplus \mathrm{id}$ for some $U \in \mathrm{GL}_2(\mathbb{Z})$ is an automorphism of A. Notice that $\mathrm{GL}_2(\mathbb{Z})$ is an infinite group, as seen by the infinite family of elements given by

$$\left\{ \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} : a \in \mathbb{Z} \right\}.$$

Problem (Problem 2)**:** Let $n \geq 3$ be an integer. Denote by S_n the symmetric group on $\{1, \dots, n\}$, and D_{2n} the dihedral group of order $2n$.

(a) Prove that for every $k \geq 3$, there exists an injective homomorphism $\varphi \colon D_{2k} \to Sk$ whose image contains a k-cycle.

(b) Prove that every element of S_n can be written as a product of two elements with order less than or equal to 2.

Solution: We write the presentation of D_{2k} as

$$D_{2k} = \left\langle r, s \middle| r^k, s^2, srsr \right\rangle.$$

(a) Viewing D_{2k} as the symmetry group of a regular k-gon, and labeling the vertices $\{1, \dots, k\}$ yields a homomorphism $D_{2k} \to S_k$ by defining the permutation in S_k by the correspondence between the vertices and the set $\{1, \dots, k\}$. This homomorphism is injective since the only element of D_{2k} that preserves every vertex of the regular k-gon is the identity permutation. Since injective homomorphisms preserve order, the image of r has order k, and the only elements of S_k that have order k are k-cycles.

(b) Notice that any element of D_{2k} is a product of either one or two reflections. For instance, the element r is given by $(rs)s$, where we observe that $(rs)^2 = rsrs = rs^2r^{-1} = e$. Taking a cycle decomposition, we have that every permutation $\sigma \in S_n$ is given by

$$\sigma = \sigma_1\sigma_2\cdots\sigma_r$$

is a product of disjoint cycles. We may assume that any of these cycles are of length at least 3, as the product of a disjoint collection of transpositions is of order 2. If we have a k-cycle defined by $(i_1, i_2, \dots, i_k)$, then we may view this k-cycle as the element r in the subgroup D_{2k} of S_k, where the action is defined on $\{i_1, i_2, \dots, i_k\}$. In particular, the supports of the cycles $\sigma_1, \dots, \sigma_r$ are disjoint. The k-cycle may then be expressed as a product of two reflections $\tau_{i,1}\tau_{i,2}$. Writing σ in this form gives

$$\sigma = \tau_{1,1}\tau_{1,2}\tau_{2,1}\tau_{2,2}\cdots\tau_{r,1}\tau_{r,2}.$$

We observe that the support of $\tau_{1,2}$ and the support of $\tau_{2,1}$ are disjoint, so the cycles commute. Similarly, we have $\tau_{i,2}\tau_{i+1,1} = \tau_{i+1,1}\tau_{i,2}$ for all $1 \leq i \leq r$. Therefore, we may write

$$\sigma = (\tau_{1,1}\tau_{2,1}\cdots\tau_{r,1})(\tau_{1,2}\tau_{2,2}\tau_{3,2}\cdots\tau_{r,2}).$$

Each of these groupings is a product of disjoint transpositions, so this yields σ as a product of two elements of order 2.

Problem (Problem 3)**:** Let $R = \mathbb{Z}\left[\sqrt{3}\right]$. You may assume without proof that R is a Euclidean domain. Let $p \in \mathbb{N}$ be a prime number with $p \equiv 5 \bmod 12$. Prove that p is a prime element of R.

Solution: We start by observing that

$$\mathbb{Z}\left[\sqrt{3}\right] \cong \frac{\mathbb{Z}[x]}{(x^2-3)},$$

where

$$a + b\sqrt{3} \leftrightarrow a + bx + \left(x^2 - 3\right).$$

Furthermore, since $\mathbb{Z}\left[\sqrt{3}\right]$ is a Euclidean domain, any prime ideals are maximal, so it suffices to show that if $p \equiv 5 \bmod 12$, then $\mathbb{Z}\left[\sqrt{3}\right]/(p)$ is a field. Writing in terms of $\mathbb{Z}[x]/\left(x^2-3\right)$,

$$\begin{aligned}\mathbb{Z}\left[\sqrt{3}\right]/(p) &\cong \frac{\mathbb{Z}[x]}{(p, x^2-3)}\\ &\cong \frac{(\mathbb{Z}/p\mathbb{Z})[x]}{(x^2-3)}.\end{aligned}$$

Therefore, it suffices to show that if $p \equiv 5 \bmod 12$, then $x^2 - 3$ is irreducible over $\mathbb{Z}/p\mathbb{Z}$. Equivalently, we will have to show that 3 is not a square mod p. Observe that since $p \equiv 5 \bmod 12$, it follows that $p \equiv 1 \bmod 4$; by the classification of primes over $\mathbb{Z}[i]$, it follows that $x^2 + 1$ is reducible mod p, or that -1 is a square mod p.

Now, suppose toward contradiction that 3 is a square mod p, Then, it would follow that $3 = (-1)(-3)$ is a square mod p. We claim that -3 is a square mod p if and only if $p \equiv 1 \bmod 3$. If there is some $t \in \mathbb{Z}/p\mathbb{Z}$ such that $t^2 = -3$, then we observe that there is then a primitive cube root of unity ω; a primitive cube root of unity has order 3 in $(\mathbb{Z}/p\mathbb{Z})^\times$, or that $3|(p-1)$, or $p \equiv 1 \bmod 3$. Yet, since we have that $p \equiv 2 \bmod 3$, it follows that -3 is not a square mod p, so 3 is not a square mod p.

Problem (Problem 4)**:** Let R be a commutative ring with 1. An ideal I of R is called irreducible if I cannot be written as $I = J \cap K$, where J and K are both ideals strictly containing I.

(a) Prove that every prime ideal is irreducible.

(b) Assume that R contains a nonzero nilpotent element. Prove that R contains an irreducible ideal which is not prime.

Solution:

(a) Let P be a prime ideal of R. Suppose toward contradiction that there exist proper ideals J and K with $P = J \cap K$ and $P \subsetneq J, K$. Choose $a \in J \setminus P$ and $b \in K \setminus P$. Then, $ab \in P$ since $ab \in J \cap K$ by absorption of multiplication. Yet, by the definition of prime ideal, it follows that either $a \in P$ or $b \in P$, which contradicts the definition of a and b.

(b) Suppose $a \in R$ is a nonzero nilpotent element. We consider the set $\mathcal{S}$ of ideals that do not contain a, ordered by inclusion. This set is nonzero since the zero ideal is in this set and a is not zero. If $\mathcal{C}$ is a chain in $\mathcal{S}$, then the union of the ideals in $\mathcal{C}$ is an upper bound in $\mathcal{S}$. If it were not the case that the union was in $\mathcal{S}$, then we would have some ideal $I \in \mathcal{C}$ with $a \in I$, which would imply that $I \notin \mathcal{S}$.

Therefore, by Zorn's Lemma, there is a maximal element M of $\mathcal{S}$. Observe that the nilradical of R is the intersection of all the prime ideals, so since M does not contain a, it follows that M is not prime.

Suppose toward contradiction that M is not irreducible. Then, there would be ideals J and K where $M \subsetneq J, K$ with $M = J \cap K$. Without loss of generality, if $a \notin J$, then it would follow that $J \in \mathcal{S}$, which would violate the maximality of M. Meanwhile, if $a \in J$ and $a \in K$, then we would have $a \in J \cap K = M$, which would violate the fact that $M \in \mathcal{S}$.

Problem (Problem 6)**:** Let K be a field of characteristic zero, and fix some algebraic closure $\overline{K}$ of K. Let L/K be a finite extension with $L \subseteq \overline{K}$, and let $\alpha \in \overline{K}$.

(a) Prove that there is a surjective K-algebra homomorphism $\rho\colon L \otimes_K K(\alpha) \to L(\alpha)$.

(b) Prove that there exists an injective K-algebra homomorphism not necessarily sending 1 to 1 $\iota\colon L(\alpha) \to L \otimes_K K(\alpha)$.

Solution:

(a) Notice that, since α is algebraic over K, we have $K(\alpha) = K[\alpha]$.

Consider a map $\phi\colon L \times K(\alpha) \to L(\alpha)$ defined by $(\ell, p(\alpha)) \mapsto \ell p(\alpha)$. We claim this map is K-bilinear. This follows from observing that

$$\begin{aligned}
\phi(\ell, p_1(\alpha) + p_2(\alpha)) &= \ell p_1(\alpha) + \ell p_2(\alpha)\\
&= \phi(\ell, p_1(\alpha)) + \phi(\ell, p_2(\alpha))\\
\phi(\ell_1 + \ell_2, p(\alpha)) &= \ell_1 p(\alpha) + \ell_2 p(\alpha)\\
&= \phi(\ell_1, p(\alpha)) + \phi(\ell_2, p(\alpha))\\
\phi(k\ell, p(\alpha)) &= k(\ell p(\alpha))\\
&= k\phi(\ell, p(\alpha))\\
&= \ell(kp(\alpha))\\
&= \phi(\ell, kp(\alpha)).
\end{aligned}$$

Therefore, ϕ descends to a K-linear map $\rho\colon L \otimes_K K(\alpha) \to L(\alpha)$ taking $\ell \otimes p(\alpha) \mapsto \ell p(\alpha)$. We show that ρ respects the multiplication operation, and that ρ is surjective. First, we observe that

$$\begin{aligned}
\rho((\ell_1 \otimes p_1(\alpha))(\ell_2 \otimes p_2(\alpha))) &= \rho(\ell_1\ell_2 \otimes p_1(\alpha)p_2(\alpha))\\
&= \ell_1\ell_2 p_1(\alpha)p_2(\alpha)\\
&= \ell_1 p_1(\alpha)\ell_2 p_2(\alpha)\\
&= \rho(\ell_1 \otimes p_1(\alpha))\rho(\ell_2 \otimes p_2(\alpha)).
\end{aligned}$$

Next, we observe that, since α is algebraic over K, α is algebraic over L, so $L(\alpha) = L[\alpha]$. We can show that any polynomial $q \in L[x]$ evaluated at α is obtained by a sum of simple tensors as follows:

$$q(\alpha) = a_n\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0$$

$$\begin{aligned} &= \rho(a_n \otimes \alpha^n) + \cdots + \rho(a_1 \otimes \alpha) + \rho(a_0 \otimes 1) \\ &= \rho\left(\sum_{i=0}^{n} a_i \otimes \alpha^i\right). \end{aligned}$$

Thus, ρ is surjective.

(b) Assume α is not contained in K or L. We note that $K(\alpha) = \frac{K[x]}{(\mu_{\alpha,K}(x))}$. By the extension of scalars, we know that $L \otimes_K \frac{K[x]}{(\mu_{\alpha,K}(x))} \cong \frac{L[x]}{(\mu_{\alpha,K}(x))}$. Since $K \subseteq L$, we know that $\mu_{\alpha,L}(x) | \mu_{\alpha,K}(x)$. In particular, this means there is some $h(x) \in L[x]$ relatively prime to $\mu_{\alpha,L}(x)$ (since $\mu_{\alpha,K}(x)$ is separable as we are working in characteristic zero) such that $\mu_{\alpha,K}(x) = \mu_{\alpha,L}(x)h(x)$.

By the Chinese remainder theorem, we have

$$\begin{aligned} \frac{L[x]}{(\mu_{\alpha,K}(x))} &\cong \frac{L[x]}{(\mu_{\alpha,L}(x))} \times \frac{L[x]}{(h(x))} \\ &\cong L(\alpha) \times \frac{L[x]}{(h(x))}. \end{aligned}$$

Therefore, we have that an element of $L(\alpha)$ injects into $L \otimes K(\alpha)$ via the sequence of isomorphisms to the first coordinate of the expression $L(\alpha) \times \frac{L[x]}{(h(x))}$.

If $\alpha \in L \setminus K$, then $L(\alpha) = L$, while L injects into $L \otimes_K K(\alpha)$ by $\ell \mapsto \ell \otimes 1$. If $\alpha \in K$, then $L = L(\alpha)$ yet again injects into $L \otimes_K K(\alpha) \cong L \otimes_K K$ by $\ell \mapsto \ell \otimes 1$.

Problem (Problem 8)**:** Let K be the splitting field of the polynomial $f(x) = \left(x^3 - 11\right)\left(x^2 + x - 1\right)$ over $\mathbb{Q}$.

(a) Find (with proof) the degree of the extension $K/\mathbb{Q}$.

(b) Determine the isomorphism class of the Galois group $\mathrm{Gal}(K/\mathbb{Q})$, and prove your answer.

Solution:

(a) Considering the embedding of $\mathbb{Q}[x] \subseteq \mathbb{C}[x]$, we observe that the roots of $f(x)$ are given by

$$A = \left\{\sqrt[3]{11}, \omega\sqrt[3]{11}, \omega^2\sqrt[3]{11}, \frac{-1+\sqrt{5}}{2}, \frac{-1-\sqrt{5}}{2}\right\},$$

where ω is a primitive cube root of unity. In particular, this means that the splitting field K of $f(x)$ is contained in $F = \mathbb{Q}\left(\omega, \sqrt[3]{11}, \sqrt{5}\right)$. Meanwhile, we have $F \subseteq K$ since we may find $\omega = \frac{\omega\sqrt[3]{11}}{\sqrt[3]{11}}$ and $\sqrt{5} = 1 + 2\frac{-1+\sqrt{5}}{2}$, so that $K = \mathbb{Q}\left(\omega, \sqrt[3]{11}, \sqrt{5}\right)$.

To compute the degree, we observe that the minimal polynomials are given by

$$\begin{aligned} \mu_{\sqrt[3]{11},\mathbb{Q}}(x) &= x^3 - 11 \\ \mu_{\omega,\mathbb{Q}}(x) &= x^2 + x + 1 \\ \mu_{\sqrt{5},\mathbb{Q}} &= x^2 - 5. \end{aligned}$$

Computing the degree, we observe that $\mu_{\sqrt[3]{11},\mathbb{Q}(\omega,\sqrt{5})} = \mu_{\sqrt[3]{11}}$, and similarly for $\mu_{\omega,\mathbb{Q}(\sqrt{5})}$ with $\mu_{\omega,\mathbb{Q}}$, so that the degree of $K/\mathbb{Q}$ is 12.

(b) First, we recall the theorem that a separable polynomial $f(x)$ is irreducible if and only if the action of the Galois group on the roots of $f(x)$ is transitive. Since f is separable (as we are working over $\mathbb{Q}$) and not irreducible, it follows that the Galois group of $f(x)$ is not transitive.

Now, we observe that $f(x)$ has two irreducible factors — namely, $p(x) = x^3 - 11$ (which is irreducible by the Eisenstein criterion) and $q(x) = x^2 + x - 1$, and in particular, the extension $K/\mathbb{Q}$ is equal to the compositum of the splitting fields of $p(x)$ and $q(x)$, which we label as L and M respectively. Since these are finite degree extensions whose intersection is equal to $\mathbb{Q}$, it follows that

$$\mathrm{Gal}(K/\mathbb{Q}) \cong \mathrm{Gal}(L/\mathbb{Q}) \times \mathrm{Gal}(M/\mathbb{Q}).$$

We start by seeing that $L = \mathbb{Q}\left(\omega, \sqrt[3]{11}\right)$. Since $p(x)$ has degree 3 and not all of its roots are real, it follows that $\mathrm{Gal}(L/\mathbb{Q}) \cong S_3$, while since M has degree 2 over $\mathbb{Q}$, it follows that $\mathrm{Gal}(M/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z}$, so that $\mathrm{Gal}(K/\mathbb{Q}) \cong S_3 \times \mathbb{Z}/2\mathbb{Z}$.

January 2021

Problem (Problem 1): Let F be a field. Let $f(x) \in F[x]$ be an irreducible separable polynomial of degree n, and let G be the Galois group of f over F, and assume that G is abelian.

(a) Prove that if $g \in G$ is any nontrivial element, then g does not fix any of the roots of f.

(b) Assume that $F \subseteq R$, and n is odd. Prove that all roots of f must be real.

(c) Now, let $F = \mathbb{Q}$. Prove that there are infinitely many n for which there exist f as above with a non-real root and cyclic G.

Solution:

(a) Since f is irreducible and separable, the Galois group G acts transitively on the roots of f, defining a faithful representation $\rho\colon G \to S_n$. Now, suppose there is some $e \neq g \in G_\theta$. Then, for any $x \in G$, we have

$$\begin{aligned} g \cdot (x \cdot \theta) &= (gx) \cdot \theta \\ &= (xg) \cdot \theta \\ &= x \cdot (g \cdot \theta) \\ &= x \cdot \theta, \end{aligned}$$

meaning that g fixes every root, implying that $e \neq g \in \ker(\rho)$, contradicting the faithfulness of the representation ρ.

(b) Observe that, by the orbit–stabilizer theorem, we have that $\{\text{roots of } f\} \cong G/H$ as sets, where H is the stabilizer of any root of f. Yet, since there are no stabilizers as discussed above, it follows that $|G| = n$. In particular, since n is odd, there are no order 2 elements.

If there were a non-real root of f, then since complex conjugation fixes $\mathbb{R}$, it follows that complex conjugation would be an order 2 element of the Galois group, which is a contradiction of the fact that n is odd.

(c) The pth cyclotomic polynomials Φ_p for odd primes p have cyclic Galois groups given by $(\mathbb{Z}/p\mathbb{Z})^\times \cong \mathbb{Z}/(p-1)\mathbb{Z}$; each of these cyclotomic polynomials admit non-real primitive roots of unity, but are contained in $\mathbb{Z}[x] \subseteq \mathbb{Q}[x]$.

Problem (Problem 2): Classify the conjugacy classes of matrices $A \in \mathrm{GL}_7(\mathbb{Q})$ such that $A^3 = -I$.

Solution: We see that $A^3 = -I$ implies that $\mu_A(x) | (x+1)\left(x^2 - x + 1\right)$.

- If $\mu_A(x) = x + 1$, then $A = -I$.
- If $\mu_A(x) = x^2 - x + 1$, then $\mu_A(x) = \Phi_6(x)$ is irreducible, but this would imply that any invariant factors are equal to $\mu_A(x)$ even though A is an element of $\mathrm{GL}_7(\mathbb{Q})$.
- If $\mu_A(x) = (x+1)\left(x^2 - x + 1\right)$, then we may consider all the possible invariant factor decompositions from here:
  - $(x+1) \supseteq (x+1) \supseteq (x+1) \supseteq (x+1) \supseteq \left(x^3 + 1\right)$;

- $(x+1) \supseteq \left(x^3+1\right) \supseteq \left(x^3+1\right)$;
- $\left(x^2-x+1\right) \supseteq \left(x^2-x+1\right) \supseteq \left(x^3+1\right)$.

These invariant factor decompositions are the only viable ones for an element of the matrix $\mathrm{GL}_7(\mathbb{Q})$ with minimal polynomial $\mu_A(x) = x^3+1$, so their corresponding rational canonical forms yield all the similarity classes for such elements.

Problem (Problem 3)**:** Let R be a commutative ring with 1, and assume $|R| > 1$. Prove that R has at least one minimal prime ideal.

Solution: Since R has a maximal ideal, the set

$$\mathcal{X} = \{P \subseteq R : P \text{ prime}\}$$

ordered by containment is non-empty. Apply Zorn's lemma with the upper bound for the chain $\mathcal{C} = \{P_i\}_{i \in I}$ given by

$$\bigcap_{i \in I} P_i$$

by showing that this is a prime ideal.

Problem (Problem 4)**:** Let G be a finite abelian group, and let p be a prime. Prove that the number of elements of order p is equal to the number of nontrivial homomorphisms from G to $\mathbb{Z}/p\mathbb{Z}$.

Solution: Write the elementary divisors decomposition of G as

$$G \cong \bigoplus_{i=1}^{r} \mathbb{Z}/p^{d_i}\mathbb{Z} \oplus S,$$

where S consists of all the elementary divisors with no factors of p. These have no elements of order p. In the group $\mathbb{Z}/p^d\mathbb{Z}$, the elements of order p are precisely the elements of the form np^{d-1} for $1 \leq n < p$. In particular, for each summand, there are $(p-1)$ elements of order p, so there are a total of $r(p-1)$ elements of order p in G.

Meanwhile, we observe that by the universal property of direct sums, any homomorphism $\varphi\colon G \to \mathbb{Z}/p\mathbb{Z}$ is given by the sum of individual homomorphisms $\varphi_i\colon E_i \to \mathbb{Z}/p\mathbb{Z}$, where E_i are summands in the elementary divisors decomposition of G. In particular, this means that any nontrivial homomorphisms $G \to \mathbb{Z}/p\mathbb{Z}$ are given by sums of nontrivial homomorphisms $\mathbb{Z}/p^{d_i}\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$. Counting the number of nontrivial homomorphisms $\mathbb{Z}/p^{d_i}\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ is equivalent to finding the number of maps of generators $[1]_{p^{d_i}} \mapsto w$, where w is a generator for $\mathbb{Z}/p\mathbb{Z}$; in particular, this means there are exactly $(p-1)$ such nontrivial homomorphisms, so we obtain $r(p-1)$ total homomorphisms $G \to \mathbb{Z}/p\mathbb{Z}$.

Problem (Problem 5)**:** Let $n \geq 2$ be an integer, $g = (1,2,\dots,n) \in S_n$, and $H = \langle g \rangle$.

(a) Prove that the centralizer of g in S_n is equal to H.

(b) Now, assume that n is prime, and let N be the normalizer of H in S_n. Prove that $|N| = n(n-1)$.

Solution:

(a) For any $\sigma \in Z_{S_n}(g)$, we have

$$\begin{aligned}\sigma g \sigma^{-1} &= g\\ &= (\sigma(1), \sigma(2), \dots, \sigma(n)).\end{aligned}$$

In particular, σ must be a cyclic permutation of $(1,2,\dots,n)$ so that it is equal to $(1,2,\dots,n)$ upon cyclic rearrangement. Thus, $\sigma = g^k$ for some k.

(b) First, we note that if $n = p$ is prime, then g^k is a p-cycle for each $1 \leq k < p$.

Since $|H| = p$, there are $(p-1)!$ elements in the conjugacy class of g. The subgroups these conjugacy classes generate are either equal to $\langle g\rangle$ or intersect at the identity with $\langle g\rangle$; since there are $(p-1)$ elements of the conjugacy class of g that generate $\langle g\rangle$, it follows there are $(p-2)!$ subgroups of S_p conjugate to $\langle g\rangle$, whence $[G : N_G(H)] = (p-2)!$ by orbit–stabilizer. Thus, $|N_G(H)| = p(p-1)$.

Problem (Problem 6)**:** In both parts of this problem, R is a commutative ring with 1, $|R| > 1$, and M is a flat R-module; that is, if $0 \to N \xrightarrow{f} N'$ is an injective map of R-modules, then $0 \to N \otimes_R M \xrightarrow{f\otimes \mathrm{id}_M} N' \otimes_R M$ is injective.

(a) Assume R is a domain. Prove that M must be torsion-free.

(b) Let R arbitrary. Prove that the following are equivalent:

  (i) for every nonzero R-module $N \neq 0$, we have $N \otimes_R M \neq 0$;

  (ii) for every maximal ideal $\mathfrak{m}$ of R, we have $\mathfrak{m}M \neq M$.

Solution:

(a) Suppose toward contradiction that M is nonzero and torsion. Observe that $R \to F$ is an injective map of R-modules, where $F = \operatorname{frac}(R)$ is the field of fractions. Therefore, it must be the case that $R \otimes M \to F \otimes M$ is an injective map; notice that $R \otimes M$ surjects onto M, so $R \otimes M \neq 0$. Yet, for any element $r \otimes m$ of $F \otimes M$, we observe that there is nonzero $s \in R$ such that $sm = 0$, while we have

$$\begin{aligned} r \otimes m &= \frac{r}{s} s \otimes m \\ &= \frac{r}{s} \otimes sm \\ &= 0, \end{aligned}$$

so any elementary tensor in $F \otimes M$ is zero, so $F \otimes M$ is the zero module. There are no injective maps from from a nonzero module to the zero module.

(b) Assume (i). Notice that if $\mathfrak{m}$ is a maximal ideal, then $R/\mathfrak{m}$ is a nonzero R-module, so that $(R/\mathfrak{m}) \otimes_R M \neq 0$. Since $R/\mathfrak{m} \otimes_R M \cong M/\mathfrak{m}M$, it follows that $M/\mathfrak{m}M \neq 0$, so $\mathfrak{m}M \neq M$.

Assume (ii). Suppose $N \neq 0$ is a nonzero R-module. If $0 \neq x \in N$ is any element, then $\langle x\rangle \cong R/\operatorname{Ann}_R(x)$ by the definition of the annihilator ideal. Write $I = \operatorname{Ann}_R(M)$. We obtain an injective map $\langle x\rangle \to N$, so since M is flat, we have $\langle x\rangle \otimes M \to N \otimes M$ is an injective map. In particular, we have $R/I \otimes M \to N \otimes M$. Notice that $R/I \otimes M \cong M/IM$. There is a maximal ideal $\mathfrak{m} \supseteq I$, and since $\mathfrak{m}M \neq M$, it follows that there is some $m \in M \setminus \mathfrak{m}M$, meaning that $m \in M \setminus IM$, so $m + IM \neq 0 + IM$, meaning that $M/IM \neq 0$. In particular, we have an injective map $M/IM \to N \otimes M$ where M/IM is nonzero, so $N \otimes M$ is nonzero.

Problem (Problem 7)**:** Recall that a field F is called perfect if F either has characteristic zero or F has characteristic $p > 0$ and every element of F is equal to a^p for some $a \in F$. Prove that F admits a finite inseparable extension if and only if F is not perfect.

Solution: Observe that if F has characteristic zero, then any irreducible polynomial over F is separable, so F cannot admit any inseparable extensions. Therefore, we will assume that the characteristic of F is equal to $p \neq 0$.

Suppose F admits no finite inseparable extensions. Then, any irreducible polynomial over F is separable, and in particular, any polynomial of the form

$$q(x) = x^p - b$$

is not irreducible, seeing as $q'(x) = 0$. In the algebraic closure $\overline{F}$, we have that $q(x) = (x-a)^p$ for some $a \in \overline{F}$, but since $q(x)$ is reducible, we have that $q(x) = x^p - a^p$, so that $a^p = b$. Thus, F is perfect.

Meanwhile, if F is perfect, we observe that if we take an inseparable polynomial $f(x) = g(x^p)$, then

$$f(x) = a_n x^{u_n p} + a_{n-1} x^{u_{n-1} p} + \cdots + a_1 x^p + a_0.$$

There are $b_0, \dots, b_n \in F$ such that $b_i^p = a_i$, so by using the Frobenius map and the fact that F has characteristic p, we have that

$$f(x) = (b_n x^{u_n} + \cdots + b_1 x + b_0)^p,$$

meaning that f is not irreducible over F. Therefore, F has no finite inseparable extensions.

Problem (Problem 8)**:** Let $p \neq 3$ be a prime, and let $R = \mathbb{F}_p[x]/\left(x^3 - 1\right)$. Describe the multiplicative group $R^\times$ as a direct product of cyclic groups.

Solution: By the Chinese remainder theorem, we have

$$\begin{aligned} R &\cong \frac{\mathbb{F}_p[x]}{(x-1)} \times \frac{\mathbb{F}_p[x]}{(\Phi_3(x))} \\ &\cong \mathbb{F}_p \times \frac{\mathbb{F}_p[x]}{(\Phi_3(x))}. \end{aligned}$$

Therefore, we concern ourselves with the reducibility of $\Phi_3(x)$. Using the reducibility criterion for cyclotomic polynomials mod p, where, assuming $p \nmid n$, we have that $\Phi_n(x)$ splits into factors with degree equal to the multiplicative order of p mod n. In particular, if $p \equiv 1$ mod 3, then $\Phi_3(x)$ splits into (distinct) linear factors, so we have

$$\begin{aligned} R &\cong \mathbb{F}_p \times \left(\frac{\mathbb{F}_p}{(x-\omega)} \times \frac{\mathbb{F}_p}{(x-\omega^2)}\right) \\ &\cong \mathbb{F}_p \times \mathbb{F}_p \times \mathbb{F}_p, \end{aligned}$$

and

$$R^\times \cong (\mathbb{Z}/(p-1)\mathbb{Z}) \times (\mathbb{Z}/(p-1)\mathbb{Z}) \times (\mathbb{Z}/(p-1)\mathbb{Z}).$$

Meanwhile, if $p \equiv 2$ mod 3, then Φ_3 admits exactly one factor of degree 2, implying that

$$R \cong \mathbb{F}_p \times \mathbb{F}_{p^2},$$

and

$$R^\times \cong (\mathbb{Z}/(p-1)\mathbb{Z}) \times \left(\mathbb{Z}/\left(p^2-1\right)\mathbb{Z}\right).$$

August 2021

Problem (Problem 1)**:** Let $n \in \mathbb{N}$, and let $A = \left(a_{ij}\right)_{i,j} \in \operatorname{Mat}_n(\mathbb{Q})$ be the matrix given by $a_{ij} = i$ for each $1 \leq i, j \leq n$. Compute:

(a) the characteristic polynomial of A;

(b) the minimal polynomial of A;

(c) the Jordan canonical form of A;

(d) the rational canonical form of A.

Solution:

(a) We start by noting that A is a rank 1 matrix. In particular, this means that the eigenvalue 0 has geometric multiplicity equal to $(n-1)$, so it has algebraic multiplicity at least $(n-1)$. Furthermore, we note that $\operatorname{Tr}(A) = \frac{n(n+1)}{2}$, so it must be the case that the algebraic multiplicity of 0 is equal to $(n-1)$

(as else we would have a trace zero matrix), so that we have characteristic polynomial of A given by

$$\chi_A(x) = x^n - \frac{n(n+1)}{2}x^{n-1}.$$

(b) By using an ansatz, we observe that A satisfies the polynomial $p(x) = x\left(x - \frac{n(n+1)}{2}\right)$. In particular, this means that the minimal polynomial of A satisfies $\mu_A(x) | x\left(x - \frac{n(n+1)}{2}\right)$. Since $A \neq 0$ and $A - \frac{n(n+1)}{2}I \neq 0$, it follows that $\mu_A(x) = x\left(x - \frac{n(n+1)}{2}\right)$.

(c) Since $\mu_A(x)$ consists exclusively of linear factors, it follows that A is diagonalizable, so the Jordan canonical form of A is given by

$$A \sim \operatorname{diag}\left(0, 0, \dots, 0, \frac{n(n+1)}{2}\right).$$

(d) Since the characteristic polynomial is the sum of all the invariant factors of A and the minimal polynomial is the largest invariant factor, it follows that the invariant factors of A consist of a chain of $(n-2)$ factors of x (corresponding to 1×1 blocks) and one 2×2 block corresponding to the minimal polynomial. This gives a rational canonical form in block diagonal

$$A \sim \begin{pmatrix} \mathbf{O}_{n-2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & \frac{n(n+1)}{2} \end{pmatrix}.$$

Problem (Problem 2)**:** Given a group G, denote by $\operatorname{Aut}(G)$ its group of automorphisms.

(a) Let $G = \mathbb{Z}/p^a\mathbb{Z} \oplus \mathbb{Z}/p^b\mathbb{Z}$ with $a \geq b > 0$. Show that $\operatorname{Aut}(G)$ is non-abelian by explicitly constructing two non-commuting automorphisms.

(b) Let G be a finite abelian group. Show that $\operatorname{Aut}(G)$ is abelian if and only if G is cyclic.

Solution:

(a) We start by considering the case of $a = b = 1$. THen, $G \cong \mathbb{F}_p^2$, so we will find elements $S, T \in \mathrm{GL}_2\left(\mathbb{F}_p\right)$ that do not commute. Considering

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$
$$T = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

we have

$$ST = \begin{pmatrix} [2] & [1] \\ [1] & [1] \end{pmatrix}$$
$$TS = \begin{pmatrix} [1] & [1] \\ [1] & [2] \end{pmatrix},$$

which are not equal regardless of the value of the prime p.

In the general case of $a \geq b > 0$, then we observe that for all $u \in \mathbb{Z}/p^b\mathbb{Z}$, we have $[u] \in \mathbb{Z}/p^a\mathbb{Z}$. Consider the transformations

$$(u, v) \overset{S}{\mapsto} (u + [v], v)$$
$$(u, v) \overset{T}{\mapsto} (u, v + [u]).$$

First, these are automorphisms since they have inverses defined by

$$(u,v) \xmapsto{S^{-1}} (u-[v],v)$$
$$(u,v) \xmapsto{T^{-1}} (u,v-[u]).$$

Then, we observe that

$$\begin{aligned} ST(u,v) &= S\left(u,v+[u]_{p^b}\right) \\ &= \left(u+[u]_{p^b}+[v],v\right) \\ TS(u,v) &= T\left(u+[v]_{p^a},v\right) \\ &= \left(u+[v]_{p^a},v+[v]_{p^a}+[u]_{p^b}\right), \end{aligned}$$

and that $ST(1,1) = ([3],1)$, while $TS(1,1) = ([2],[3])$.

(b) Write the elementary divisors decomposition of G by

$$G = \bigoplus_{i=1}^{n} \mathbb{Z}/p_i^{d_i}\mathbb{Z}.$$

If G is not cyclic, then there is a repeated factor in the elementary divisors decomposition, since else the Chinese remainder theorem (as distinct primes are coprime) gives

$$G \cong \mathbb{Z}/\left(p_1^{d_1}\cdots p_n^{d_n}\right)\mathbb{Z}.$$

Assuming G is not cyclic, we have that

$$G \cong \mathbb{Z}/p^a\mathbb{Z} \oplus \mathbb{Z}/p^b\mathbb{Z} \oplus Q,$$

where Q is some finite abelian group. The automorphisms S and T discussed in part (a) can be extended by setting them equal to the identity on Q, yielding non-commuting automorphisms $S \oplus \mathrm{id}_Q$ and $T \oplus \mathrm{id}_Q$ in $\mathrm{Aut}(G)$.

Meanwhile, if G is cyclic, we have that $G \cong \mathbb{Z}/n\mathbb{Z}$, so that $\mathrm{Aut}(G) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, which is abelian.

Problem (Problem 3)**:**

(a) Let p be a prime, and let G be a group of order p^3. Prove that any two elements x and y of G that are conjugate in G commute.

(b) Give an example of a group G with order 16 and two elements x, y of G which are conjugate in G but do not commute.

Solution:

(a) We start by observing that either G is abelian, in which case there are no elements conjugate to each other and all elements commute, or $|Z(G)| = p$. Notice that this follows from the fact that if $|Z(G)| = p^2$, then $|G/Z(G)| = p$, implying that $G/Z(G)$ is cyclic, or G is abelian, implying $|Z(G)| = p^3$.

By the class equation, we have that

$$\underbrace{|G|}_{=p^3} = \underbrace{|Z(G)|}_{=p} + \underbrace{\sum_{j=1}^{r}\left[G : Z_G\left(a_j\right)\right]}_{=p(p^2-1)}$$

for some collection of representatives $\left\{a_j\right\}_{j=1}^{n}$ of conjugacy classes. Notice that $Z_G\left(a_j\right) \leq G$, so by Lagrange's theorem, $\left|Z_G\left(a_j\right)\right| = p$ or $\left|Z_G\left(a_j\right)\right| = p^2$. Notice that this yields

$$\left[G : Z_G\left(a_j\right)\right] = \begin{cases} p^2 & \text{if } \left|Z_G\left(a_j\right)\right| = p \\ p & \text{if } \left|Z_G\left(a_j\right)\right| = p^2. \end{cases}$$

Since p^2 does not divide $p\left(p^2-1\right)$, it follows that $\left|Z_G\left(a_j\right)\right| = p^2$ for all conjugacy class representatives a_j. Thus, $Z_G\left(a_j\right)$ is abelian for each representative a_j.

Notice that, since $\left[G : Z_G\left(a_j\right)\right] = p$, it follows that $Z_G\left(a_j\right)$ is normal in G by the fact that any subgroup of index p where p is the smallest prime that divides G is normal. Therefore, if x and y are in the same conjugacy class, then we have some $g \in G$ such that $gxg^{-1} = y$, while since $Z_G(x)$ is normal,

$$\begin{aligned} Z_G(x) &= g(Z_G(x))g^{-1} \\ &= Z_G\left(gxg^{-1}\right) \\ &= Z_G(y). \end{aligned}$$

In particular, this means that x and y are in the same centralizer, so x and y commute since the centralizers are abelian.

January 2022

Problem (Problem 2)**:** Prove that there exist precisely 3 isomorphism classes of groups G that contain a subgroup H which is infinite cyclic of index 2.

Solution: We start with the abelian case. We observe that G is then a finitely generated abelian group, since if t denotes the generator for H and g is such that $gH \neq H$, then $\langle t, g\rangle = H$ as H has index 2. Therefore, by the classification theorem, we may write

$$G \cong \mathbb{Z}^r \oplus T,$$

where T is a finite abelian group. Since H has finite index, we must have $r = 1$. Furthermore, $H \cap T = \{e\}$ since H has no torsion elements. Therefore, we have $|G/\mathbb{Z}| \leq |G/H| = 2$, whence $G \cong \mathbb{Z}$, where $H = 2\mathbb{Z}$ has index 2 in $\mathbb{Z}$, or $G \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, which has torsion elements and thus is not isomorphic to $\mathbb{Z}$.

In the non-abelian case, we start by observing that H is normal in G, which implies that conjugation yields a non-trivial homomorphism in $\operatorname{Aut}(H)$. Since $\operatorname{Aut}(H) \cong \{-1, 1\}$, it follows that inversion is the only non-identity element of $\operatorname{Aut}(H)$, and any element of G not in H must act on H by inversion. For each such $g \notin H$, we observe that $g^2H = H$, so that $g^2 \in H$. Writing $g^2 = t^k$, we have that $gt^kg^{-1} = t^{-k}$, so that $g^2 = t^{-k}$, implying that $k = 0$, so that g has order 2. This yields the unique nontrivial semidirect product $\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$.

Problem (Problem 4)**:** Let $R = \mathbb{Z}\left[\sqrt{-11}\right]$, and let $I = \left(3, 1+\sqrt{-11}\right)$ be the ideal of R generated by 3 and $1 + \sqrt{-11}$.

(i) Prove that I is maximal.

(ii) Prove that I is not principal.

Solution:

(i) Via an evaluation homomorphism taking $p(x) \mapsto p\left(\sqrt{-11}\right)$, we observe that the map $\mathbb{Z}[x] \to \mathbb{Z}\left[\sqrt{-11}\right]$ yields an isomorphism

$$\frac{\mathbb{Z}[x]}{(x^2+11)} \cong \mathbb{Z}\left[\sqrt{-11}\right].$$

We call the left-hand side R'. Pulling back the ideal $\left(3, 1+\sqrt{-11}\right)$ into R' yields the ideal

$$I' = \frac{(3, 1+x) + \left(x^2+11\right)}{(x^2+11)}$$

in R'. Therefore, by the third isomorphism theorem, we have

$$\begin{aligned}
\mathbb{Z}\left[\sqrt{-11}\right]/\left(3, 1+\sqrt{-11}\right) &\cong R'/I' \\
&\cong \frac{\frac{\mathbb{Z}[x]}{\left(x^2+11\right)}}{\frac{(3,1+x)+\left(x^2+11\right)}{\left(x^2+11\right)}} \\
&\cong \frac{\mathbb{Z}[x]}{(3, 1+x)} \\
&\cong \frac{\frac{\mathbb{Z}[x]}{(3)}}{\frac{(3,1+x)}{(3)}} \\
&\cong \frac{(\mathbb{Z}/3\mathbb{Z})[x]}{(1+x)} \\
&\cong \mathbb{Z}/3\mathbb{Z},
\end{aligned}$$

where the final isomorphism emerges from the fact that $(x+1)$ is the kernel of the evaluation map $p(x) \mapsto p(-1)$. Therefore, the ideal I is maximal since the quotient by I yields a field.

(ii) Define a norm $N\colon \mathbb{Z}\left[\sqrt{-11}\right] \to \mathbb{N}$ given by

$$N\left(a + b\sqrt{-11}\right) = a^2 + 11b^2.$$

Notice that this norm is the restriction of the traditional norm $|z|^2 = z\overline{z}$ on $\mathbb{C}$, so in particular it is multiplicative. Suppose toward contradiction that there exists $q = a + b\sqrt{-11}$ such that $\left(3, 1+\sqrt{-11}\right) = (q)$. By the definition of the ideal (q), we have that $q|3$ and $q|1+\sqrt{-11}$, so that $N(q)|9$ and $N(q)|12$. Yet, this means that $N(q)|3$, so either $N(q) = 1$ or $N(q) = 3$. If $N(q) = 1$, then $q = 1$, implying that the ideal is not proper, which is a contradiction to what we have shown in (i). Therefore, $N(q) = 3$. Yet, there do not exist $a, b \in \mathbb{Z}$ such that $a^2 + 11b^2 = 3$, so we reach a contradiction.

Problem (Problem 5)**:** Let V be a finite-dimensional vector space over an arbitrary field F, and $T\colon V \to V$ an F-linear map. Prove that the following two conditions on T are equivalent:

(i) the characteristic polynomial and minimal polynomial of T coincide;

(ii) there exists a T-cyclic vector.

Solution: Consider an $F[x]$-module structure on V given by $x \cdot v = Tv$. Then, the invariant factors decomposition yields

$$V \cong \bigoplus_{i=1}^{m} \frac{F[x]}{(a_i(x))},$$

where $a_1|a_2|\cdots|a_m$. Notice that the minimal polynomial of T is equal to a_m, while the characteristic polynomial of T is equal to the product of all the invariant factors.

If the minimal polynomial and characteristic polynomial coincide, it follows that there is one invariant factor. In particular, this means that $V \cong F[x]/(\mu_T(x))$, so that V is a cyclic $F[x]$-module with cyclic vector given by $\overline{1} \in \frac{F[x]}{(\mu_T(x))}$.

In the reverse direction, if V admits a $F[x]$-cyclic vector, then $V = F[x]\cdot v$, so we have that $V \cong F[x]/\operatorname{Ann}_{F[x]}(v)$. Since v is $F[x]$-cyclic, it follows that any element of the annihilator ideal of v is the annihilator for all elements of V. In particular, this means that $\operatorname{Ann}_{F[x]}(v)$ consists of all the polynomials in T that yield the zero operator — that is, $\operatorname{Ann}_{F[x]}(v) = (\mu_T(x))$. Therefore, there is exactly one summand in the invariant factors decomposition of V, so this summand is both the minimal polynomial and the characteristic polynomial for T.

Problem (Problem 6): In each part, determine whether the statement is true in all cases or false in at least one case, and prove your claim.

(a) Let R be a commutative domain with 1, and M an R-module. If $x \in M$ and $y \in M$ are torsion elements, then $x + y$ is a torsion element.

(b) If K and L are fields, then $K \otimes_{\mathbb{Z}} L$ is nonzero.

(c) If R is a commutative ring with 1, and every R-module is free, then R is a field.

(d) If R is a commutative ring with 1, and M is a finitely generated R-module, then every submodule of M is finitely generated.

Solution:

(a) Let $m, n \in R \setminus \{0\}$ be such that $mx = 0$ and $ny = 0$. Then, $mn \neq 0$ since R is a domain, and we have

$$\begin{aligned}(mn)(x+y) &= (mn)x + (mn)y\\ &= n(mx) + m(ny)\\ &= 0,\end{aligned}$$

so that $x + y$ is torsion. This statement is true.

(b) Consider $K = \mathbb{F}_p$ and $L = \mathbb{Q}$. Observe that for any elementary tensor, the $\mathbb{Z}$-balancedness of the tensor product gives

$$\begin{aligned}a \otimes \frac{b}{c} &= a \otimes p\frac{b}{cp}\\ &= ap \otimes \frac{b}{cp}\\ &= 0,\end{aligned}$$

meaning that $\mathbb{F}_p \otimes \mathbb{Q} = 0$ since every elementary tensor is zero.

(c) Let $I \subseteq R$ be an ideal not equal to R. Then, R/I is a nontrivial R-module, so it is free; furthermore, $R \to R/I$ is surjective, meaning that $\operatorname{rank}(R/I) \leq \operatorname{rank}(R)$, since the rank of a free module over a commutative ring is well-defined. Since $\operatorname{rank}(R) = 1$, it follows that $\operatorname{rank}(R/I) = 0$ or $\operatorname{rank}(R/I) = 1$, while since $I \neq R$, we have that R/I is nontrivial, meaning that $\operatorname{rank}(R/I) = 1$, so $R/I \cong R$. Thus, $I = \{0\}$, so the only ideals of R are either $\{0\}$ or R, whence R is a field.

(d) If $R = \mathbb{Z}[x_1, x_2, x_3, \dots]$ is the polynomial ring with integer coefficients in infinitely many indeterminates, then $M = R$ is a free R-module over itself (hence finitely generated), while the ideal $(x_1, x_2, x_3, \dots)$ is an infinitely generated ideal (hence infinitely-generated R-submodule) of M.

Problem (Problem 7): Let p be a prime, and let F be a field of order p^6. Let

$$\operatorname{Prim}(F) = \left\{\alpha \in F : \mathbb{F}_p(\alpha) = F\right\}.$$

(a) Find an explicit formula for $|\operatorname{Prim}(F)|$.

(b) Let $\operatorname{Irr}_6(p)$ denote the set of all monic irreducible polynomials of degree 6 over $\mathbb{F}_p$. Find a simple relation between $|\operatorname{Irr}_6(p)|$ and $|\operatorname{Prim}(F)|$.

Solution:

(a) Any element of $F \cong \mathbb{F}_{p^6}$ that is not in any proper subfield of F is a generator for F. Since $\mathbb{F}_{p^6}$ contains subfields of the form $\mathbb{F}_{p^m}$ with $m|6$, it follows that the largest proper subfield of F is given by $\mathbb{F}_{p^3}$, so that there are $p^6 - p^3$ such α with $\mathbb{F}_p(\alpha) = F$.

(b) Since $\mathbb{F}_p$ is perfect, any irreducible polynomial over $\mathbb{F}_p$ is separable. Furthermore, we recall that the Galois group of $\mathbb{F}_{p^6}/\mathbb{F}_p$ is generated by the Frobenius endomorphism, where $\mathrm{fr}(x) = x^p$, and $\mathrm{fr}^6 = \mathrm{id}$. Therefore, if $\alpha \in \mathrm{Prim}(F)$, then the polynomial

$$p(x) = \prod_{i=1}^{6}\left(x - \mathrm{fr}^i(\alpha)\right)$$

is invariant under the action of the generator fr, so that $p(x) \in \mathbb{F}_p[x]$. Since $\mathrm{Gal}\left(\mathbb{F}_{p^6}/\mathbb{F}_p\right)$ acts transitively on the roots of $p(x)$, it follows that $p(x)$ is irreducible. To find all such irreducible polynomials of degree 6, it follows that each of these has degree 6, so that $|\mathrm{Irr}_6(p)| = \frac{1}{6}|\mathrm{Prim}(F)|$.

Problem (Problem 8)**:** Let F be a field, and let $f(x) \in F[x]$ be a separable irreducible polynomial of degree n.

(a) Let $\alpha \neq \beta$ be distinct roots of f in some field extension K/F. Prove that $[F(\alpha,\beta) : F] \leq n(n-1)$.

(b) Let α and β be as in (a). Prove that

$$[F(\alpha+\beta) : F] \leq \binom{n}{2}.$$

(c) Assume $F = \mathbb{Q}$ and n is prime. Give an explicit example of f and α and β satisfying the above conditions such that (a) holds. Prove that your example has the required properties.

Solution:

(a) We have that

$$[F(\alpha,\beta) : F] = [F(\alpha,\beta) : F(\alpha)][F(\alpha) : F].$$

Since f is separable and irreducible (and without loss of generality, monic), it follows that f is the minimal polynomial for α, meaning that $[F(\alpha) : F] = n$. Meanwhile, we observe that $\mu_{\beta,F(\alpha)} | \frac{f(x)}{(x-\alpha)} \in F(\alpha)[x]$, so since $\frac{f(x)}{(x-\alpha)}$ has degree $n-1$, it follows that $[F(\alpha,\beta) : F(\alpha)] \leq (n-1)$. This gives the desired result.

(b) We observe that the minimal polynomial of $(\alpha+\beta)$ over F is given by the product of all $(x-\theta_i)$, where the θ_i are the $\mathrm{Gal}(K/F)$-orbit of $(\alpha+\beta)$. Since $\alpha \neq \beta$, we have that $\sigma(\alpha) \neq \sigma(\beta)$ for all $\sigma \in \mathrm{Gal}(K/F)$, meaning that there are most $\binom{n}{2}$ distinct Galois conjugates of $(\alpha+\beta)$.

(c) We have that $p(x) = x^n - 2$, with splitting field given by $\mathbb{Q}\left(\zeta_n, \sqrt[n]{2}\right)$ where ζ_n is a primitive nth root of unity, satisfies

$$\left[\mathbb{Q}\left(\zeta_n, \sqrt[n]{2}\right) : \mathbb{Q}\right] = \left[\mathbb{Q}\left(\zeta_n, \sqrt[n]{2}\right) : \mathbb{Q}\left(\sqrt[n]{2}\right)\right]\left[\mathbb{Q}\left(\sqrt[n]{2}\right) : \mathbb{Q}\right],$$

with the latter term equal to n seeing as $p(x)$ is irreducible by the Eisenstein criterion (and separable since $\mathbb{Q}$ is characteristic zero). Meanwhile, we observe that, since n is prime, the minimal polynomial of ζ_n is given by

$$\Phi_n(x) = x^{n-1} + x^{n-2} + \cdots + x + 1,$$

and $\Phi_n(x)$ remains as such over $\mathbb{Q}\left(\sqrt[n]{2}\right)$ as there are no roots of $\Phi_n(x)$ that are real-valued. Thus, we have that

$$\left[\mathbb{Q}\left(\zeta_n, \sqrt[n]{2}\right) : \mathbb{Q}\left(\sqrt[n]{2}\right)\right] = n-1.$$

August 2022

Problem (Problem 1)**:** Let R denote the commutative ring $\mathbb{Z}\left[\sqrt{-5}\right]$.

(a) Construct two non-associate factorizations of 6 into products of irreducible elements.

(b) Using the factorizations from (a), construct an ideal $\mathfrak{a} \subseteq R$ which is not principal.

(c) Show that the square $\mathfrak{a}^2 = \mathfrak{a}\mathfrak{a}$ of the ideal from part (b) is principal.

Solution:

(a) We claim that $6 = (2)(3) = \left(1+\sqrt{-5}\right)\left(1-\sqrt{-5}\right)$ is a factorization into non-associate irreducibles. To see that these are irreducible, observe that if we define a norm $N\colon R \to \mathbb{N}$ given by $N\left(c+d\sqrt{-5}\right) = c^2+5d^2$, then this norm is inherited from the standard norm on $\mathbb{C}$, so it is multiplicative. In particular, this means that $N(ab) = N(a)N(b)$ for any $a,b \in R$. If a were invertible, then $N\left(aa^{-1}\right) = N(a)N\left(a^{-1}\right) = 1$, so $N(a) = N\left(a^{-1}\right) = 1$, and in particular, we must have that $a = 1$ or $a = -1$ in R.

Regarding the factorizations of 6, we have that $N(2) = 4$ and $N(3) = 9$. Given a factorization ab of 2, we have $N(ab) = N(a)N(b) = 4$, so either $N(a) = 4$, meaning $N(b) = 1$, or $b = \pm 1$, $N(a) = 2$, which is not possible, or $N(a) = 1$, or $a = \pm 1$. Therefore, it follows that either a or b is a unit, so that 2 is irreducible. A similar argument follows for 3, substituting the instances of 2 with 3 and the instances of 4 with 9.

Furthermore, we have $N\left(1 \pm \sqrt{-5}\right) = 6$, so if there is a factorization $ab = 1 \pm \sqrt{-5}$, then $N(ab) = N(a)N(b) = 6$, so if $N(a) = 6$, then $b = 1$ by the same argument as with 2, and $N(a) \neq 2, 3$, and if $N(a) = 1$, then $a = \pm 1$. In particular, we have that $1 \pm \sqrt{-5}$ is irreducible. Since the only units of R are ± 1, it follows that these factorizations are not associate.

(b) Consider the ideal $\mathfrak{a} = \left(3, 1+\sqrt{-5}\right) = (a,b)$. Suppose there were some $c \in R$ such that $(c) = (a,b)$. Then, we have $(a) \subseteq (c) = (a,b)$, meaning that there is some $q \in R$ such that $qc = a$. Since a is irreducible, it follows that either q is a unit or c is a unit, while since $(a,b) \neq R$, we have q is a unit. Thus, we have that $(c) = (a)$, so we have that $(b) \subseteq (a,b) = (c) = (a)$, implying that $a|b$, but b is irreducible and a is irreducible, implying that a and b are associates, but 3 and $1+\sqrt{-5}$ are not associates. Thus, $\left(3, 1+\sqrt{-5}\right)$ is not principal.

(c) We observe that $\mathfrak{a}^2 = \left(9, 3+3\sqrt{-5}, -4+2\sqrt{-5}\right)$. We start by observing that

$$\left(3+3\sqrt{-5}\right) - \left(-4+2\sqrt{-5}\right) = 7+\sqrt{-5},$$

so that $\mathfrak{b} = \left(9, 7+\sqrt{-5}\right) \subseteq \mathfrak{a}^2$. Furthermore, we have $9 \in \mathfrak{b}$, and

$$\begin{aligned} 2\left(7+\sqrt{-5}\right) - 9 &= -4+2\sqrt{-5} \\ 3\left(7+\sqrt{-5}\right) - (2)(9) &= 3+3\sqrt{-5}, \end{aligned}$$

so that $\mathfrak{b} = \mathfrak{a}^2$. Similarly, we observe that that $2-\sqrt{-5} = 9 - \left(7+\sqrt{-5}\right) \in \mathfrak{b}$, so that $\mathfrak{c} = \left(2-\sqrt{-5}\right) \subseteq \mathfrak{a}^2$. Meanwhile, we have

$$\begin{aligned} \left(2+\sqrt{-5}\right)\left(2-\sqrt{-5}\right) &= 9 \\ \left(1+\sqrt{-5}\right)\left(2-\sqrt{-5}\right) &= 7+\sqrt{-5}, \end{aligned}$$

so that $\mathfrak{a}^2 \subseteq \mathfrak{c}$. In particular, this means that $\mathfrak{a}^2$ is principal.

Problem (Problem 2)**:** For a complex $n \times n$ matrix $A = \left(a_{ij}\right)_{i,j}$, we let $\overline{A}$ denote the matrix $\left(\overline{a_{ij}}\right)_{i,j}$, where the bar denotes complex conjugation. Let χ_A denote the characteristic polynomial.

(a) If A and $\overline{A}$ are conjugate matrices, prove that the characteristic polynomial χ_A has real coefficients.

(b) Conversely, if A is diagonalizable and χ_A has real coefficients, show that A and $\overline{A}$ are conjugate.

(c) Give an example of a non-diagonalizable complex matrix A such that χ_A has real coefficients, but A and $\overline{A}$ are *not* conjugate.

Problem (Problem 3):

(a) Let G be a group, $H \subseteq G$ a normal subgroup. Assume that $Z(H) = \{e\}$, and that every automorphism of H is inner. Show that G is the direct product $H \times C_G(H)$, where $C_G(H)$ is the centralizer of H in G.

(b) Show that there is no group G such that the commutator subgroup $[G,G]$ is isomorphic to the symmetric group S_3.

Solution:

(a) Let G act on H by automorphisms — that is, $g \mapsto \iota_g$. Suppose $g \in G$ acts non-trivially. Then, since every automorphism of H is inner, it follows that there is $h \in H$ such that $gxg^{-1} = hxh^{-1}$ for all $x \in H$. This gives that $x = \left(g^{-1}h\right)x\left(g^{-1}h\right)^{-1}$, or $g^{-1}h \in Z(H)$. Since $Z(H) = \{e\}$, it follows that $g = h$.

We thus have that $G \to \operatorname{Aut}(H) \cong H$ is surjective with kernel given by $C_G(H)$, giving $G/C_G(H) \cong H$. Since $C_G(H)$ is normal and $C_G(H) \cap H = Z(H) = \{e\}$, and the second isomorphism theorem gives

$$\begin{aligned}\frac{HC_G(H)}{C_G(H)} &\cong \frac{H}{C_G(H) \cap H}\\ &\cong H,\end{aligned}$$

we have that $HC_G(H) \cong G$, so $G \cong H \times C_G(H)$ by the classification for direct products.

(b) Suppose it were the case that $[G,G] = S_3$. Since S_3 satisfies the assumptions in (a), it follows that $G \cong [G,G] \times C_G([G,G])$. Since direct products and commutators commute, it follows that

$$S_3 \cong [[G,G],[G,G]] \times [C_G([G,G]), C_G([G,G])].$$

Since $G/[G,G] \cong C_G(H)$, where $H = [G,G]$, it follows that $C_G(H)$ is abelian, so it is equal to the commutator subgroup. Meanwhile, the commutator of S_3 is given by $A_3 \cong \mathbb{Z}/3\mathbb{Z}$. Therefore, this implies that

$$S_3 \cong \mathbb{Z}/3\mathbb{Z} \times C_G(H),$$

implying that S_3 is abelian, which is not the case.

Problem (Problem 4): Let R be a commutative ring with identity. We say that a finitely generated R-module P is *projective* if P is a direct summand of a free R-module. That is, there is some $d \geq 1$ and an R-module Q such that $P \oplus Q \cong R^d$.

(a) Suppose P is a finitely generated projective R-module. Show that for every epimorphism of R-modules $\varphi\colon M \to N$, an arbitrary R-module homomorphism $\alpha\colon P \to N$ lifts to a homomorphism $\beta\colon P \to M$ such that $\varphi \circ \beta = \alpha$.

(b) Show that if P_1 and P_2 are finitely generated projective modules, then their tensor product $P_1 \otimes_R P_2$ is a finitely generated projective R-module.

January 2023

Problem (Problem 1):

(a) Let G be a finite group, and let $H \subsetneq G$ be a proper subgroup. Prove that

$$G \neq \bigcup_{g \in G} gHg^{-1}.$$

(b) Let G be a transitive subgroup of the symmetric group S_n. Prove that there exists $g \in G$ that has no fixed points on $\{1, \dots, n\}$.

Solution:

(a) We observe that there are $[G : N_G(H)]$ distinct conjugates of H, giving

$$\left|\bigcup_{g \in G} gHg^{-1}\right| \leq [G : N_G(H)](|H| - 1) + 1.$$

Since $|N_G(H)| \geq |H|$, we have $[G : N_G(H)] < [G : H]$, so that

$$\begin{aligned}\left|\bigcup_{g \in G} gHg^{-1}\right| &\leq [G : H](|H| - 1) + 1\\ &= |G| - [G : H] + 1\\ &< |G|\end{aligned}$$

since H is a proper subgroup of G.

(b) If G is a transitive subgroup of S_n, then for each $i \in \{1, \dots, n\}$, $\mathrm{Stab}_G(i) \subsetneq G$ is a proper subgroup. If $\sigma \in \mathrm{Stab}_G(i)$, then σ admits a (full) cycle decomposition including length-one cycles with a copy of (i). Furthermore, note that if $x \in G$ is any permutation, then $x\sigma x^{-1}$ has a cycle decomposition that fixes $x(i)$.

Since G acts transitively, for any $j \in \{1, \dots, n\}$, there is $x \in G$ such that $x(1) = j$. We thus have that the set of all elements in G that fixes any point is given by

$$\bigcup_{x \in G} x(\mathrm{Stab}_G(1))x^{-1}.$$

Since this does not equal all of G, it follows that there is some $g \in G$ such that g is not in this set — that is, g does not fix any point in $\{1, \dots, n\}$.

Problem (Problem 2): Let $R = \mathbb{Z}\left[\sqrt[3]{2}\right]$.

(a) Prove that R is a subring of $\mathbb{R}$.

(b) Prove that the evaluation homomorphism $\mathbb{Z}[x] \to R$, given by $p(x) \mapsto p\left(\sqrt[3]{2}\right)$ yields a ring isomorphism

$$R \cong \frac{\mathbb{Z}[x]}{(x^3 - 2)}.$$

(c) Using the isomorphism from (b) and the third isomorphism theorem, can you determine if the ideals (5) and (7) are prime or maximal?

Solution:

(a) By direct computation, it can be shown that R is closed under subtraction and multiplication.

(b) We start by observing that the map $p(x) \mapsto p\left(\sqrt[3]{2}\right)$ gives that $\left(x^3-2\right) \subseteq \ker(\varphi)$. Note that since $q(x) = x^3-2$ is irreducible by Eisenstein's criterion, we have that if $\ker(\varphi)$ were principal, then $\ker(\varphi) = \left(x^3-2\right)$ as we would have $\ker(\varphi) = (p(x)) \supseteq (q(x))$, meaning that $p(x)|q(x)$, and since $q(x)$ is irreducible, we have $p(x) = \pm q(x)$ as the only units in $\mathbb{Z}[x]$ are ± 1.

Furthermore, we note that if $\mathbb{Z}[x] \ni f(x) \in \ker(\varphi)$, then $f(x) \in \mathbb{Q}[x]$; by Gauss's lemma, $q(x)$ is monic and irreducible over $\mathbb{Q}[x]$, meaning it is the minimal polynomial for $\sqrt[3]{2}$, meaning we must have $f(x) = g(x)\left(x^3-2\right)$ for some $g(x) \in \mathbb{Q}[x]$. We will show that $g(x) \in \mathbb{Z}[x]$. Observe that we may write

$$\begin{aligned} f(x) &= cu(x) \\ g(x) &= \frac{a}{b}v(x) \end{aligned}$$

with $u(x), v(x) \in \mathbb{Z}[x]$ with content 1, $a, b, c \in \mathbb{Z}$, and $\gcd(a,b) = 1$. Therefore, we have

$$bcu(x) = av(x)\left(x^3-2\right).$$

Since taking the content of polynomials respects multiplication, it follows that $bc = a$, so that $\frac{a}{b} = c$ is an integer. In particular, this means that $g(x) \in \mathbb{Z}[x]$. Therefore, it follows that $\ker(\varphi) = \left(x^3-2\right)$.

Problem (Problem 3)**:** Let F be a field (possibly finite). Prove that the polynomial ring $F[x]$ has infinitely many prime ideals.

Solution: We observe that irreducibles are prime in $F[x]$ since $F[x]$ is a principal ideal domain (hence a unique factorization domain). Therefore, we will show that there are infinitely many irreducible polynomials over $F[x]$.

Suppose toward contradiction that there are finitely many (monic) irreducible polynomials $p_1, \dots, p_n \in F[x]$. Then, it follows that the polynomial

$$q(x) = \prod_{i=1}^{n} p_i(x) + 1$$

is reducible. In particular, $q(x)$ admits a factorization in terms of the polynomials $p_1, \dots, p_n$. Without loss of generality, we may assume that $p_1(x)|q(x)$, we have that there is some $b(x)$ such that $p_1(x)b(x) = q(x)$. Therefore, we have

$$p_1(x)(b(x) - p_2(x)p_3(x)\cdots p_n(x)) = 1.$$

Yet, this implies that $p_1(x)$ is invertible, which contradicts the assumption that $p_1(x)$ is irreducible. Thus, there are infinitely many irreducible polynomials in $F[x]$, yielding infinitely many prime ideals.

Problem (Problem 4)**:** Let $f: \mathbb{Z}^n \to \mathbb{Z}^m$ be a $\mathbb{Z}$-module homomorphism, and let $K = \ker(f)$. Prove that there exists a submodule $M \subseteq \mathbb{Z}^n$ such that $\mathbb{Z}^n = K \oplus M$.

Solution: We have the exact sequence given by

$$0 \to K \to \mathbb{Z}^n \to P \to 0,$$

where $P = \operatorname{im}(f) \subseteq \mathbb{Z}^m$. Since $P \subseteq \mathbb{Z}^m$ is a submodule of a finitely generated module over the principal ideal domain $\mathbb{Z}$, it follows that P is a free module. Writing

$$P \cong \bigoplus_{i=1}^{k} \mathbb{Z}e_i,$$

then we will show that P is projective. If $\varphi\colon M \to N$ is a surjection, and $\psi\colon P \to N$ is a map, then we may consider an extension taking $e_i \mapsto m_i \in \varphi^{-1}(\psi(e_i))$ that, by the universal property, yields a $\mathbb{Z}$-module homomorphism $\widetilde{\psi}\colon P \to M$ satisfying $\varphi \circ \widetilde{\psi} = \psi$.

In particular, we have that $\mathbb{Z}^n \xrightarrow{f} P \to 0$ is a surjection that admits a section $\iota\colon P \to \mathbb{Z}^n$ satisfying $f \circ \iota = \mathrm{id}_P$. Therefore, we have that the exact sequence

$$0 \to K \to \mathbb{Z}^n \to P \to 0$$

splits, giving $\mathbb{Z}^n \cong K \oplus P$.

Problem (Problem 6)**:** Let $K/\mathbb{Q}$ be a Galois extension with Galois group $\mathrm{Gal}(K/\mathbb{Q}) = S_n$ for some $n \geq 5$. Assume that K contains a primitive root of unity ζ_d. Prove that $d \leq 6$.

Solution: We observe that if $\zeta_d \in K$ is a primitive root of unity, then K admits the subfield $\mathbb{Q}(\zeta_d)$. Since $\mathbb{Q}(\zeta_d)/\mathbb{Q}$ is a Galois extension, it follows that $\mathrm{Gal}(K/\mathbb{Q}(\zeta_d))$ is a normal subgroup of S_n. The only normal subgroups of S_n are given by $\{e\}$, A_n, and S_n. If $\mathrm{Gal}(K/\mathbb{Q}(\zeta_d)) \cong \{e\}$, then it would imply that $\mathbb{Q}(\zeta_d) \cong K$, but this is not possible since $\mathrm{Gal}(\mathbb{Q}(\zeta_d)/\mathbb{Q}) \cong (\mathbb{Z}/d\mathbb{Z})^\times$, which is not isomorphic to S_n since it is abelian while S_n is not. If $\mathrm{Gal}(K/\mathbb{Q}(\zeta_d)) \cong S_n$, then $\mathbb{Q}(\zeta_d) \cong \mathbb{Q}$, so $d = 1$. If $\mathrm{Gal}(K/\mathbb{Q}(\zeta_d)) \cong A_n$, then $\mathbb{Q}(\zeta_d)/\mathbb{Q}$ has degree equal to $[S_n : A_n] = 2$, meaning that we get $d = 3, 4, 6$.

Problem (Problem 7)**:**

(a) Let L/K be a field extension, and let $f(x) \in K[x]$. Prove that there is an isomorphism of L-algebras given by

$$L \otimes_K \frac{K[x]}{(f(x))} \cong \frac{L[x]}{(f(x))}.$$

(b) Suppose L/K is a finite Galois extension of degree n. Prove that there is an isomorphism of L-algebras

$$L \otimes_K L \cong L^n.$$

Solution:

(a) We consider the K-bilinear map $\left(\ell, \overline{g(x)}\right) \mapsto \overline{\ell g(x)}$. We claim that this map is well-defined. If $\overline{g_1(x)} = \overline{g_2(x)}$ in $K[x]/(f(x))$, then $g_1(x) = f(x)h(x) + g_2(x)$ for some $h(x) \in K[x]$, and we have

$$\begin{aligned} \phi\left(\ell, \overline{g_1(x)}\right) &= \overline{\ell(g_1(x))} \\ &= \overline{\ell(g_2(x) + f(x)h(x))} \\ &= \overline{\ell g_2(x)} + \overline{\ell f(x)h(x)} \\ &= \phi(\ell, g_2(x)). \end{aligned}$$

This yields a well-defined K-linear map $\varphi\colon L \otimes_K \frac{K[x]}{(f(x))} \to \frac{L[x]}{(f(x))}$ by the universal property. By using the L-algebra structure on the left-hand side, we have φ is a map of L-algebras.

To show surjectivity, we observe that if $n = \deg(f)$, then every element of $L[x]/(f(x))$ is represented by a polynomial of the form

$$a(x) = \sum_{i=0}^{n-1} a_i x^i,$$

with $a_i \in L$. Then, we have that

$$\varphi\left(\sum_{i=0}^{n-1} a_i \otimes \overline{x}^i\right) = \sum_{i=0}^{n-1} a_i \overline{x}^i$$

$$= \overline{a(x)}.$$

To show injectivity, we count dimensions over L. Observe that $1 \otimes \overline{x}^i$ for $i = 0, \dots, n-1$ forms an L-basis for $L \otimes_K K[x]/(f(x))$, giving that $L \otimes_K K[x]/(f(x))$ is dimension n over L. Similarly, $\overline{x}^i$ for $i = 0, \dots, n-1$ forms an L-basis for $L[x]/(f(x))$, so that the dimensions of both sides are equal, yielding that φ is injective.

(b) Since L/K is a finite Galois extension, it is a finite separable extension, so it is generated by a single element $\alpha \in L$ such that $K(\alpha) = L$. Since $K(\alpha) = \frac{K[x]}{(\mu_{\alpha,K}(x))}$, we get that

$$L \otimes_K \frac{K[x]}{(\mu_{\alpha,K}(x))} \cong \frac{L[x]}{(\mu_{\alpha,K}(x))}.$$

Since, over L, we have that $\mu_{\alpha,K}(x)$ splits into distinct linear factors, Chinese Remainder Theorem gives

$$\frac{L[x]}{(\mu_{\alpha,K}(x))} \cong \prod_{i=1}^{n} \frac{L[x]}{(x-\alpha_i)},$$

where the α_i are distinct roots of $\mu_{\alpha,K}(x)$ in L. Therefore, we have the series of isomorphisms

$$\begin{aligned} L \otimes_K L &\cong L \otimes_K K(\alpha) \\ &\cong L \otimes_K \frac{K[x]}{(\mu_{\alpha,K}(x))} \\ &\cong \frac{L[x]}{(\mu_{\alpha,K}(x))} \\ &\cong \prod_{i=1}^{n} \frac{L[x]}{(x-\alpha_i)} \\ &\cong L^n. \end{aligned}$$

August 2023

Problem (Problem 1): Let G be a finite group, and let $g \in G$ be an element different from e that lies in every nontrivial subgroup $H \subseteq G$.

(a) Show that G is a p-group for some prime p, and moreover, the order of g equals p.

(b) Prove that if the group G is abelian then it must be cyclic.

(c) Provide an example of a non-abelian (hence non-cyclic) finite group that contains a nontrivial element that lies in every nontrivial subgroup.

Solution:

(a) Since g is contained in every nontrivial subgroup, $\langle g \rangle \leq H$ for all nontrivial subgroups H. Then, every subgroup of G is divisible by the order of g, which is not equal to 1. Letting $p | o(g)$ be a prime, we may find an element h of $\langle g \rangle$ such that $o(h) = p$ by Cauchy's theorem.

Suppose there were a prime $q \neq p$ such that q divides the order of G. Then, we may write $|G| = q^r m$ for some $r \geq 1$. Yet, this would give that G has a q-Sylow subgroup Q, and we would have $\langle g \rangle \leq Q$, but this would imply that $o(h) | q^r$, but $p \nmid q$. Thus, G is a p-group.

Observe then that g is an element of all the subgroups $\langle g \rangle, \langle g^2 \rangle, \dots, \langle g^{o(g)-1} \rangle$. In particular, this means that $\gcd(o(g), k) = 1$ for all $1 \leq k \leq o(g) - 1$, so that $\varphi(o(g)) = o(g) - 1$. In particular, this means that $o(g) = p$ for some prime p since $\varphi(k) = k - 1$ if and only if k is prime.

(b) If G is an abelian p-group, then if G is not cyclic, the structure of finite abelian groups gives at least two non-trivial factors $G \cong G_1 \times G_2$ with G_1 and G_2 abelian p-groups, and $G_1G_2 = G$. Yet, the assumptions on g say that $e \neq g \in G_1 \cap G_2 = \{e\}$, yielding a contradiction.

Problem (Problem 2)**:** Let G be a simple group of order 60.

(a) Use the Sylow theorems to show that G has a subgroup H of index 6.

(b) Use the action of G on the coset space G/H by left multiplication to construct an embedding $\phi\colon G \to S_n$. Then, show that the image $\phi(G)$ of this embedding is contained in $\Gamma := A_6$.

(c) Thinking of G as a subgroup of Γ, use the action of G on the coset space Γ/G by left multiplication to embed G into A_5, then conclude that $G \cong A_5$.

Solution:

(a) Writing $60 = 2^2 \cdot 3 \cdot 5$. We observe that the number of 5-Sylow subgroups divides 12 and equals 1 mod 5, implying there are 6 5-Sylow subgroups since G is simple. Since the number of 5-Sylow subgroups is equal to the index of the normalizer $[G : N_G(P_5)]$ of some 5-Sylow subgroup P_5, it follows that $N_G(P_5)$ is our desired subgroup H.

(b) Label the cosets $\{H, g_1H, \dots, g_5H\}$, and let $G \curvearrowright G/H$ by taking $g \cdot (g_iH) = (gg_i)H$. This yields a permutation representation $\phi\colon G \to \mathrm{Sym}(G/H) \cong S_6$. The action is non-trivial, so since G is simple, $\ker(\phi) = \{e\}$, meaning this is an embedding.

Next, since $\mathrm{sgn}\colon S_6 \to \mathbb{Z}/2\mathbb{Z}$ is a nontrivial homomorphism with kernel A_6, we get that $\mathrm{sgn} \circ \phi\colon G \to \mathbb{Z}/2\mathbb{Z}$ is a homomorphism whose kernel must be equal to G, as else we would have an index 2 subgroup of $\phi(G) \cong G$ that would be normal. Therefore, $G \subseteq \ker(\mathrm{sgn}) = A_6$.

(c) If G acts on the coset space Γ/G, then we observe that G is its own stabilizer, meaning we may view the action via a permutation representation as $\rho\colon G \to \mathrm{Sym}((\Gamma/G) \setminus \{G\}) \cong S_5$. This map is yet again injective by the simplicity of G, and by an analogous argument to the previous part, we must have that $G \subseteq A_5$ yet again. Since G and $\phi(G)$ have the same cardinality with $\phi(G) \subseteq A_5$, it follows that $G \cong \phi(G) = A_5$.

Problem (Problem 3)**:** Let R be a commutative ring with 1, and let $I, J \subseteq R$ be two ideals.

(a) Consider R, I, and J as R-modules. Show that the map $f\colon I \oplus J \to R$ taking $(x, y) \mapsto x + y$ is a homomorphism of R-modules.

(b) Assume that $I + J = R$. Show that $I \oplus J \cong R \oplus IJ$.

Solution:

(a) Observe that

$$\begin{aligned} f((x_1, y_1) + (x_2, y_2)) &= x_1 + x_2 + y_1 + y_2 \\ &= x_1 + y_1 + x_2 + y_2 \\ &= f(x_1, y_1) + f(x_2, y_2) \\ f(r(x, y)) &= rx + ry \\ &= r(x + y) \\ &= rf(x, y). \end{aligned}$$

(b) If $I + J = R$, then we observe that $IJ = I \cap J$. The inclusion $IJ \subseteq I \cap J$ is immediate, while we observe that $1 = a + b$ for some $a \in I$ and $b \in J$, so that if $z \in I \cap J$, then $z = za + zb \in IJ$, whence $IJ = I \cap J$.

We observe that $f(x, y) = 0$ if and only if $x = -y$, meaning that $I \cap J$ embeds into $I \oplus J$ via the map $x \mapsto (x, -x)$, giving rise to the exact sequence

$$0 \to IJ \to I \oplus J \xrightarrow{f} R \to 0.$$

Since R is a projective R-modules, the map f admits a splitting, so that $R \oplus IJ \cong I \oplus J$.

Problem (Problem 4)**:** Let $A = J_n(\lambda)$ be a Jordan block of size $n \geq 1$ with eigenvalue λ over a field K. Find all A-invariant subspaces of the n-dimensional vector space $V \cong K^n$.

Solution: Viewing V as a $K[x]$-module with the action $x \cdot v = Av$, we obtain that, by the definition of A, there is one summand in the elementary divisors decomposition of A, given by

$$V \cong \frac{K[x]}{(x-\lambda)^n}.$$

The right-hand side admits a basis given by

$$\left\{\overline{(x-\lambda)}^{n-1}, \overline{(x-\lambda)}^{n-2}, \dots, \overline{(x-\lambda)}, \overline{1}\right\} = \{e_1, e_2, \dots, e_n\}.$$

Observe that we have

$$\begin{aligned} xe_1 &= \overline{(x-\lambda)}^n + \lambda\overline{(x-\lambda)}^{n-1} \\ &= \lambda e_1 \\ xe_2 &= \overline{(x-\lambda)}^{n-1} + \lambda\overline{(x-\lambda)}^{n-2} \\ &= e_1 + \lambda e_2 \\ &\vdots \end{aligned}$$

yielding a complete flag of invariant subspaces given by

$$\{0\} \subseteq \operatorname{span}\{e_1\} \subseteq \operatorname{span}\{e_1, e_2\} \subseteq \cdots \subseteq \operatorname{span}\{e_1, \dots, e_{n-1}\} \subseteq V,$$

yielding $(n+1)$ such invariant subspaces. Meanwhile, we observe that under the $K[x]$-module interpretation, the invariant subspaces are precisely the submodules of $\frac{K[x]}{(x-\lambda)^n}$, and the submodules are the ideals of this ring, which are the ideals of $K[x]$ containing $(x-\lambda)^n$. Since $(q(x)) \supseteq ((x-\lambda)^n)$ if and only if $q(x) | (x-\lambda)^n$, and $(x-\lambda)$ is irreducible, it follows that any such ideal is some ideal of the form $(x-\lambda)^k$, implying that these are indeed all the invariant subspaces.

Problem (Problem 5)**:** Identify the following tensor products:

(a) $\mathbb{Q}(x) \otimes_{\mathbb{Q}[x]} \mathbb{Q}(x)$ as a $\mathbb{Q}[x]$-module;

(b) $\mathbb{Q}\left(\sqrt{2}\right) \otimes_{\mathbb{Q}} \mathbb{Q}\left(\sqrt{2}\right)$ as a $\mathbb{Q}$-algebra.

Solution:

(a) By using the fact that elementary tensors in $\mathbb{Q}(x) \otimes_{\mathbb{Q}[x]} \mathbb{Q}(x)$ are $\mathbb{Q}[x]$-balanced, we have that any elementary tensor has an equivalent representative given by

$$\begin{aligned} \frac{a(x)}{b(x)} \otimes \frac{c(x)}{d(x)} &= \frac{a(x)c(x)}{b(x)} \otimes \frac{1}{d(x)} \\ &= \frac{a(x)c(x)d(x)}{b(x)d(x)} \otimes \frac{1}{d(x)} \\ &= \frac{a(x)c(x)}{b(x)d(x)} \otimes \frac{d(x)}{d(x)} \\ &= \frac{a(x)c(x)}{b(x)d(x)} \otimes 1. \end{aligned}$$

In particular, using $\mathbb{Q}[x]$-linearity, we have that any element of $\mathbb{Q}(x) \otimes_{\mathbb{Q}[x]} \mathbb{Q}(x)$ is represented by an elementary tensor of the form $\frac{p(x)}{q(x)} \otimes 1$. Therefore, we get an isomorphism $\mathbb{Q}(x) \otimes_{\mathbb{Q}[x]} \mathbb{Q}(x) \to \mathbb{Q}(x)$ given by taking $\frac{p(x)}{q(x)} \otimes 1 \mapsto \frac{p(x)}{q(x)}$. That it is injective follows from the fact that $\frac{p(x)}{q(x)} = 0$ if and only if

$p(x) = 0$ if and only if $\frac{p(x)}{q(x)} \otimes 1 = 0$. That it is surjective follows immediately from the definition of $\frac{p(x)}{q(x)} \otimes 1$.

(b) Using the isomorphism $\mathbb{Q}\left(\sqrt{2}\right) \cong \frac{\mathbb{Q}[x]}{(x^2-2)}$, we use the isomorphism $L \otimes_K \frac{K[x]}{(p(x))} \cong \frac{L[x]}{(p(x))}$ for any field extension $L \supseteq K$ and the Chinese Remainder Theorem to obtain

$$\begin{aligned}\mathbb{Q}\left(\sqrt{2}\right) \otimes_{\mathbb{Q}} \mathbb{Q}\left(\sqrt{2}\right) &\cong \mathbb{Q}\left(\sqrt{2}\right) \otimes_{\mathbb{Q}} \frac{\mathbb{Q}[x]}{(x^2-2)} \\ &\cong \frac{\mathbb{Q}\left(\sqrt{2}\right)[x]}{(x^2-2)} \\ &\cong \frac{\mathbb{Q}\left(\sqrt{2}\right)[x]}{\left(x-\sqrt{2}\right)} \times \frac{\mathbb{Q}\left(\sqrt{2}\right)[x]}{\left(x+\sqrt{2}\right)} \\ &\cong \mathbb{Q}\left(\sqrt{2}\right) \times \mathbb{Q}\left(\sqrt{2}\right).\end{aligned}$$

Problem (Problem 6): Let p be an odd prime, and consider the polynomial $f(x) = x^4 - x^2 + 1$ in $\mathbb{F}_p[x]$.

(a) For which p is $f(x)$ separable?

(b) Show that for $p > 3$, the polynomial $f(x)$ splits into linear factors over the extension $\mathbb{F}_{p^2}$ of $\mathbb{F}_p$. What happens for $p = 3$?

(c) Find the minimal odd prime p for which $f(x)$ splits into linear factors over $\mathbb{F}_p$.

Solution:

(a) We start by observing that for $p = 2$, $f'(x) = 0$, so that $\gcd(f, f') \neq 1$. If $p > 3$, then $\gcd(f'(x), f(x)) = 1$ as $f'(x) = 4x^3 - 2x \neq 0$. Meanwhile, if $p = 3$, then $f'(x) = 4x^3 - 2x \equiv x^3 + x$. Since $f(x)\left(x^2+1\right) = x^6 + 1$, we have that $f(x)\left(x^2+1\right) - \left(x^2+1\right)^3 = 0$, or that $f(x) = \left(x^2+1\right)^2$, so that $\gcd(f(x), f'(x)) \neq 1$.

(b) Notice that $f(x)\left(x^2+1\right) = x^6+1$, meaning that $f(x)\left(x^2+1\right)\left(x^6-1\right) = x^{12}-1$. Taking the factorization

$$x^{12} - 1 = (\Phi_{12}(x))(\Phi_4(x))(\Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_6(x)),$$

we then find that $f(x) = \Phi_{12}(x)$. In particular, to show that f splits into linear factors over $\mathbb{F}_{p^2}$ for any $p > 3$, we show that for any $p > 3$, $\mathbb{F}_{p^2}$ admits a primitive 12th root of unity. Note that if $p > 3$, then either $p \equiv 1 \bmod 3$ or $p \equiv -1 \bmod 3$, and similarly, since p is odd, either $p \equiv 1 \bmod 4$ or $p \equiv -1 \bmod 4$. Taking squares, we have that $p^2 \equiv 1 \bmod 3$ and $p^2 \equiv 1 \bmod 4$, meaning that $3 | \left(p^2-1\right)$ and $4 | \left(p^2-1\right)$, so that $12 | \left(p^2-1\right)$. Since $\left|\mathbb{F}_{p^2}^\times\right| \cong \mathbb{Z}/\left(p^2-1\right)\mathbb{Z}$, it follows that, since $12 | \left(p^2-1\right)$, we have a unique subgroup of order 12 in $\mathbb{F}_{p^2}^\times$, so that $\mathbb{F}_{p^2}$ admits a primitive 12th root of unity.

If $p = 3$, then we may write

$$\begin{aligned}f(x)\left(x^2+1\right) &= x^6 + 1 \\ &= \left(x^2+1\right)^3,\end{aligned}$$

so that

$$\left(x^2+1\right)\left(f(x) - \left(x^2+1\right)^2\right) = 0,$$

giving that $f(x) = \left(x^2+1\right)^2$. Notice that -1 is not a square mod 3, so that $f(x) = (x-\omega)^2(x+\omega)^2$ over $\mathbb{F}_{3^2}$, where ω is a primitive cube root of unity.

(c) We want to find the smallest odd prime such that $\mathbb{F}_p$ contains a primitive 12th root of unity. That is, the smallest odd prime p such that $12 | p - 1$. This gives that 13 is the smallest such odd prime.

Problem (Problem 7)**:** Use cyclotomic extensions to construct two distinct cyclic Galois extensions of $\mathbb{Q}$ of degree 7, and show that there exists a Galois extension of $\mathbb{Q}$ with Galois group isomorphic to $\mathbb{Z}/7\mathbb{Z}\times\mathbb{Z}/7\mathbb{Z}$.

Solution: Consider ω to be a primitive 29th root of unity. Then, $\mathbb{Q}(\omega)/\mathbb{Q}$ is a cyclic Galois extension of degree 28 with Galois group $G = (\mathbb{Z}/29\mathbb{Z})^{\times}$. In particular, we have a unique order 4 subgroup $H \leq G$, giving a fixed field $\mathbb{Q}(\omega) \supseteq K \supseteq \mathbb{Q}$ with $[G : H] = [K : \mathbb{Q}] = 7$.

Similarly, we may use a primitive 43rd root of unity ν to construct a cyclic Galois extension $\mathbb{Q}(\nu)/\mathbb{Q}$ of degree 42 with Galois group $G' = (\mathbb{Z}/43\mathbb{Z})^{\times}$. There is a unique order 6 subgroup $H' \leq G'$ yielding a fixed field $\mathbb{Q}(\nu) \supseteq L \supseteq \mathbb{Q}$ with $[G' : H'] = [L : \mathbb{Q}] = 7$.

Notice that both K and L are Galois over $\mathbb{Q}$ since the groups G and G' are abelian, so all their subgroups are normal.

We claim now that $\mathbb{Q}(\omega) \cap \mathbb{Q}(\nu) = \mathbb{Q}$. For this, observe that we have

$$\begin{aligned}[][\mathbb{Q}(\omega\nu) : \mathbb{Q}] &\leq [\mathbb{Q}(\omega, \nu) : \mathbb{Q}]\\ &= \frac{[\mathbb{Q}(\omega) : \mathbb{Q}][\mathbb{Q}(\nu) : \mathbb{Q}]}{[\mathbb{Q}(\omega) \cap \mathbb{Q}(\nu) : \mathbb{Q}]}\\ &\leq [\mathbb{Q}(\omega) : \mathbb{Q}][\mathbb{Q}(\nu) : \mathbb{Q}]\\ &= (28)(42).\end{aligned}$$

Notice that $\omega\nu$ is a primitive root of unity with order $(29)(47)$, yielding a degree equal to $\varphi((29)(47)) = (28)(42)$. Therefore, we must have that $[\mathbb{Q}(\omega) \cap \mathbb{Q}(\nu) : \mathbb{Q}] = 1$. In particular, this means that $K \cap L = \mathbb{Q}$. Since $KL/\mathbb{Q}$ is Galois as $K/\mathbb{Q}$ and $L/\mathbb{Q}$ are Galois, and we have $K \cap L = \mathbb{Q}$, it follows that the isomorphism $\operatorname{Gal}(KL/\mathbb{Q}) \cong \operatorname{Gal}(K/\mathbb{Q}) \times \operatorname{Gal}(L/\mathbb{Q})$, so that $\operatorname{Gal}(KL/\mathbb{Q}) \cong \mathbb{Z}/7\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$.

Problem (Problem 8)**:** Let K be a field of characteristic $p > 0$, and let L/K be a cyclic Galois extension of degree p. Let σ be a generator of the Galois group $\operatorname{Gal}(L/K)$. We consider σ as a linear transformation of L as a vector space over K.

(a) Find the eigenvalues of σ.

(b) Identify the Jordan canonical form of σ.

(c) Use the result from (b) to show there exists $\alpha \in L$ such that $\sigma(\alpha) = \alpha + 1$.

Solution:

(a) We see that $\sigma^p = 1$; since $\left\{1, \sigma, \sigma^2, \dots, \sigma^{p-1}\right\}$ are linearly independent (by the linear independence of characters), it follows that $\mu_\sigma(p) = x^p - 1 = (x-1)^p$. Therefore, there is one eigenvalue of σ, namely 1.

(b) We may view L as a $K[x]$-module where x acts by σ. Taking the elementary divisors form of the corresponding finitely generated module, we get that

$$L \cong \bigoplus_{i=1}^{r} \frac{K[x]}{(x-1)^{d_i}},$$

where $d_1 + \cdots + d_r = p$. To identify the number of Jordan blocks, we notice that the largest Jordan block has size $\max(d_1, \dots, d_r)$, which is also equal to the degree of the minimal polynomial $(x-1)^p$. Yet, since $d_1, \dots, d_r$ add up to p, it follows that there is exactly one Jordan block, namely giving

$$L \cong \frac{K[x]}{(x-1)^p}.$$

In particular, this means that σ has a Jordan canonical form given by one Jordan block with size p and eigenvalue 1.

(c) Let $\theta \in L$ be such that

$$\begin{aligned}\operatorname{Tr}(\theta) &= \sum_{i=0}^{p-1} \sigma^i(\theta)\\ &= 1.\end{aligned}$$

Then, if we define

$$\beta = \sum_{i=0}^{p-1} i\sigma^i(\theta),$$

we get

$$\begin{aligned}\sigma(\beta) &= \sum_{i=0}^{p-1} i\sigma^{i+1}(\theta)\\ &= \sum_{i=0}^{p-1}(i-1)\sigma^i(\theta)\\ &= \sum_{i=0}^{p-1} i\sigma^i(\theta) - \sum_{i=0}^{p-1}(\theta)\\ &= \beta - 1,\end{aligned}$$

so that $\alpha = -\beta$ is our desired element.

January 2024

Problem (Problem 1)**:** Let $R \subseteq \mathbb{Q}$ be a subring with 1.

(a) Show that if $\frac{a}{b} \in R$ with $a, b \in \mathbb{Z}$, $b \neq 0$, and $\gcd(a,b) = 1$, then $\frac{1}{b} \in R$.

(b) Show that R coincides with the localization $\mathbb{Z}_S$ of $\mathbb{Z}$ with respect to some multiplicative set $S \subseteq \mathbb{Z}$ which should be described explicitly.

(c) For a prime p, define

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, p \nmid b \right\}.$$

Show that every $\mathbb{Z}_{(p)}$ is a maximal proper subring of $\mathbb{Q}$, and conversely, that every maximal proper subring $R \subseteq \mathbb{Q}$ is of the form $\mathbb{Z}_{(p)}$.

Solution:

(a) We find $x, y \in \mathbb{Z}$ such that $xa + yb = 1$. Then, since $b \neq 0$, we have that $x\left(\frac{a}{b}\right) + y = \frac{1}{b}$. Since $x, y \in \mathbb{Z}$, they are equal to some integer multiple of 1, whence x and y are both elements of R, meaning that $\frac{1}{b} \in R$.

(b) Consider the set defined by

$$S = \left\{ b \in \mathbb{Z} : \frac{1}{b} \in R \right\}.$$

Then, we observe that the set S is multiplicatively closed, since $1 \in S$ as $1 \in R$ and, for any $b, d \in S$, we have that $\frac{1}{b}, \frac{1}{d} \in R$, whence $\frac{1}{bd} \in R$ since R is a subring of $\mathbb{Q}$, so that $bd \in S$.

Consider a map $\mathbb{Z} \to R$ given by inclusion. Then, we observe that, if $b \in S$, then since $\frac{1}{b} \in R$, we have $b\left(\frac{1}{b}\right) = 1$, so that $b \in R^\times$. Since $S \subseteq R^\times$, the universal property of localization yields a map $\mathbb{Z}_S \to R$ given by $\frac{a}{b} \mapsto \frac{a}{b}$. This map is surjective since for any $\frac{a}{b} \in R$, we have that $\frac{a}{b} \in \mathbb{Z}_S$ maps to $\frac{a}{b} \in R$, while $0 = \frac{a}{b} \in R$ if and only if $a = 0$ if and only if $\frac{a}{b} = 0$ in S. Thus, this yields our desired isomorphism.

(c) We start by observing that, if we consider $\mathbb{Z}_{(p)}$ as a localization $\mathbb{Z}_S$, then each element $b \in S$ not equal to 1 defines a factorization into primes $p_1^{e_1} \cdots p_r^{e_r}$, none of which are equal to p. In particular, since the only restriction on $b \in S$ is that $p \nmid b$, we have that the set of all primes in S is all primes except for p. This yields maximality of the subring $\mathbb{Z}_{(p)}$, since any $\mathbb{Q} \supseteq R \supsetneq \mathbb{Z}_{(p)}$ would define a set of primes that must include p, hence yielding all of $\mathbb{Q}$ as the corresponding localization set S would be equal to all of $\mathbb{Z}$ not equal to 0.

If $R \subseteq \mathbb{Q}$ is a maximal subring, we observe that the corresponding localization set S is a maximal multiplicatively closed subset that is not equal to all of $\mathbb{Z}_{>0}$. Consider the smallest element q of $\mathbb{Z}_{>0} \setminus S$. We claim that q is prime. If not, then we would have some composites $b, c \in \mathbb{Z}_{>0}$ such that $bc = q$. Yet, since q is minimal, it follows that $b, c \in S$, implying that $bc = q \in S$, contradicting the definition of q. We claim that q does not divide any element of S; if it were the case that there were some multiple aq in S, then we would have $\frac{1}{aq} \in R$, meaning that $\frac{a}{aq} = a\left(\frac{1}{aq}\right) = \frac{1}{q}$ is in R; therefore, q does not divide any element of S, so $R \subseteq \mathbb{Z}_{(q)}$, where we define

$$\mathbb{Z}_{(q)} = \left\{\frac{a}{b} : a, b \in \mathbb{Z}, q \nmid b\right\}.$$

Since R is a maximal subring, it follows that $R = \mathbb{Z}_{(q)}$.

Problem (Problem 2)**:** Let K be a field.

(a) Let $K[x], K[y], K[x,y]$ be rings of polynomials in x, y, and both x and y respectively. Show that the natural inclusions $K[x] \to K[x,y]$ and $K[y] \to K[x,y]$ induce a ring homomorphism $\varphi\colon K[x] \otimes_K K[y] \to K[x,y]$, and that this homomorphism is actually a ring isomorphism.

(b) Let $K(x)$, $K(y)$, and $K(x,y)$ be the fields of rational functions in their respective variables (i.e., the fields of fractions of their respective polynomial rings). Show that the inclusions $K(x) \to K(x,y)$ and $K(y) \to K(x,y)$ give rise to a ring homomorphism $\psi\colon K \to K(x) \otimes_K K(y) \to K(x,y)$. Derive, using ψ, that $K(x) \otimes_K K(y)$ is not a field.

Problem (Problem 3)**:** Let G be a finite group of order n. Let $X = \{(x,y) \in G \times G : xy = yx\}$. Show that X has size nh, where h is the number of conjugacy classes of G.

Solution: Let C be a conjugacy class. Notice that $|C| = [G : C_G(g)]$ for any $g \in C$, meaning that $|G| = |C||C_G(g)|$.

Observe that for each $g \in C$, the section

$$\begin{aligned} X_g &= \{h \in G : (g,h) \in X\} \\ &= \{h \in G : gh = hg\} \\ &= \left\{h \in G : g = hgh^{-1}\right\} \\ &= C_G(g). \end{aligned}$$

Therefore, the set of all $(g,h) \in X$ with $g \in C$ has cardinality equal to $|C||C_G(g)| = |G|$. Therefore, since conjugacy classes are disjoint, it follows that $|X| = nh$.

Problem (Problem 4)**:** Let $T\colon V \to V$ be a linear transformation of an n-dimensional vector space V over a field K, and let $\chi_T(x)$ and $\mu_T(x)$ be the characteristic and minimal polynomials of T in $K[x]$ respectively.

(a) Prove that if $\deg(\mu_T) < n$, then V has a T-invariant subspace different from $\{0\}$ and V.

(b) Prove that V does not have any T-invariant subspaces different from $\{0\}$ and V if and only if $\chi_T(x)$ is irreducible in $K[x]$.

Solution:

(a) Define a $K[x]$-module structure on V by taking $x \cdot v = Tv$. The submodules are equivalent to the T-invariant subspaces.

Since V is a finitely generated $K[x]$-module, the structure theorem in invariant factors yields

$$V \cong \bigoplus_{i=1}^{n} \frac{K[x]}{(a_i(x))},$$

where we have $a_1(x)|a_2(x)|\cdots|a_n(x) = \mu(x)$ are nonzero, non-unit monic polynomials. The product of all the invariant factors has degree equal to n, and is also equal to the characteristic polynomial. Since $\deg(\mu(x)) < n$, it follows that there is a nontrivial invariant factor $a(x)$ in the decomposition not equal to $\mu(x)$. The corresponding $K[x]$-submodule $\frac{K[x]}{(a(x))}$ is not equal to V or $\{0\}$, so that V has a nontrivial invariant subspace.

(b) Suppose V has no nontrivial invariant subspaces. There is then one summand in the invariant factors decomposition, as otherwise the summand would be a $K[x]$-submodule of V not equal to $\{0\}$ or V. Therefore, we have

$$V = \frac{K[x]}{(\chi(x))}.$$

Any submodule of the ring $K[x]/(\chi(x))$ is an ideal of the ring; by the third isomorphism theorem, there is a one-to-one correspondence between the ideals of the quotient ring $K[x]/(\chi(x))$ and the ideals of $K[x]$ containing $(\chi(x))$. Since there are no nontrivial invariant subspaces, it follows that the only ideals of $K[x]$ containing $(\chi(x))$ are equal to $(\chi(x))$ and $K[x]$, implying that $(\chi(x))$ is a maximal ideal. Therefore, $\chi(x)$ is irreducible.

Now, suppose that $\chi(x)$ is irreducible. The only monic polynomials that divide $\chi(x)$ are equal to 1 or $\chi(x)$. In particular, since all the invariant factors of T divide $\chi(x)$, and are nonzero, non-unit monic polynomials, it follows that there are no invariant factors of T not equal to $\chi(x)$, so that

$$V \cong \frac{K[x]}{(\chi(x))}.$$

As discussed in the previous section, the ideals of $K[x]/(\chi(x))$ are precisely the $K[x]$-submodules of V, which are precisely the invariant subspaces of V. Since there are no ideals in $K[x]/(\chi(x))$ not equal to either $\{0\}$ or V, as $\chi(x)$ is irreducible implying that $K[x]/(\chi(x))$ is a field, it follows that V has no nontrivial invariant subspaces.

Problem (Problem 5)**:** Let $f(x) \in K[x]$ be a nonconstant monic polynomial in one variable over a field K, and let $f = p_1^{a_1}\cdots p_r^{a_r}$ be its factorization into a product of powers of distinct irreducible monic polynomials. Find necessary and sufficient conditions in terms of r and $a_1,\dots,a_r$ for the quotient ring $K[x]/(f)$ to be:

(a) a field;

(b) a local ring which is not a field;

(c) a direct product of fields.

Solution:

(a) Since $K[x]$ is a principal ideal domain, irreducibles are prime, and prime ideals are maximal. Therefore, $K[x]/(f)$ is a field if and only if f is irreducible, which holds if and only if $r = 1$ and $a_1 = 1$ in the factorization $f = p_1^{a_1}\cdots p_r^{a_r}$.

(b) The ring $K[x]/(f)$ is a non-field if f has a nontrivial factorization into irreducibles. Since any ideal of $K[x]/(f)$ corresponds to an ideal $(q) \supseteq (f)$, which is given by a polynomial $q|f$. The maximal ideals of $K[x]/(f)$ then correspond to irreducible factors $p|f$. Therefore, if $K[x]$ is a local ring, there is then only one irreducible factor of f, and if $K[x]$ is not a field, then this irreducible factor must have exponent greater than 1. That is, $r = 1$ and $a_1 > 1$.

(c) Since, for any $i \neq j$, $p_i, p_j | f$, and $a_i, a_j \geq 1$, $\gcd\left(p_i^{a_i}, p_j^{a_j}\right) = 1$, we have that the ideals $\left(p_i^{a_i}\right)$ and $\left(p_j^{a_j}\right)$ are coprime. Therefore, by Chinese Remainder Theorem, we have

$$K[x]/(f) \cong \bigoplus_{i=1}^{r} \frac{K[x]}{\left(p_i^{a_i}\right)}.$$

Assuming that $K[x]/(f)$ is a field thus implies that each of the ideals $\left(p_i^{a_i}\right)$ are maximal, meaning that each $a_i = 1$. Therefore, we have that $K[x]/(f)$ with $r \geq 1$ and $a_i = 1$ for each $i = 1, \dots, r$.

Problem (Problem 6)**:** Let L/K be a finite Galois extension of degree n. Show that for every prime p dividing n, there exists an intermediate subfield $K \subseteq M \subseteq L$ such that $[M : K]$ is prime to p.

Solution: Let $G = \operatorname{Gal}(L/K)$. Then, if $p | [L : K] = |G| = n$, we may write $|G| = p^k m$ with $\gcd(m, p) = 1$. There exists a p-Sylow subgroup $H \leq G$. Taking $M = L^H$ as the fixed field, then we have $\operatorname{Gal}(L/M) = H$, so that

$$\begin{aligned} [M : K] &= \frac{[L : K]}{[L : M]} \\ &= m. \end{aligned}$$

In particular, we have $\gcd([M : K], p) = 1$. Since, for any prime $p | [L : K]$, there exists a p-Sylow subgroup, the corresponding fixed field has $[M : K]$ is prime to p.

August 2024

Problem (Problem 1)**:** Let s, n be positive integers, and let I_n denote the $n \times n$ identity matrix. Consider the following matrix

$$A_{s,n} = \begin{pmatrix} 0 & I_n & 0 & \cdots & 0 & 0 \\ 0 & 0 & I_n & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & I_n \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix},$$

an $s \times s$ matrix of $n \times n$ blocks. Find the Jordan canonical form of $A_{s,n}$ over $\mathbb{C}$.

Solution: First, we observe that $\mu_A(x) = x^s$ since the $n \times n$ blocks are preserved whenever A is multiplied by itself, and we have that A^{s-1} is given by a single copy of I_n in the $(1, s)$ block of A. Meanwhile, we also observe that $\chi_A(x) = x^{sn}$. Therefore, viewing $\mathbb{C}^{sn}$ as a $\mathbb{C}[x]$-module with x acting via A, we have that $\mathbb{C}^{sn}$ in elementary divisors form is of the form

$$\mathbb{C}^{sn} \cong \left(\bigoplus_{i=1}^{r} \frac{\mathbb{C}[x]}{x^{d_i}} \right) \oplus \frac{\mathbb{C}[x]}{x^s},$$

where the latter term denotes the size of the largest Jordan block, and we have $d_1 + \cdots + d_r + s = sn$ with each $d_i \geq 1$. To find the number of Jordan blocks with eigenvalue 0 (or, since $\chi_A(x)$ has only eigenvalue zero, the whole Jordan block structure of A), we only need to compute the geometric multiplicity of 0. Observe that if $(v_1, \dots, v_s)$ is an ordered collection of s n-dimensional vectors, then A acts via taking $(v_1, \dots, v_s) \mapsto (v_2, \dots, v_s, 0)$. In particular, this means that any collection of the form $(v, 0, \dots, 0)$ is mapped to 0 by A, so that $\dim \ker(A) = n$.

Since there are n total Jordan blocks, it follows that each of $d_1, \dots, d_r = s$, that $r = n - 1$, and that $\mathbb{C}^{sn}$ decomposes in elementary divisors form as

$$\mathbb{C}^{sn} \cong \left(\frac{\mathbb{C}[x]}{x^s}\right)^{\oplus n}.$$

Therefore, the Jordan canonical form for A is given in block diagonals by $A \sim \operatorname{diag}(J_s(0), \dots, J_s(0))$, where there are n total blocks.

Problem (Problem 2): Let $\mathbb{F}_q$ be a finite field with q elements. Denote by $\mathrm{GL}_n(\mathbb{F}_q)$ the group of invertible $n \times n$ matrices with coefficients in $\mathbb{F}_q$, and by $\mathrm{Gr}(k, n)$ the set of all k-dimensional subspaces in $\mathbb{F}_q^n$.

(a) Find the cardinality of the group $\mathrm{GL}_4(\mathbb{F}_q)$.

(b) Show that the natural action of $\mathrm{GL}_n(\mathbb{F}_q)$ gives rise to a transitive action of $\mathrm{GL}_n(\mathbb{F}_q)$ on $\mathrm{Gr}(k, n)$.

(c) Find the cardinality of the set $\mathrm{Gr}(3, 4)$.

Solution:

(a) Any element of $\mathrm{GL}_4(\mathbb{F}_q)$ is determined by selecting four linearly independent vectors in $\mathbb{F}_q^4$. Counting these yields

$$\left|\mathrm{GL}_4(\mathbb{F}_q)\right| = \left(q^4 - 1\right)\left(q^4 - q\right)\left(q^4 - q^2\right)\left(q^4 - q^3\right).$$

(b) We may view $\mathrm{Gr}(k, n)$ as the collection of ordered sets of k linearly independent n-dimensional vectors (with a suitable notion of equivalence). We will show that $\mathrm{GL}_4(\mathbb{F}_q)$ acts transitively on all these sets, so that it acts transitively on $\mathrm{Gr}(k, n)$.

Given any two sets $(v_1, \dots, v_k)$ and $(w_1, \dots, w_k)$ in $\mathrm{Gr}(k, n)$, we may extend each of these to ordered bases $\left(v_1, \dots, v_k, v'_{k+1}, \dots, v'_n\right)$ and $\left(w_1, \dots, w_k, w'_{k+1}, \dots, w'_n\right)$. We may define a linear transformation T taking $v_i \mapsto w_i$ and $v'_j \mapsto w'_j$ for each $i = 1, \dots, k$ and $j = k+1, \dots, n$. Taking the matrix of this linear transformation defines an element of $\mathrm{GL}_4(\mathbb{F}_q)$.

(c) Since $\mathrm{GL}_4(\mathbb{F}_q)$ acts transitively on $\mathrm{Gr}(3, 4)$, we may use orbit–stabilizer to find the number of elements in $\mathrm{Gr}(3, 4)$.

A matrix A preserves $\operatorname{span}\{e_1, e_2, e_3\}$ if and only if A is block upper-triangular of the form

$$A = \begin{pmatrix} B & v \\ 0 & t \end{pmatrix},$$

where $B \in \mathrm{GL}_3(\mathbb{F}_q)$, $v \in \mathbb{F}_q^3$, and $t \in \mathbb{F}_q \setminus \{0\}$. The total number of these matrices is then given by $q^3(q-1)\left(q^3 - 1\right)\left(q^3 - q\right)\left(q^3 - q^2\right)$. Thus, we have

$$\begin{aligned} |\mathrm{Gr}(k, n)| &= \frac{\left|\mathrm{GL}_4(\mathbb{F}_q)\right|}{(q^3 - 1)(q^3 - q)(q^3 - q^2)\left(q^4 - q^3\right)} \\ &= (q+1)\left(q^2 + 1\right). \end{aligned}$$

Problem (Problem 3): Let S be a principal ideal domain, and let R be a subdomain of S that is also a PID. Let $a, b \in R$. Show that the GCD of a and b in R and the GCD of a and b in S differ only by a unit in S.

Solution: Let d be the GCD in S and d' the GCD in R. We will show that d' is the GCD in S. First, observe that we may write

$$d = s_1 a + s_2 b$$

$$d' = r_1 a + r_2 b$$

where $s_1, s_2 \in S$ and $r_1, r_2 \in R$. Since $d|a$ and $d|b$, we have that $d|(r_1 a + r_2 b) = d'$ in S. Similarly, since $d'|a$ and $d'|b$, it follows that $d'|(s_1 a + s_2 b) = d$ in S. Therefore, since $d|d'$ and $d'|d$, we have that $(d) = (d')$, so that d and d' necessarily differ by a unit, as S is a unique factorization domain.

Problem (Problem 4)**:** Fix a prime number p. Let $\mathbb{F}_p$ be a field of order p, and let $\mathbb{F}_q$ be a finite field of order $q = p^n$ elements. Show that, as commutative algebras, there is an isomorphism

$$\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathbb{F}_q \cong \mathbb{F}_q^{\times n}.$$

Solution: There is a separable irreducible polynomial $q(x)$ in $\mathbb{F}_p[x]$ with degree n such that

$$\mathbb{F}_q \cong \frac{\mathbb{F}_p[x]}{(q(x))}.$$

Using the tensor product isomorphism of rings given by $L \otimes_K \frac{K[x]}{(f(x))} \cong \frac{L[x]}{(f(x))}$ for any extension L/K, we may use the Chinese Remainder Theorem with

$$\begin{aligned}
\mathbb{F}_q \otimes_{\mathbb{F}_p} \mathbb{F}_q &\cong \mathbb{F}_q \otimes_{\mathbb{F}_p} \frac{\mathbb{F}_p[x]}{(q(x))} \\
&\cong \frac{\mathbb{F}_q[x]}{(q(x))} \\
&\cong \prod_{i=1}^{n} \frac{\mathbb{F}_q}{(x - \alpha_i)} \\
&\cong \mathbb{F}_q^{\times n}.
\end{aligned}$$

Problem (Problem 5)**:** If G is a group of order 483, show that $G \cong H \times K$, where H is a group of order 23 and K is a group of order 21.

Solution: Using the factorization $|G| = 3 \cdot 7 \cdot 23$. There exists a 23-Sylow subgroup K, and this 23-Sylow subgroup is normal since there are either 1 or 24 such Sylow subgroups, and the number of these Sylow subgroups must divide 21. Similarly, there is exactly one 7-Sylow subgroup, since the number of 7-Sylow subgroups must divide $3 \cdot 23$ and is congruent to 1 mod 7. This 7-Sylow subgroup is normal. Call it P. If Q is any 3-Sylow subgroup, then $H = PQ$ is a subgroup of G since P is normal.

We note that $H \cap K = \{e\}$ since all non-trivial elements of K are of order 23, while any element of H is of order either 3, 7, or 21. Furthermore, this means that $|HK| = (21)(23) = 483 = |G|$, so $HK = G$. Finally, observe that there are no nontrivial homomorphisms $H \to \operatorname{Aut}(K) \cong \mathbb{Z}/22\mathbb{Z}$. Therefore, we have that $G \cong K \times H$, where K is a group of order 23 and H is a group of order 21.

Problem (Problem 6)**:** Let D be the dihedral group of order 8. The generators and relations for D are given by

$$D = \left\langle s, t | s^2, t^4, stst \right\rangle.$$

(a) List all conjugacy classes of D.

(b) Describe all the 1-dimensional irreducible representations of D over $\mathbb{C}$.

(c) Compute the character table of D.

Solution:

(a) Computing all the conjugacy classes of D is found by brute force. Specifically, we see that the following are all the conjugacy classes of D:

$$C_1 = \{e\}$$

$$\begin{aligned} C_2 &= \left\{t^2\right\} \\ C_3 &= \left\{s, st^2\right\} \\ C_4 &= \left\{t, t^3\right\} \\ C_5 &= \left\{st, st^3\right\}. \end{aligned}$$

(b) Letting $\varphi\colon G \to \mathbb{C}^\times$ be a 1-dimensional representation, we have that $\varphi(t)^4 = 1$, $\varphi(s)^2 = 1$, and $\varphi(t) = \varphi(t)^3$. Combined, this gives that $\varphi(s)^2 = 1$ and $\varphi(t)^2 = 1$, so that there are exactly 4 1-dimensional representations, found by selecting the values of $\varphi(s)$ and $\varphi(t)$ to be equal to either $+1$ or -1.

(c) Since there are five conjugacy classes, there are five (equivalence classes of) irreducible representations. The sum of squared dimensions of these representations (equal to the character on the identity conjugacy class) equals the order of the group, so the final irreducible representation is order 2.

Labeling these representations $\rho_1, \dots, \rho_5$ and corresponding characters $\chi_1, \dots, \chi_5$, we observe that $\rho_5 \cong \rho_5 \otimes \rho_j$ for any $j = 1, \dots, 4$ since the dimension of tensor products of irreducible representations is multiplicative. In particular, for any conjugacy class not equal to $\left\{t^2\right\}$, we have that χ_5 evaluates to 0.

Finally, we have that $\langle \chi_5, \chi_5 \rangle = 1$ by the irreducibility criterion, so $\left|\chi_5\left(t^2\right)\right|$, and since $\langle \chi_1, \chi_5 \rangle = 0$ by the first orthogonality relation, we have that $\chi_5\left(t^2\right) = -2$. This gives the following character table.

	$\{e\}$	$\left\{t^2\right\}$	$\left\{t, t^3\right\}$	$\left\{s, st^2\right\}$	$\left\{st, st^3\right\}$
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1
χ_5	2	-2	0	0	0

Problem (Problem 7)**:** Consider the field extension $\mathbb{Q}\left(\sqrt{8 + 2\sqrt{15}}\right)$ over $\mathbb{Q}$.

(a) Determine the degree of the extension.

(b) Show that this is a Galois extension.

(c) Compute the Galois group for this extension.

Solution:

(a) We start by seeking a simplified form of $\sqrt{8 + 2\sqrt{15}}$. This can be seen by the fact that $\left(\sqrt{3} + \sqrt{5}\right)^2 = 8 + 2\sqrt{15}$, so that we are dealing with $\mathbb{Q}\left(\sqrt{3} + \sqrt{5}\right)$. Observe now that $\mathbb{Q}\left(\sqrt{3} + \sqrt{5}\right) = \mathbb{Q}\left(\sqrt{3}, \sqrt{5}\right)$, since we have

$$\frac{2}{\sqrt{5} + \sqrt{3}} = \sqrt{5} - \sqrt{3},$$

meaning that $\sqrt{5} \in \mathbb{Q}\left(\sqrt{3} + \sqrt{5}\right)$ and $\sqrt{3} \in \mathbb{Q}\left(\sqrt{3} + \sqrt{5}\right)$, so that $\mathbb{Q}\left(\sqrt{3}, \sqrt{5}\right) \subseteq \mathbb{Q}\left(\sqrt{3} + \sqrt{5}\right) \subseteq \mathbb{Q}\left(\sqrt{3}, \sqrt{5}\right)$.

Next, we see that

$$\left[\mathbb{Q}\left(\sqrt{3}, \sqrt{5}\right) : \mathbb{Q}\right] = \left[\mathbb{Q}\left(\sqrt{3}, \sqrt{5}\right) : \mathbb{Q}\left(\sqrt{5}\right)\right]\left[\mathbb{Q}\left(\sqrt{5}\right) : \mathbb{Q}\right].$$

Observe that any element of $\mathbb{Q}\left(\sqrt{5}\right)$ is of the form $a + b\sqrt{5}$ for some $a, b \in \mathbb{Q}$. If it were the case that $\sqrt{3} \in \mathbb{Q}\left(\sqrt{5}\right)$, then we would have

$$\sqrt{3} = a + b\sqrt{5},$$

or that

$$3 = a^2 + 5b^2 + 2ab\sqrt{5},$$

implying that $\sqrt{5} = 3 - a^2 - 5b^2$, or that $\sqrt{5}$ is rational, which is not the case. Therefore, each of these extensions is of degree 2, so that $\left[\mathbb{Q}\left(\sqrt{3}, \sqrt{5}\right) : \mathbb{Q}\right] = 4$.

(b) Observe that $\mathbb{Q}\left(\sqrt{3}, \sqrt{5}\right)$ is the splitting field for the separable polynomial $p(x) = \left(x^2 - 3\right)\left(x^2 - 5\right)$. Therefore, the extension is Galois.

(c) There are three subfields of $\mathbb{Q}\left(\sqrt{3}, \sqrt{5}\right)$, given by $\mathbb{Q}\left(\sqrt{3}\right), \mathbb{Q}\left(\sqrt{5}\right), \mathbb{Q}\left(\sqrt{15}\right)$. These correspond to three subgroups of the Galois group of $\mathbb{Q}\left(\sqrt{3}, \sqrt{5}\right)$ over $\mathbb{Q}$. Since the extension is degree 4, there are three subgroups of the Galois group, meaning that by the classification of groups of order 4, we must have that the Galois group is equal to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Problem (Problem 8)**:** Let $R = \mathbb{Q}[x]/\left(x^2 - 9\right)$, and let M be an R-module. Show that there is a direct sum decomposition of R-modules $M \cong M_+ \oplus M_-$, where $x \in R$ acts on $M_\pm$ as ± 3.

Solution: By the Chinese Remainder Theorem, we have

$$\begin{aligned} R &= \frac{\mathbb{Q}[x]}{((x+3)(x-3))} \\ &\cong \frac{\mathbb{Q}[x]}{(x-3)} \oplus \frac{\mathbb{Q}[x]}{(x+3)}, \end{aligned}$$

where we see that $6 \in (x-3) + (x+3)$, so that $(x-3)$ and $(x+3)$ are coprime ideals. Observe that, for any element $p(x) \in R$, we have that p is determined via $(p(3), p(-3))$ in the Chinese Remainder Theorem decomposition. Let $\varphi\colon R \to \frac{\mathbb{Q}[x]}{(x-3)} \oplus \frac{\mathbb{Q}[x]}{(x+3)}$ be the corresponding isomorphism.

We observe that, over $\frac{\mathbb{Q}[x]}{(x-3)} \oplus \frac{\mathbb{Q}[x]}{(x+3)}$, M gains a module structure by taking the ring operation to be defined by $(a,b) \cdot m \equiv \varphi^{-1}(1,0)(am) + \varphi^{-1}(0,1)(bm)$.

$$\begin{aligned} M_1 &= \left\{\varphi^{-1}(1,0) \cdot m : m \in M\right\} \\ M_2 &= \left\{\varphi^{-1}(0,1) \cdot m : m \in M\right\}, \end{aligned}$$

seeing that $M_1 \cap M_2 = \{0\}$ as any $m \in M_1 \cap M_2$ is equal to

$$\begin{aligned} m &= \varphi^{-1}(1,0)m \\ &= \varphi^{-1}(0,1)\left(\varphi^{-1}(1,0)m\right) \\ &= \varphi^{-1}((0,1)(1,0))m \\ &= 0, \end{aligned}$$

while $M_1 + M_2$ follows from the fact that $m = \varphi^{-1}(0,1)m + \varphi^{-1}(1,0)m$.

To find the action of $x \in R$ on this direct sum decomposition, it suffices to find polynomial representatives $p_+(x)$ and $p_-(x)$ such that $\varphi(p_+(x)) = (1,0)$ and $\varphi(p_-(x)) = (0,1)$. Since the quotient ring R has representatives given by degree 1 polynomials, we find that these representatives in R can be given by

$$\begin{aligned} p_+(x) &= \frac{1}{6}x + \frac{1}{2} \\ p_-(x) &= -\frac{1}{6}x - \frac{1}{2}. \end{aligned}$$

Inside M, we see that this yields $M_1 = p_+ M$ and $M_2 = p_- M$. Taking the action of x on M then yields $x \cdot (p_+ m) = (xp_+)m$; observe that $\varphi(x \cdot p_+) = ((xp_+)(3), (xp_+)(-3)) = (3,0)$, so that $(xp_+)m = 3p_+ m$. Similarly, we have $x \cdot (p_- m) = (xp_-)m$, and $\varphi(xp_-) = ((xp_-)(3), (xp_-)(-3)) = (0,-3)$, whence $(xp_-)m = -3p_- m$.

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Problem (Problem 2): Consider a group action $G \times X \to X$.

(a) Show that G naturally extends to an action on X^k away from the multi-diagonal, i.e. on the set

$$\left\{(x_1, \dots, x_k) \in X^k : x_i \neq x_j \text{ whenever } i \neq j\right\}.$$

(b) The group action of G on X is said to be k-transitive if the action on X^k away from the multi-diagonal is transitive. Prove that the natural action of the alternating group A_n on $[n] = \{1, \dots, n\}$ is $(n-2)$-transitive.

Solution:

(a)

(b) Given tuples $(a_1, \dots, a_{n-2})$ and $(b_1, \dots, b_{n-2})$ away from the multi-diagonal, we want to find some $\sigma \in A_n$ such that $\sigma(a_i) = b_i$ for each i.

Write c_1, c_2 for the elements of $[n]$ not in $(a_1, \dots, a_{n-2})$, and write d_1, d_2 for the elements of $[n]$ not in $(b_1, \dots, b_{n-2})$. Define the permutation

$$\sigma = (a_1, b_1)(a_2, b_2) \cdots (a_{n-2}, b_{n-2})(c_1, d_1)(c_2, d_2).$$

If σ is even, then we are done. Else, if σ is not even, then the permutation $\sigma' = (d_1, d_2)\sigma$ does not affect any codomain symbol in $(b_1, \dots, b_{n-2})$, so σ' still takes $a_i \mapsto b_i$ for each $i = 1, \dots, n-2$, whence A_n acts $(n-2)$-transitively on $[n]$.

Problem (Problem 4): Let $\phi \colon \mathbb{Z}^r \to \mathbb{Z}^r$ be a homomorphism of free abelian groups given by multiplication by a square matrix $A \in \operatorname{Mat}_r(\mathbb{Z})$.

(a) Show that $\operatorname{im}(\phi)$ is of finite index in the target if and only if $\det(A) \neq 0$.

(b) Show that when the index is finite, it is equal to $|\det(A)|$.

Solution:

(a) Write $N = \operatorname{im}(\phi)$. Then, since $N \leq \mathbb{Z}^r$ is a submodule of a finitely generated free module over a principal ideal domain, the compatible basis theorem yields the existence of $y_1, \dots, y_r \in \mathbb{Z}^r$ and nonzero elements $a_1, \dots, a_k \in \mathbb{Z}$ with $(a_1) \supseteq (a_2) \supseteq \cdots \supseteq (a_k)$ such that

$$\begin{aligned} \mathbb{Z}^r &= \langle \{y_1, \dots, y_r\} \rangle \\ N &= \langle \{a_1 y_1, \dots, a_k y_k\} \rangle. \end{aligned}$$

Notice that the compatible bases are determined by the matrices $P, Q \in \operatorname{GL}_r(\mathbb{Z})$ in the Smith Normal Form decomposition $\operatorname{SNF}(A) = PAQ$.

Suppose N has finite index in $\mathbb{Z}^r$. Then, we must have that $\mathbb{Z}^r/N$ is a finite abelian group; by the compatible basis theorem, since

$$\mathbb{Z}^r/N \cong \mathbb{Z}^{r-k} \oplus \left(\bigoplus_{i=1}^{k} \mathbb{Z}/(a_i) \right),$$

it follows that $r = k$ (as else this would contradict the finitude of the index of N), so that the coefficients in the Smith Normal Form of A are given by $a_1, \dots, a_r$, all of which are nonzero (i.e., the compatible basis for N has the same cardinality as the rank of M). Since

$$\begin{aligned} \det(\operatorname{SNF}(A)) &= \det(PAQ) \\ &= \det(P)\det(A)\det(Q) \\ &= u \det(A) \end{aligned}$$

for some unit u of $\mathbb{Z}$; since $u = \pm 1$, it follows that $\det(A) = \pm(a_1 \cdots a_k) \neq 0$.

Meanwhile, if N has infinite index in $\mathbb{Z}^r$, then we have

$$\mathbb{Z}^r/N = \mathbb{Z}^{r-k} \oplus \left(\bigoplus_{i=1}^{k} \mathbb{Z}/(a_i)\right),$$

with $r > k$. In particular, this means that there are $k < r$ invariant factors $a_1, \dots, a_k$, and the Smith Normal form of A is given by $\operatorname{diag}(a_1, \dots, a_k, 0, \dots, 0)$. In particular, this means that $\det(A) = 0$.

(b) Recalling that if $N \leq \mathbb{Z}^r$ has finite index, then the Smith Normal Form of A has determinant given by $\det(\mathrm{SNF}(A)) = a_1 \cdots a_k$, and that $\det(A) = \pm\det(\mathrm{SNF}(A))$. Furthermore, the index of N is given by the total number of elements in $\mathbb{Z}^r/N$, which is equal to the absolute value of the product $a_1 \cdots a_k$ of the invariant factors since N is a direct sum of cyclic groups, whence the index is equal to $|\det(A)|$.

Problem (Problem 6)**:** Let A be a finitely generated abelian group such that $A \otimes_{\mathbb{Z}} \mathbb{Z}/25\mathbb{Z} \cong A$. Classify all such groups up to isomorphism.

Solution: First, we establish the isomorphism that $R/I \otimes_R R/J \cong R/(I+J)$, where R is a commutative unital ring. For this, observe that any simple tensor of $R/I \otimes_R R/J$ can be written using the R-balancedness of the tensor product as

$$\begin{aligned}(a+I) \otimes (b+J) &= a(1+I) \otimes (b+J)\\ &= (1+I)(ab+J).\end{aligned}$$

In particular, since any element of $R/I \otimes_R R/J$ is a sum of simple tensors all of the form $(1+I)(t+J)$, it follows that any element of $R/I \otimes_R R/J$ is of the form $(1+I)(t+J)$.

Defining the R-balanced map $\phi\colon R/I \times R/J \to R/(I+J)$ by $(a+I, b+J) \mapsto (ab) + (I+J)$ yields an R-module homomorphism $R/I \otimes_R R/J \to R/(I+J)$ taking $(1+I) \otimes (t+J) \mapsto t + (I+J)$. Since this map admits the inverse $t + (I+J) \mapsto (1+I) \otimes (t+J)$, it follows that $R/I \otimes_R R/J \cong R/(I+J)$.

Using the classification of finitely generated abelian groups in elementary divisors form as well as the fact that tensor products distribute over direct sums yields that $A \otimes \mathbb{Z}/25\mathbb{Z}$ is given by

$$\left(\mathbb{Z}^k \oplus \bigoplus_{i=1}^{r} \mathbb{Z}/\left(p_i^{d_i}\right)\mathbb{Z}\right) \otimes \mathbb{Z}/25\mathbb{Z} \cong (\mathbb{Z}/25\mathbb{Z})^k \oplus \bigoplus_{i=1}^{r} \mathbb{Z}/\left(\left(p_i^{d_i}\right) + (25)\right).$$

In order for such an isomorphism $A \otimes_{\mathbb{Z}} \mathbb{Z}/25\mathbb{Z} \cong A$ to hold, we cannot have any free part as the right-hand side is a torsion $\mathbb{Z}$-module, and by the definition of the greatest common denominator, we have $\left(p_i^{d_i}\right) + (25) = \left(\gcd\left(p_i^{d_i}, 25\right)\right)$ in $\mathbb{Z}$. Since $\gcd\left(p_i^{d_i}, 25\right) = p_i^{d_i}$ if and only if $p_i = 5$ for each i and $d_i = 1, 2$, it follows that the isomorphism $A \otimes_{\mathbb{Z}} \mathbb{Z}/(25) \cong A$ holds if and only if A is of the form

$$A \cong (\mathbb{Z}/5\mathbb{Z})^{\oplus n} \oplus (\mathbb{Z}/25\mathbb{Z})^{\oplus m}$$

for some integers $n, m \geq 0$.

Problem (Problem 8)**:** Let F be a field of characteristic $p > 0$, and consider the polynomial $f(x) = x^p - x - c$. Let α be a root of $f(x)$ in an algebraic closure $\overline{F}$ of F.

(a) Prove that $f(x)$ has roots $\alpha, \alpha+1, \dots, \alpha+p-1$.

(b) Prove that $F(\alpha)/F$ is a Galois extension.

(c) Prove that $f(x)$ is either irreducible or splits into linear factors in $F[x]$.

Solution:

(a) Suppose α is a root of $f(x)$. Then, if $0 \leq k \leq p-1$, the Frobenius endomorphism yields

$$\begin{aligned} f(\alpha + k) &= (\alpha + k)^p - (\alpha + k) - c \\ &= (\alpha^p - \alpha - c) + (k^p - k). \end{aligned}$$

Since $k \in \mathbb{F}_p$, we have $k^p = k$, so that $f(\alpha + k) = 0$.

(b) Notice that $\alpha + k \in F(\alpha)$ for each $0 \leq k \leq p-1$ since there is a copy of $\mathbb{F}_p$ inside every field of characteristic p. In particular, this means that $F(\alpha)/F$ is a normal extension as all the roots of $f(x)$ are in $F(\alpha)$. The extension is also separable since $\alpha + k \neq \alpha + \ell$ whenever $k \neq \ell$, and there are exactly p roots of f given by $\alpha + k$ for each $0 \leq k \leq p-1$. Therefore, $F(\alpha)/F$ is separable. Since $F(\alpha)/F$ is normal and separable, it is Galois.

(c) Suppose $\alpha \in F$. Then, all of $\alpha + k$, $0 \leq k \leq p-1$, are in F, meaning that $f(x)$ splits into linear factors over $F[x]$. If $\alpha \notin F$, then all of $\alpha + k$, $0 \leq k \leq p-1$, are *not* in F, so that none of the roots of $f(x)$ are contained in F. In particular, this means $F[x]$ is irreducible.